WOMEN HEALTH

VOL - I

Dr. Mamta Bansal

WOMEN HEALTH

VOL-1

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BFC PUBLICATIONS

BFC PUBLICATIONS

Published in India by BFC Publications Private Limited

CP-61, Viraj Khand, Gomti Nagar,

Lucknow-226010, Uttar Pradesh, India

www.bfcpublications.com

ISBN – 978-93-5632-569-2

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Foreword

I am delighted to write the foreword for this book **"Women Health-Vol 1**" edited **by Dr. Mamta Bansal.**

Dr. Mamta Bansal is serving as Assistant Professor at Maharaja Agrasen University and has been teaching health and environment associated courses in various departments like, Applied Sciences, Pharmacy, Engineering and Management. She has a very rich experience in her field, a virtuous command over the subject and practical experience in her field.

Today's women are sharing equal responsibility at homes and at the workplace too. Despite advances and progress in medical facilities and the eradication of social disparity in society, the basic requirements for the overall well-being of women are still a dream, particularly in developing nations. Many women in the world are still not able to explore their full potential because of persistent health issues, inadequate health and education systems, and persistent inequalities. Progressing, protecting and promoting the women's health leads to healthy family and healthy society. So, it has become important that every woman should get awareness and education to improve their health conditions. In this book all aspects related to women's health throughout their life span is nicely explained section wise in the form of individual chapters. The data and evidences indicate key health issues that affect girls and women. It offers a comprehensive analysis of the state of women and health around the globe.

The present book will surely educate the girls and women at local, regional and global level. The readers will have positive feelings after reading this book. I appreciate all efforts taken by authors from different related fields for the successful completion of the book. I wish editor of the book to have spirit to write more good work in future to benefit the society.

Prof. (Dr.) R.K.Gupta

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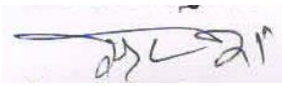
Foreword

In today's scenario, women health has become an urgent priority worldwide. Women are biologically, mentally and physically different from men and therefore have different health issues in their lifespan. Therefore, they need specific attention, devotion and consideration. Most of the women issues require proper care, knowledge and awareness instead of medical treatment. The present book is structured around different stages of life course of women. The book will help the readers to understand women health and what women can do to be healthier. It is designed to provide the rapid information and motivation to improve the overall well-being of girls and women. I appreciate the efforts of Editor Dr. Mamta Bansal for the accomplishment of the book. I am looking forward to its timely publication.



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PREFACE

Women health is a multidisciplinary subject related to girls, women and society that promotes better and balanced understanding of health. The title of the book "Women Health- Vol I" appears to be encouraging and promising to learn more about health of a woman. The concept of women health does not only deal with etiology of diseases and mortality but also discuss overall personality of a woman beyond the narrow focus. This broader approach is more multifaceted with woman-oriented approach than the disease-oriented approach. Emphasis is given on the issues and challenges and on how to tackle and overcome the challenges which sometime is not easily accessible to the readers.

Book coverage

Women Health-Vol I is designed both as an introduction to the subject and as a textbook for students pursuing their under-graduation and post-graduation studies in this subject matter. For the convenience, book is divided into three sections and thirteen chapters. Each chapter of the book centres on the specified objectives. References are provided at the end of each chapter. Simple illustrations, formulae, tables and figures are added wherever needed, to aid the understanding of concepts. A concise conclusion to highlight the main points is given at the end of every chapter.

First section is 'Understanding Women Health' that has four chapters focusing on basic health of women of different areas and different phase of life. Women play many roles in their life and focus more on the health of their partners and children, neglecting their own health. In fact, a number of health related problems among women actually be prevented or minimized with proper knowledge, awareness and timely care. Problems in adolescents are far different form the problems in older women which are separately discussed as individual chapters. In the same way, problems associated with rural women are different from the challenges of urban women which are again discussed separately. Gender discrimination is still

a major issue that persist in many areas around world, along with inequities in distribution of benefits provided by the government. It is imperative that research advances and treatment benefits should reach to masses to maintain good health.

Second section is 'Burden of Diseases' that comprises of five chapters that explain the concept of menstrual cycle, polycystic ovarian syndrome, cancers, HIV/AIDS and mental disorders. This part deals with the increase in the pattern of diseases in recent times. Along with the illnesses mentioned in the chapters, today's girls and women are also facing mental health issues like depression, suicidal tendencies, insomnia, anxiety, that require focused approach to deal with. The need of the hour is to adopt the easiest way to remain healthy which is 'Prevention'. Prevention means early diagnose, timely healthcare provision and effective treatment of the disease.

Third section of the book is 'Enhancing Women Health' comprises four chapters that talk about total well-being and promoting health of the women. Individual chapter explains regarding health promoting behaviour among women, role of social media to enhance women health, managing stress and legislative aspects of women health. This section of the book is dedicated to the goal of promoting health and quality of life throughout their lives through proper information, knowledge, awareness and motivation.

It is really important that women are educated by all health aspects at home and at the workplace too. A healthy woman means a healthy family and hence healthy society.

I shall be grateful to the readers for genuine suggestions for the improvement of the book, so that it may serve the audience even better. Thanks to all those whose contributions are cited in the book and my sincere apology for any typo error in text.

"Hope every woman get a little bit better everyday"

- Dr. Mamta Bansal

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CHAPTER 1

ADOLESCENCE PROBLEMS AMONG GIRLS

Dr. Mamta Bansal, Ms. Prabhjot Kaur, Ms. Silky Bensal

It is better to be violent, if there is violence in our hearts, than to put on the cloak of nonviolence to cover impotence. Mahatma Gandhi

Adolescence is a Latin word which means 'to grow to maturity'. It is a transitional period between puberty and adult in human development extending mainly over the teen years. It has a broader meaning which includes mental, social and emotional maturity. It is a period of physical and psychological development from the onset of puberty to complete growth and maturity. It is usually a time of good health, growth and development. This phase of life is depicted by different names such as 'Critical period', 'Awkward period' or 'Transitional period'. Simultaneously, it is a time of risk for them like health – related behaviours, social and mental dilemma, substance use alcohol, drugs, smoking, injuries etc. Adolescence is considered as most rapid phase of human development. It is also a period of special attention. This period is something 'In-between' phase where an adolescent girl/boy is no longer a dependent, biologically immature, socially incompetent and psychologically weak, as well not yet independent, grown-up, stanch and adult. It is very difficult to describe female adolescence accurately and affirmatively as it encompasses a part of adult and a part of a child. Many health professionals now use the word 'Adolescent turmoil' to describe both normal and disturbed adolescents.

Stages of adolescence period

Adolescence is a period of life during which physical, psychological, emotional and social maturity develops. Adolescence typically happens in three stages:

Early adolescence: It starts from 10 to 14 years and characterized by a sexual maturation changes in body image, change in behaviour like the beginning of the shift from dependence on parents to independent behaviour.

Middle adolescence: It begins between the ages of 15 and 17 and is marked by the rising significance of peer-group values, a sense of identity, growing independence, and a growing distance from parents. This time is frequently when conflicts with parents reach their height.

Late adolescence: It is a final stage of adolescence starts from 18 to 21 years. Adolescent during this period become more responsible for work and relationships.

Key facts

- Over 1.5 million adolescents and young adults aged 10–24 died in 2020, about 5000 per day.
- The major cause of deaths includes Injuries (including road traffic injuries and drowning), violence, self-harm, suicide and family conditions.
- Half of all mental health disorders and depressions in adolescence start by age 14, but in most cases it is undetected and evens untreated.
- Early onset of substance use is associated with higher risks of developing dependence and other problems during adult life.

Challenges among adolescents

It is the period of life with lowest mortality rates and maximum physical, social and emotional changes. At precisely the time when they are most in need of support that could help protect them from health risks and changes occurring in their body. But in many countries, and numerous part of counties girls are not given due attention and support they need to deal with the changes. Even their living area, lack adequate housing conditions, hygienic environment and healthy surroundings for their healthy overall development (Fig. 1). In these conditions girls are not able to grow congruously from childhood to adulthood. They also face significant burden of diseases along with various other mental health affairs. For example, in developed countries like America, Europe and Western Pacific regions, the female adolescents are suffering from depressive disorders, schizophrenia disorders and other neuropsychiatric conditions. The highest rate of mortality in adolescent girls are found in Africa and South-East Asia which is in the form of suicides, injuries like road accidents, burns. Communicable

diseases including HIV/AIDS and respiratory tract infections are the cause of deaths in low-income countries especially in Africa (Table 1). In some settings, adolescent girls are pressured into early marriage and childbearing. According to a data 38% of girls in developing countries, markedly in Asia, marry before the age of 18 and 14% before the age of 15.

Puberty

Puberty is a period during which adolescents reach sexual maturity and become capable of reproduction. On average, puberty begins in girls around ages 10-11 and ends around at the ages of 17-19. As compared to 19th century, in 21st century, the average age to attain the puberty is lower. During this period, the girls experience a range of changes to their bodies. It is initiated by hormones from the brain to the gonads of the girls. These changes are external and internal as well in terms of size, shape, structure and composition along with psychological and social changes. Girls grow rapidly in the first half of puberty and they stop growing when puberty is completed. It is initiated by hormones from the brain to the gonads of the girls. The major landmark of puberty for girls is menarche (the onset of menstruation), which starts between ages 12-13. In addition to menarche, other changes include breast development, the appearance of pubic hair, hair growth under the arms and on the legs. Puberty which starts earlier than usual is called **precocious puberty**, whereas, it is starts late than usual called **delayed puberty**.

Sexual debut and adolescent pregnancy

Sexual debut is commonly defined as having had first sexual intercourse at or before age 14 years. It is strongly associated with health of a girl in the form of sexual transmitted infections including HIV (Human Immuno Deficiency Virus), unwanted pregnancies and long term mental and physical health consequences. In some cases, these kinds of activities ultimately culminate in prostitutions. Other related outcomes are acceptance of violence, male dominance, withdrawal from schools and fear towards marriages. Almost everywhere in the world, girls' first sexual activity occurs between the ages of 15-19 years. Early sexual debut, in many cases, associated with intimidation or violence. The younger the girl more is a chance that her sexual debut is forced. This issue is now considered as important public health issue owing to its association with adverse ill-health outcomes. Moreover, girls at their first exposure to sexual activity don't

know the use of any kind of contraceptives. The reason seems to be lack of information among young girls and also hesitation because gynaecological services in developing countries majorly focus on married women.

Adolescent pregnancies are higher is Africa and Asia, mostly in rural areas and among less educated population. In developing countries, complications in adolescent pregnancies and birth of a child are the primary cause of death among young girls aged between 15-19 years. Out of total maternal deaths which are around 15%, 26% occurs among adolescents in Africa alone. The health of such infants is also pitiable and survival rates of such babies also lessen. For example, perinatal deaths are 50% higher among babies born to young women below 20 years of age than then those born to mothers aged 20-29 years. The live infants are underweight, which again is a reason for various other diseases and deaths in early childhood. As adolescent pregnancies are usually unwanted pregnancies, rate of abortions also increases in developing countries especially in Africa. These abortions are generally unsafe, may lead to deaths or other consequences like, haemorrhage, reproductive tract infections and infertility among these young girls.

Table 1: Ten leading causes of deaths in female adolescent aged 10-19 years in World in 2021.

Rank	Cause	Death (000s)	Percentage
1.	Lower respiratory infections	57	8.9
2.	Road traffic injuries	37	5.6
3.	Self-inflicted injuries	36	5.5
4.	Tuberculosis	29	4.5
5.	Fires	24	3.8
6.	Drowning	23	3.5
7.	Meningitis	21	3.3
8.	HIV/AIDS	20	3.0
9.	Malaria	17	2.7
10.	Diarrheal diseases	14	2.2

Risk-taking behaviour of youth during adolescence

Adolescence is a time of maximum energy, growth, development and minimum mortality. In Indian language '*Iska khoon garam hai'* or '*jyada khoon ubal rha hai'* (to be very angry all of sudden) denoted to their risk-taking behaviour. The newly emerging physical, hormonal and mental modifications make them prone to certain abuses like drugs and alcohol consumption, drive hasty or attraction towards opposite gender. Other adolescent ill health includes unwanted pregnancies, sexual transmitted infections, stress, depressions etc. this population also suffers higher risk of suicide than the rest of populations.

Sexual transmitted infections

Sexually transmitted infections (STIs) also known as sexually transmitted diseases (STDs) are infections that are passed from person to person through sexual contact. In many cases, health problems related to STDs are more severe for women than men. These problems become more severe in case of pregnant women as it can further transfer to new born babies. The highest rates in STIs are in young girls aged 15-24. More than 1 million people are infected with STIs every day. The different causes of STDs are bacteria, parasites, yeast and viruses but the maximum caused by bacteria and viruses. There are more than 20 types of STDs, the major one are discussed below

Chlamydia: It is a bacterial disease caused by *Chlamydia trachomatis*. It can infect both males and females. Females get this disease in the cervix, rectum or throat whereas males can get in the urethra (inside the penis), rectum or throat.

Genital Herpes: Genital herpes is a viral disease caused by a herpes simplex virus (HSV). It causes sores on genital/vaginal or rectal area, buttocks and thighs. Newly born babies can also get infection through mothers during childbirth.

Gonorrhea: it is a bacterial disease can infect genital tract, mouth, or anus during vaginal, oral or anal sex with an infected partner. A pregnant woman can also pass it to her baby during childbirth. Symptoms are not very clear. It can cause pain during urination, bleeding between periods and discharged from vagina. Untreated situation may leads to pelvic inflammatory disease that can cause complicated pregnancies and infertility.

HIV/AIDS: HIV stands for human immunodeficiency virus and AIDS stands for acquired immunodeficiency syndrome. AIDS is the final stage of infection with HIV. Not everyone with HIV develops AIDS. This virus destroys white blood cells and harm immune system. HIV mostly spreads through unprotected sex with an infected person. Other possible reasons may include sharing drug needles or through contact with the blood of an infected person. A woman can also pass it to her baby during childbirth.

HPV: HPV stands for Human Papilloma Viruses (HPV) responsible for causing lumps in genitalia or if untreated may cause various cancers like cervix, anal, vaginal cancers through sexual contact with an infected partner.

Syphilis: It is a bacterial disease which infects genital area, lips, mouth or anus from sexual contact with a person suffering from it. It can also pass from mother to baby during pregnancy.

Trichomoniasis: This disease is caused by parasite through sexual contact without any early noticeable symptoms. It may cause vaginitis in woman with following symptoms:

- Yellow-green or gray discharge from the vagina
- Discomfort during sex
- Vaginal odor
- Painful urination
- Itching in or near the vagina

Fig. 1: Challenges among adolescents

Mental health in adolescence

The discussion on mental health is highly ignored issue especially in developing countries. Although mental health illnesses are associated with significant burden of morbidity and disability. Here, gender also become critical determinants of mental health illness, as it is more seriously contemplated for male a compare to female. But it is more prevalent and persistent problem in women than men. Among girls adolescents this illness is commonly due to stress, depression, anxiety and somatic complaints. But more interesting here is, the suffer girls or patients seem reluctant to seek any professional help. Moreover, they actually don't compute it as a problem or disorder. But now these kinds of illness and disorders are now considered as serious public health problem. No doubt more research and awareness is much needed.

Following are the common factors which affect mental health of an adolescent girl:

a. Confused behaviour: Today's adolescent girls are exceedingly confused between whether they are still a child or a grownup. Sometimes they want to be 'Own bosses' or sometimes they need pampering and care. During this period they become more argumentative and reluctant to certain norms of society like fashion, drinking, family rituals, dressing sense, coming late at home and many more.

b. Peer pressure: Adolescents in this age start spending more time with their friends rather than families. Girls start expecting more trust and loyalty in friendship. They are more emotional revelation and sharing of thoughts and feelings with their friends. Girls are more influenced by their peers in terms of dresses, music, hair style, go out for picnic etc. They feel more supported and protected by their peers at this age rather than their family members.

c. Sexual maturity: Mental health of adolescents highly depends upon pubertal change experienced by them. Pubertal changes have two aspects like pubertal timing and pubertal status. Early or late maturation and personality adjustments are sometimes a disadvantage and influence badly on their mental development. For example, early puberty sometimes related with earlier and greater sexual experience, which in turn, are associated with the risks of premarital pregnancies and abortions. On the other hand, they are

also ambiguous towards menstrual cycle. Sometimes, they are happy to become a 'Complete women' and capable of reproduction and on the other side they are disconcerted about the inconvenience, discomfort and embarrassment.

d. Career opportunity: In this age, selecting a right career is also become a reason for mental health issues. They feel parental pressure and peers competitions which disturbs their mental status. Parents and society are sometime reluctant for girls to send away from the home and spending too much money on them. This thinking of parents interferes with their career and it affects the mental status of girls. Nowadays, scenario has changed and more and more girls are joining and completing lengthy professional courses with the help of their families. At one side they feel pleased but on other side delay their marriage and may indirectly interfere with their sexuality. It in turn causes lots of metal health issues.

e. Violence: Girls adolescents are many times exposed to violence like childhood sexual abuse, parental domestic violence, corporal punishment at school, bullying, and sexual coercion. Lack of attention and care in such conditions lead to serious repercussions on their mental status and their journey to adulthood.

f. Nuclear families: Now- a- days most children and their advices are built around the small, nuclear family. The families that prays, plays, camps together, party together and stays together. Not long ago, families were used to be much larger, frequently with many relatives living around including maids, nannies, gardeners, gate keepers etc. In today's scenario small family and great mobility have created unusual problems for adolescents. Teenagers are on their road to independence, go through a stage where their parents seem hopelessly stupid and out-dated. During this stage of life, no parental advice is generally acceptable. In years past, other members or cousins of the extended family could substitute for the parents for genuine advice for their problems. Now teenagers are thrown and trust on their peers for advice and suggestions. The peers are also trying to cope with the very same problems. Certain well-to-do families use summer camps, picnics and boarding school to help ease them through this difficult period. Quite often the adolescents get along well at their school and in other homes may be their friends and cousins house.

g. Addiction to substance: Alcohol and tobacco consumption has increased among female adolescents since last decades. Data received from several European countries, low- middle- and high- income countries; indicating that the percentage of girls drinking alcohol and smoking tobacco has been amplified as compared to boys. Use of tobacco can be smoking cigarettes and tobacco products other than cigarettes like pipes, smokeless tobacco, chewing tobacco or *bidis*. Girls are more vulnerable to psychological effects of substance use because of the difference in their body weight and body water contents as compared to boys. As a result, they are more likely to suffer the consequences of substance use like violence, injuries and sexual coercion.

h. Obesity: Poor habits of eating and physical inactivity among female adolescents are leading to premature deaths, disability, overweight and obesity. Obese adolescents tend to grow up to be obese adults and are thus exposed to a higher risk of diseases, such as osteoarthritis, diabetes and cardiovascular diseases, at a younger age than those who are not obese. Data from 20 low- and middle-income countries show that around 12% of school-going 13–15-year-old girls are overweight. Physical activity is necessary for every adolescent not only to avoid weight gain but also in improving mental health like stress, depression and anxiety. Adolescents doing physical activities adopt healthy life style by avoiding substance use like alcohol, drugs and tobacco and show higher academic performance.

Enhancing health of adolescents

Understanding and dealing with the adolescent health issues in a positive manner can help adolescent girls to be more social, responsible and caring. The main role at this stage is related to their parents, teachers and friends as teenagers mostly have trust on them. Following is the list of solution to the issues faced by adolescent girls.

1. Physical changes
   - We need to explain them and make them aware the body changes are normal for every teenagers and every adolescent goes through it.
   - We need to help them to accept those changes.
   - Enable them to stay healthy and fit through a nutritious diet and exercise.

2. Emotional changes
   - Encourage them to exercise as physical activity helps keep the serotonin (creates good feelings and happiness) levels up.
   - We should have habit to listen our daughters. So let them talk. Most importantly listen to them without judging and avoid giving them advice when they are not ready for it.
   - Parents can share their adolescence experiences.
   - Girls can be indulge in a creative activity so that they can channelize their emotions.
3. Behavioural changes
   - At this stage girls have mixed feelings of sadness, loneliness, angry, jealous and excitement. Help them to understand their changes and decisions.
   - Adolescent are sensitive and they may not take criticism in a positive way. So parents sometime need to intervene if girls are falling into bad society and substance use. Parents can also help them by telling what decision they would do if they are at their places.
   - Parents need to keep ayes on their routine pattern of sleep, moods and foods.
4. Educational challenges
   - Parents and teacher can help girls to take appropriate decision for education and help them to build carrier well.
   - Girls are involved in household chores. But during exams parents could cut down their household chores to enable them to focus on their studies.
   - Girls can be provided with proper nutrition to help them get more strength.
5. Social issues, dating and relationships
   - Your child may seem to spend more time outside than with you. Accept that your adolescents are discovering a whole new world. Just let them know you are there when they need you.
   - Try to Share your dating and social life experiences in school to make them more friendly and help to gain their trust.

5. Addiction to social media or cyberspace
   - At this age teens are more creative and excites. If they spend lots of time on internet and mobiles, do not assume that are wasting time. Rather could be doing more productive things on the internet other than surfing the net.
   - Enrol them in activities that encourage them to interact with others. Parents should also spend more quality time with their girls. They should have family activities and get together so that will they spend less time at the computer.
   - Make cyber rules and boundaries for adolescent child at home. Limit the use of the mobile phone to a few hours in a day, and avoid bringing the phone to the bedroom as it is likely to affect a person's sleep.

WHO (World Health Organization) Initiatives

WHO in 2017 published a report on adolescent health titled "Global accelerated action for the health of adolescents". (https://www.who.int/news-room/fact-sheets/detail/adolescents-health-risks-and-solutions)

This report aims to assist governments in deciding what and how they plan to the health needs of adolescents in their countries. This reference report helps in planning, implementing, monitoring and evaluation of adolescent health programs. Teams from over 68 countries have been trained to promote adolescent health.

To improve adolescent health globally, WHO, in collaboration with UNAIDS, UNESCO, UNFAP, UNICEF, UN Women, the World Bank Group and the World Food Programme (WFP), has established the Global Action for Measurement of Adolescent health (GAMA) Advisory Group *(Kindly see list of abbreviation at page 185 for full form)*. GAMA provides technical guidance to WHO and UN partner to explain a core set of adolescent health indicators, for harmonizing efforts to adolescent health measurement and proper reporting. WHO carries out a range of functions to improve the adolescent health, including

- production of the evidence and practical based guidelines to support adolescent health services;

- making recommendations to governments on adolescent health and adolescent-responsive health systems including the provision of high quality, age-appropriate health services for adolescents;
- documenting progress in adolescent health and development;
- raising awareness of health issues for young people among the general public and other interested stakeholders; and
- advocating with governments and working with youth-led and youth serving organizations to support establish national level structures

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CHAPTER 2

HEALTH ISSUES AMONG WOMEN LIVING IN REMOTE AREAS

Dr. Suninder Kaur Sodhi, Dr. Sunil Rani, Dr. Kavita Choudhary

Definition of rural

Rural areas in the Indian context, refers to the *villages* in India. Usually agriculture is the main source of occupation among people living in the rural areas. A town with a maximum population of 15,000 is considered rural in nature. The National Sample Survey Organization (NSSO) defines 'rural' as an area with a population density of up to 400 per square kilometer with about 75% of its male population involved in activities related to farming, having a Panchayat i.e. a village council, but no municipal board. In India, more than 70% of the population resides in the rural areas and considering the low quality of health care services, mortality due to disease is on the increase. Infant mortality is also very high since children die before reaching the age of five due to either lack of proper nutrition, vaccinations, or illness. 700 million people living in rural areas lack basic health services and have to often travel miles before reaching the nearest medical facility. The cost of treatment also eludes the pocket of these people. Lack of quality infrastructure, facilities and services hinders the right to good health. Dearth of qualified Medical and Para-medical staff is also a concern in many developing and under developed countries.

> ***Health Care is the Right of Every Individual, irrespective of caste, race, sex, age or occupation.***

Social inequities and health

Social Inequity is the inequality of status, power and wealth among families, societies, and community members. It is often accompanied by discrimination, lack of freedom and at times social exclusion. A difference in the quality of health care and health services across the country is health inequity. The cause of disparities in providing health care and treatment among various pockets of the society should be looked into and health policies should reach out to the most vulnerable including women and children to build equity. A social commitment is required to ensure

effectiveness of the policies. Health is a fundamental human right and no one should be barred from it. Reducing social and health equity will reap benefits all aspects of health and life per se. Social and Health equity will minimize equalities making societies vibrant, healthy, positive and powerful. It will give us the equality of opportunity to enjoy life. Differences in health status such as disease or disability will not be a hindrance in getting health benefits as a fundamental human right, and provide the opportunity to enjoy life and pursue one's life plans. Promoting health and protecting the right to natural health, needs to be supplemented with essential health information for everyone to make informed choices. The obstruction of the right to information is a crime against humanity as it fosters inequity and inequality and decays societies. Low-cost prevention, treatment and elimination of diseases can help the poor and the needy. People in the rural areas, including the most vulnerable such as the elderly, infants and women need to be focused for building equity. The World Health Organization's effort "Health for All by the Year 2000" focused more on administrative healthcare changes but failed make inroads and reach the most vulnerable, to reach its goal. The dissemination of information about the need for optimum health, and ways to get it should be re enforced with full potential to create a sustainable healthy society. Health is an investment and a driver for development of societies. Efforts should be made to provide affordable access to sustainable, good quality health care for all. The right to education, health and Life should be for each one of us. Health systems should be strengthened and policies should reach out to the rural areas.

Health for All by the Year 2020

A Sustainable World must be a Healthy World

In 2010, the Department of Health and Human Services launched “Healthy people 2020” in the USA, whose main 4 goals are:

- Attain high-quality, longer lives free of preventable disease, disability, injury, and premature death;
- Achieve health equity, eliminate disparities, and improve the health of all groups;
- Create social and physical environments that promote good health for all; and
- Promote quality of life, healthy development, and healthy behaviours across all life stages.

(Centre for Disease Control
https://www.cdc.gov/nchs/healthy_people/hp2020.htm)

Status of the woman living in remote areas

All women in the world whether it is urban or remote, plays a central role in society and has ensured stability, progress, and long-term development of nations. In Indian mythological history, there are many good qualities in Indian women. A good Indian woman embodies the sanctity of nation and goddesses. A woman is said to be good if she plays all the roles of mother, sister, wife, and daughter-in-law as well as brave through the extremities of life. Indian society has many expectations from a woman especially in remote areas like being sacrificial for every need of a man and her family over her own desires, being devoted to her husband, caretaker of the house, observing every religious ritual, being silent in front of men, and should cover her body, should be a good cook and arrange the house. All these expectations create social pressure on women and lead to many thought processes.

Women play simultaneously a multiplicity of roles and many women juggling with these multiple roles. The situation can be understood by the following statements of a woman *"There is not enough of me. Everybody needs so much of me. I constantly feel as if one person or one group of people has one of my arms, and another group has the other arm, and they are pulling in different directions And then two other groups have my feet and then phone rings" Faye Crosby.* Indeed, women today have become the busiest cohort in human history. As the women play multiple roles at a time it would produce role overload or psychological distress and inadequate social support in these roles may contribute to immunosuppression and disease vulnerability.

Society blames mothers for every pathology of their kids yet they have little or nil control over that and it creates pressure on women to be the "perfect" parent. A mother is expected to be nurturing every time and selfless; she feels great guilt if unable to achieve this ideal condition. So, the role of the mother is the major source of psychological distress for women. Bringing up of women in Indian society is in the way that they perceive other's feelings as their own and takes other's burden as their own. In this way, women's well-being is dependent on their surroundings and their relationship with others. The health of Indian women is linked to their status in society. Women are considered an economic burden and their contributions to families are generally overlooked.

Health of women in remote areas

The problems associated in rural areas differ from that of urban areas. While access to health services is not easily available, other issues such as identification of health issue and going to medical practitioners for access to treatment and services is an altogether alien concept. Neighbours, Dai and older women act as health providers and advice givers in times of health issues. Most of the issues are tackled within the community, by intake of various types of foods and fruits, ointments made from certain plants etc. for curing any physical and mental ailments.

Outside help and going to a proper doctor or hospital is often considered a last resort. Usually when she comes for medical help the person is critically sick. Money i.e. paying exorbitant bills is also an underlying concern among people living in rural areas earning meagre wages. Even though the government has set up schemes and subsidized rates for treatment of people living in rural areas and for those live below poverty line. People by and large remain unaware of these schemes and free health facilities. As a result, rural women experience poorer health outcomes and have less access to health care than their urban counterparts. There is also a dearth of women medical practitioners and limited health care providers, which also act as a barrier in health services reaching out to the poor.

Indian rural health care faces a crisis with nearly 86% of all the medical visit made by ruralizes; majority travelling more than 100 km to avail health care facility of which 70-80% is born out of pocket landing them in even more poverty.

Health disparities in rural women

In India, as in other parts of the world, rural areas is a heterogeneous mix and health care officials should be aware of it and also advocate for reducing health disparities in rural women. India records the highest maternal and infant deaths worldwide accounting for 1/5th of all global maternal mortalities. 21% of children who die before they reach the age of five, in the world, due to lack of proper health care are Indians.

Maternal and infant mortality rates have shown declining trends when maternal and child health services are promoted such as such as ante-natal care (ANC), institutional delivery and child immunization.

Use of maternal and child health services are relatively low despite the fact that most primary healthcare in public health facilities is available free of charge. This is due to considerable socioeconomic inequity within and across the Indian states (Pallikadavath *et al.,* 2004; Vora *et al*., 2009; Pathak *et al.,* 2010; Sanneving *et al.,* 2013; Joe 2014).

Efforts in improving health status of women in rural areas

The Indian government launched National Rural Health Mission (NRHM) with an aim to strengthen the primary healthcare system and address these longstanding inequalities in all aspects of health care, treatment and service.

Government efforts

Anganwadis

The concept of anganwadi centres having workers was started by the government of India in 1975 to provide child care services in rural areas as part of the ICDS program (Integrated Child development Services) in an attempt to combat malnutrition and hunger among children. These centres catered to the basic health care activities such as contraceptive counselling and its supply, immunization, education on health and nutrition, provide basic medicines and supplements, provide health checkups and referral services, as well as pre-school activities.

National Rural Health Mission (NRHM)

Launched in 2005, NRHM strives to provide accessible, affordable and quality health care to the rural population, especially the vulnerable groups. Since its launch, NRHM has made inroads into the interior of the country and have started many programs and schemes for providing quality and timely health services to the deprived. It aims to encourage pregnant women of low socio economic backgrounds to use institutional maternal and child health care services. NRHM has a set of core strategies which includes:

- To Increase public health funding
- To decentralize village and district level health planning and management
- To strengthen the public health service delivery infrastructure at village, primary and secondary levels
- To promote the non-profit sectors to increase social participation and community empowerment (Planning Commission of India 2001)

Some of the successful programs of NRHM:

- NRHM employed 'Accredited Social Health Activists' (ASHA) at the grass-roots (village) level to support the use of services and ensure outreach.
- The 'Janani Suraksha Yojana' (JSY), a cash-transfer program, provides financial support to enable women from lower socio-economic groups to give birth in a health facility.

Public health efforts should focus more among those at the far end of receiving services. All efforts need to be channelized in remote areas where serves are minimal or non-existent. People have to travel a minimum of 25 kms. to reach the nearest health centre or dispensary. The medicine inventory and treatment facilities are also limited in these areas. Basic facilities like a fridge to store medicines and other instruments, sterilizing equipment, and also basic amenities like electricity to operate these appliances are not present. Problems abound; their consequences manifold, as people who do not get these services when needed, bear the brunt. The economy of the country should focus more on making inroads in remote areas. NGOs and other voluntary national and international organizations have tried to make inroads, by mobile vans, providing syndromes treatment and services to the needy. People need to be aware of their health rights, preventive and curative health promotion efforts can bring about a positive and dynamic change. Sustained awareness interventions for demand generation will improve the health status of people living in remote communities.

- Ashray Social Welfare Organization, an NGO, in collaboration with Global Giving (a global crowd funding community) has ventured to provide healthcare to Rural Indian Women by giving free medical services to distinctly vulnerable population of 30,000, residing in a cluster of 14 sparsely connected and one of the most backward areas of Gujarat state in India through Mobile Medical Unit.
- Mainly maternal, reproductive, nutritional and menstrual hygiene care is offered to women, adolescent girls and children in the community.
  https://www.globalgiving.org/projects/provide-healthcare-in-rural-india-and-save-lives/

Another aspect of health, neglected in many societies, is oral health and hygiene. Health disparity is a universal phenomenon and so is oral health. Oral health education is minimal among people in the rural areas. Oral health in pregnant women is of even more concern as oral disease burden of rural antenatal women can lead to serious consequences. Adequate access to oral healthcare is a persistent rural health issue. Major reasons are lack of awareness, inequitable distribution of service providers, cost of care, unavailability of services and no budgetary allocation or oral health policy at national or regional level in India. Hence, oral health services are way beyond the reach of our rural population and primary prevention of oral-dental problems through education, service availability and empowering women population is the need of the hour. Dental caries is a common childhood infection. Integrating dental strategies into the primary health care approach can help reduce oral health problems in rural areas.

Primary Health Care (PHC) for women living in the rural areas

Primary care is a universal, systemic and ideological reform of care models today. Primary care provides a gateway to individual needs and problems and their community, having first hand contact, comprehensive care, proper coordination and follow-up, having longitudinal, family centralization and community orientation as its attributes. PHC's are state-owned rural health care facilities in India. These are essentially single-physician clinics usually with diagnostic facilities, 4-6 beds, facilities for minor surgeries, too. They are the cornerstones of rural health services, acting as referral units for sub centres and they refer out cases to community health centres (CHCs).

PHC is a basic functional unit providing accessible and affordable health services. Their main focus is on antenatal services, deliveries, birth control programs, infant immunizations, and anti epidemic programs. It provides 24X7 services. It is the first contact point between the villager and the medical officer providing preventive, curative and health promotion services to people.

Oral health among woman in remote areas

Oral health is state of being free form mouth and facial pain, oral and throat cancer, oral infection and sores, gum diseases, tooth carries, tooth loss and other diseases that limit a person's capacity in biting, chewing, smiling, speaking and psychological wellbeing.

Why do teeth need cleaning?

Our mouths are full of small microorganisms called *bacteria.* Some are harmless, some are beneficial called probiotics as they aid in the digestion of foods, some bacteria protect our teeth and gums but some bacteria, cause tooth decay and gum disease. The bacteria along with mucus and other particles constantly form a sticky, colourless "plaque" on teeth. Proper Brushing of teeth and flossing can help clean the plaque from the teeth. However, *Plaque* that is not removed can harden and form "*tartar*". Brushing isn't able to clean tarter because it is quite stiff and rigid. Professional cleaning by a dentist/dental hygienist can remove tartar, also called *calculus*. If plague and calculus are left untreated and not cleaned, they release toxins that can lead to inflammation of the gums called "gingivitis." In gingivitis, the gums become red, swollen and can bleed easily.

Gingivitis is a mild gum disease that can be reversed with daily brushing and flossing, and routine cleaning by a dentist/dental hygienist/orthodontist. Regular teeth cleaning by a professional is called 'Scaling'/Periodontal scaling. A prophylaxis refers to scaling and polishing of the teeth in order to prevent oral diseases. When gingivitis is not treated, it can advance to "periodontitis" which means "inflammation around the tooth". Periodontal disease range from simple gum inflammation to major damage to the soft tissue and bone that support the teeth and in worst cases, teeth are lost**.** In periodontitis, gums pull away from the teeth and form spaces called "pockets" that become infected. (Refer *Periodontal (Gum) Disease: Causes, Symptoms, and Treatments* NIH Publication No. 13-1142 September 2013 https://www.nidcr.nih.gov; www.colgate.com)

> The body's immune system fights the bacteria as the plaque spreads and grows below the gum line. Bacterial toxins and the body's natural response to infection start to break down the bone and connective tissue that hold teeth in place. If not treated, the bones, gums, and tissue that support the teeth are destroyed. The teeth may eventually become loose and have to be removed. (National Institute of Dental and Craniofacial research)

Good eating habits and proper teeth cleaning, brushing and oral hygiene can help keep the teeth healthy. Healthy teeth are a gateway to total health and well-being. In remote areas, where proper facilities for oral hygiene are not available, such as water and proper toothpaste, floss etc., one should avoid eating starchy and sugary foods to protect teeth.

Why is it important for women to know about oral health?

Good oral health habits help protect against tooth infections such as decay, gum disease and other oral health issues. Women are more susceptible to changing hormone levels from puberty, certain types of birth control, pregnancy and menopause, and these hormone changes can impact oral health. Morning sickness associated with antenatal stage, usually around the 6th week of pregnancy also impacts oral hygiene, as women tend to pay less attention to teeth and avoid good brushing and cleanliness. Infections in our teeth can affect the rest of our body. Untreated oral problems sometimes lead to fatal infections such as:

- Chronic gum disease can lead to heart disease - Gum disease is a bacterial infection and if the bacteria enter the blood stream and get attached to fatty deposits in the heart blood vessels, it can cause blood clots leading to heart attacks. (See Bleeding Gums- Heart Disease www.colgate.com; Periodontal disease and heart health by Kelli Miller https://www.webmd.com)
- Chronic gum disease can lead to diabetes- Severe gum disease can negatively affect your blood sugar control and increase your chances of suffering from other common long-term complications of diabetes. The inflammation, which occurs in the gums, escapes into the bloodstream and upsets the body's defense system, which in turn affects blood sugar control. (Refer *Diabetes and Gum disease* www.diabetes.co.uk)
- Research published in the journal of clinical periodontology in 2011 found an association between gum disease and male fertility. Many with gingivitis and periodontitis also had low sperm counts and diminished serum quality. The association of periodontitis and woman's chance of conception and efficacy of infertility treatment cannot be undermined.

Difficulty in chewing food results in poor nutrition and in turn physical development. Decayed teeth and chronic bad breath lead to psychological problems and higher risk for social stigma. Position of teeth affects speech and personality. Healthy teeth are a gateway to a healthy body. However, people neglect their teeth for many reasons, of which lack of awareness and lack of proper facilities, and social factors play a major role.

Women more at risk for oral health problems

There are phases in a woman's life during which they are more susceptible to oral health. Puberty, mensuration cycle, pregnancy, while taking birth control pills and during menopause, the hormones estrogens and

progesterone can increase the blood flow in gums and alter the way gum tissue reacts. This may lead to inflammation of the gums and may also increase gum bleeding.

Barriers to keep good oral health of women of remote areas

Only 10% of dentist serves 70% of Indian population residing in rural areas, which accounts to 1 dentist per 250,000 people. Dental care utilization is the percentage of the population who access dental services over a specified period of time. Dental patients only visit the dentist when in severe pain and follow-up visits are not done in most cases. Adequate knowledge of the way the individuals use health services and the factors predictive of this behaviour is crucial in improving oral health outcomes (Gambhir *et al.*, 2013; Gambhir and Gupta 2016). Myths about oral health abound. Some of the common myths and misconceptions about oral health are:

- Cleaning teeth with fingers is better than brush
- Extraction affects the eyesight
- Spacing between teeth is a sign of good luck.
- Teeth pain is God's punishment/past sin.
- Do not take medicine when you are fasting.

A survey done in Sunam District of Punjab, India in 2012 among 1664 individuals highlighted the misconceptions with 65.3% of the respondents believing that after cleaning, teeth get loosen whereas about 50% of the sample believed that extraction of upper teeth affects eye sight. Nearly 74% of our subjects believe that there is no need to visit a dentist until all the permanent teeth of children erupt and an approximately 30% of the participants persist in believing that diseases of oral cavity (like carcinomas) are due to past sins. Rural people have cultural beliefs and taboos limiting them in the uptake of dental treatment as compared to urbanites, was also seen in the study. To improve health among people in the Indian subcontinent, there is a need to understand the beliefs and attitudes, to understand the local and traditional Indian system of medicine, awareness and education about health and oral health, traditional eating habits and addictions such as tobacco etc.

In tribal women, some traditions can worsen oral health

Mentawai, a tribe of some 64,000 people, live on islands in West Sumatra in Indonesia. They are known for their distinctive tattoos and the practice of sharpening their teeth with chisels. Staining the teeth i.e. teeth blackening is common among Asian tribal women. Using chemicals for

whitening can be harmful and cause gum swellings. 'Teeth staining' is common among tribal people. Alternatives to tooth brush and paste being used in remote areas are:

- Clove – easily available, used to relieve dental pain.
- Charcoal or brick powder– cost effective, used instead of toothpaste.
- Neem stem/sticks – as brush

Who can improve oral health and remove misconceptions:

- Dental professionals
- Primary health workers
- School teachers
- Dental health planners and policy makers
- Government
- Local civil society
- Non-government organizations

Ways to Improve Oral health and Hygiene

There are many ways we can improve such as regular cleaning and checkups. The government can train more dentists to reach these adverse pockets of the country. People should be made aware and educated about oral health and government should main more sustainable efforts in this regard. A databank can be created and updated at regular intervals. Material and equipment cost should be subsidized so that private health practitioners can also reduce their treatment costs thereby reaching out more to the poor and the needy.

Water fluoridation in areas where needed can go a long way in reaching out to the masses as fluoride can decrease cavities Proper water fluoridation i.e. 0.7-1.2 parts per million (ppm) has been accepted as a safe, effective, and inexpensive method of preventing tooth decay. Adding fluoride to municipal drinking water also is an efficient strategy to reduce the inequalities in dental disease CDC included water fluoridation in its list of 10 great public health achievements of the 20th century. (Refer American dental health association website *CDC Releases Guidelines on Fluoride Use to Prevent Tooth Decay 2001* https://www.adha.org/.)Social determinants, alcohol and tobacco consumption, bad eating habits increase the burden of disease worldwide.

Providing free treatment, free demonstrations at schools, clinics, and other public places of proper brushing and oral health awareness programs should be increased. Mobile vans for free oral checkups and supplying free vitamin supplements to pregnant women can help in reducing dental health problems.

Barriers in Utilization of health services: Social determinants

People suffer from stigmatization, which is a neglected social issue and is a Barrier to health service utilization. **Stigmatization** associated with perceptions, concerns about confidentiality and a reluctance to ask for support are major barriers to health service utilization. The area of greatest concern is mental health, which affects all physical and oral health as well. Parenting issues, gender issues, children are neglected, and many other social determinants come into play which act as barriers in health service utilization. . Improving access may involve providing a greater awareness of what services are available and allowing women to access services in a way that does not require them to label themselves or be labeled by others.

Advanced Healthcare delivery by Gram Vaani Technologies

Gram Vaani started in 2009. It has been implemented in 7 Indian states, beside in other countries such as Pakistan, Namibia, South Africa and Afghanistan. Its intent is to have bottom up rather than top-down flow of information. People in the rural areas have access to mobile technology, and it helps them in asking for help when needed. Making optimum use of technology, rural radio stations are able to manage and share content over mobiles and the web. People have alerted episodes of corruption and discrepancies through their mobiles. Gram Vaani built innovative voice applications for organizations working in health care sector such as ASHA and anganwadi workers, (ANWs) to automate and manage their processes efficiently.

- Gram Vaani provides cutting- edge solutions to automate processes and cater to health care sector, social sector, and corporate organizations for connecting with the most vulnerable and difficult to reach pockets of the society.
- In partnership with the White Ribbon Alliance for Safe Motherhood, for a program of Merck for Mothers, Gram Vaani is working to upgrade the quality of maternity healthcare in India. (www.gramvaani.org)

Making inroads in the interiors of the country is not easy. People in the villages have a completely different lifestyle and to be able to touch their heart and soul needs time, and persistent effort. The Shakti Project was born out of this realization, and has been successful in reaching out to the remote areas of the country. Hindustan Unilever, the Indian subsidy of the Dutch conglomerate made inroads via their 'Shakti Project'.

> "Shakti" Project by Hindustan Unilever has become a case study for business schools and evolved beyond its original goals. "The objectives of Project Shakti are to create income-generating capabilities for underprivileged rural women by providing a small-scale enterprise opportunity, and to improve rural living standards with greater awareness of health and hygiene," says Dalip Sehgal, executive director of the Shakti initiative.

Conclusion

Being in good health and well being is the greatest blessing of life. One should constantly strive to keep fit and have all family members in good health. Basic determinants of good health are good food intake, maintaining cleanliness and staying happy. However, factors like social beliefs, customs and values also affect health of a person. There is a dire need to understand the social determinants, which account for low utilization of health services. A multi prolonged strategy is needed to address myths, stigma, and educate people about their health needs. Focus should be on the collective consciousness of societies to eradicate any misconceptions and to understand the connections between women living in rural areas and health services. Understanding the social settings is a necessary challenge for promoting accessible humane care, and to be able to make the search for care a daily practice without any pain or pressure.

People in rural areas need facilities at affordable charges. Rural areas lack basic facilities like cleanliness; lack of proper sanitation, standing water and open tunnels of dirty water spreads many contagious and infectious diseases, which can be avoided by government's timely intervention. Many NGOs are already working and are being aided by the government sector to reach out to the women, children and the needy in remote areas but statistics have shown that the no of health providers' vis-à-vis the population needing

health care is inadequate. Targeting deprived populations and designing public health programs by linking maternal and child healthcare components are critical for universal access to healthcare. NRHM is making concerted efforts in this area and have received some success. Dearth of clean water in rural areas, and lack of adequate drinking water, and other water and sanitation facilities also adds to the poor health of the community. In remote areas, people still have more faith on traditional healers, dais, witchcraft, and hermits etc. more than on professional health workers. All these factors should be taken into account and providing better facilities should encompass the goal of health inclusion. Barriers to reaching out to health services abound. NGOS, Social workers, health professionals, Dental health professionals, private and government sector should work together for this cause. A joint effort is required to achieve the goal of health for all by 2020. Persistent efforts, continuous education, and active participation of society in its own overall health, oral health and hygiene, is of paramount importance. Understanding the underlying issues, barriers, careful planning and implementation of policies and programs, providing effective and cost-effective treatment in close proximity is the call of the times in Indian society.

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CHAPTER 3

HEALTH PROBLEMS IN OLDER WOMEN

Dr Kavita Choudhary, Dr Uttam Kumar

Ageing is a normal physiological process and it should not be treated as a disease. In simple words the meaning of ageing is growing old. Now a day's word 'senescence' is used more frequently than ageing. 'Senescence' as deterioration in the vitality or lowering of the biological efficiency or feebleness of the body and mind, associated with the process of aging, such as decline in sexual prowess, diminution in the endocrine activity, loss of elasticity of blood vessels and rise in blood pressure. Due to continuous activity for many decades; decline in physical activity of elderly can be anticipated. Not only physical wellbeing but emotional, sexual and mental health is also equally important. We live in a society where the youth is valued and the elderly are given less value. Elderly women also have to work hard to preserve their self worth in the family and society. In most countries women out lives men and are involved in less risky behaviour such as alcoholism, rash driving and smoking. Life expectancy at 65 years of age for both sexes is 19.3 years and for men and women 18 and 20.5 years respectively. So as the women lives longer than the men they are more prone to have more chronic diseases and more morbidity.

India has the second largest numbers of elderly population after China. The number of 60 years and older was 100 million in 2013 and will increase to 198 million in 2030. Some special characteristics of Indian older population are as follows:

(a) Majority of elderly population belongs to rural areas
(b) Nearly 51% of elderly population are females
(c) Increase in number of older olds (>80years)
(d) Majority of elderly are below poverty line

Health issues of older women are different from that of men. Woman attains menopause at and around 50 years of age and faces some unique and different health issues (Fig. 1). At the time of menopause, ovaries stop

producing sex hormones that are progesterone and estrogens. Like ageing; menopause is also a normal physiological process and should not be treated as a disease. Despite being a natural phenomena menopause has the potential to trigger severe symptoms in some females. Estrogen and progesterone hormones have effects on several body systems like genitourinary, musculoskeletal, heart, brain and skin. Due to lack of these hormones women faces some symptoms like hot flashes, night sweets, dryness of skin, mouth and vagina, urinary disturbances, mood alterations, irritability, depression and dull aching joints. Other than that menopause can affect cognitive functions such as concentration and can lead to short term memory loss. Symptoms of menopause can start even years before and after actual menopause. Woman need to understand the symptoms of menopause clearly; because some of the symptoms of menopause can mimic serious medical conditions that need urgent attention. Hormone replacement therapy (HRT) can be effective to overcome the symptoms of menopause to some extent.

Challenges faced by older women:

"No country sends its soldiers to war to protect their country without seeing to it that they will return safely and yet mankind for centuries has been sending women to battle to renew the human resource without protecting them."

1. **Malnutrition**: India has one of the highest rates of malnutrition in women among developing countries and it is seen in various studies that malnutrition increases for women as they enter adulthood. Traditionally, women eat after all the family members in Indian society; they automatically get less food. There is food discrimination in Indian societies and it is not due to poverty or scarcity but it is a perception that men need better and more food because they are bread earners. Maternal malnutrition has been associated with an increased risk of maternal mortality and birth defects.

2. **Suicide**: The Suicide rate in India is five times higher than that of the developed world and it is higher in women as compared to men. The most common reasons for women's suicide are directly related to:
   - Depression
   - Anxiety
   - Gender discrimination

3. **Domestic violence**: Domestic violence is defined as acts of physical, psychological, and sexual violence against women and is currently viewed as a hidden epidemic by WHO. As per the (National Family Health Survey) NFHS III, 31% percent of all women reported having been the victims of physical violence in the last year. Alcoholism and dowry are the main reasons for domestic violence in India
4. **Lack of self-care**: For long women in Indian society are trained in such a way that they should care for others over themselves and so women were too busy caring for others to care for themselves. They are less likely to join any program related to health and wellness.
5. **Difference in socialization**: In the Indian societal system boys are free to spend time from school to home but it is not so with the same age girls and they are not free to experiment with their cycle as boys do. Again, in Indian societies, most of the girls are helping their mothers at the house and take care of younger sisters and brothers. Though these are small observations but have a big impact on the developing minds of young girls. Similarly, girls are not allowed to travel alone at distant places and are not considered safe for them; this social norm also impacts the minds of women and in adverse conditions too they have to be dependent on others and become victims many a time.

Health problems of older women can be divided into 3 categories:
(A) Pathological problems
(B) Psychological problems
(C) Social problems

(A). Pathological problems: Includes nearly all body systems

1) ***Central Nervous System***: like all other parts of the body; brain also goes through the process of ageing.
   Common neurological health issues are as follows:
   Neuropathy: Symptoms of neuropathy includes numbness, weakness and tingling sensations. Neuropathy affects mainly hands and feet and it can be due to some systemic diseases like diabetes and peripheral vascular diseases. Some medications like Isoniazid can also cause neuropathy.

   Alzheimer's disease: Alzheimer's disease is a type of dementia and the cause is also not clear so far. It affects basic functioning of brain cells and interferes with memory.

Parkinson's disease: Parkinson's disease occurs due to death of nerve cells in the brain. Symptoms include involuntary abnormal body movements which are progressive.

Myopathy: Myopathy affects muscles of the body especially upper arms and thighs. Muscle weakness is predominant symptom of Myopathy.

Amyotrophic Lateral Sclerosis: It is otherwise known as ALS or *Lou Gehrig's disease*. This chronic disease affects motor nerves of the body and deteriorates the muscle function.

Stroke: Stroke can be life threatening and occurs when any blood vessel in brain get thromboses or bursts. Symptoms include weakening of half portion of the body, deviation of the angle of mouth and speech difficulty.

Myasthenia Gravis: It is a chronic auto immune disorder and it affects the conduction of nerve impulse from nerves to muscles and makes the muscles inactive. The main symptom is muscle weakness which becomes worse at the time of muscles activity.

Muscular Dystrophy: This disease has a genetic basis and involves muscles near the trunk. Further weakness progresses to severe muscle contraction. Slowly patient becomes depended on others to move his/her body.

2) ***Respiratory and Cardiovascular system***: Heart attacks are one of the leading cause of morbidity and mortality among woman older than 60. Previously women were at low risk of developing heart attacks but now women make the half of all deaths due to heart attacks. Many women don't understand that they can have heart attack and so they don't seek medical care.

   Heart attacks are difficult to diagnose among females as they perceive different set of symptoms than the male. Chest pain is the most typical symptom of heart attack among males but females don't have chest pain but have pain in other parts of the body like abdomen and back. These symptoms of heart attack in females can easily mimics with symptoms of heartburns. In males heart attacks can present as pain in left upper arm and in lower jaw but in females can feel pain in both the upper arms,

both sides of the neck and jaw. During heart attack women can feel some other symptoms like nausea, vomiting, light headedness, dizziness and generalized weakness. These other symptoms can easily be confused with flu or other viral infections. Sometimes women can feel shortness of breath, cold sweats and anxiety at the time of heart attack and can easily be confused with symptoms of menopause.

The numbers of people affected by coronary artery disease increases with age in both males and females and four out of five people who die due to coronary artery diseases are 65 or older. As the age increases the blood vessels becomes harder and less flexible and blood flow through these vessels becomes difficult. Fatty deposits across the vessel walls make the flow of blood more slow. Along with these things and sedentary life style, smoking, alcoholism and diabetes increases the risk for heart diseases among elderly.

3) ***Gastrointestinal System***: There are some basic structural differences in men's and women's gastrointestinal tract. Due to these differences women have some different set of symptoms for common diseases of gastrointestinal system. Even women may present with different unique set of diagnosis.

Heartburns: women are more sensitive to physical and emotional irritants. So are more prone for developing heartburns as compared to men and they feel more pain than men. Many things can precipitate heartburns like stress, lack of sleep, spicy foods, overeating, alcohol intake, long fasting, obesity and lying just after food.

Gastroparesis (Delayed gastric emptying): women have slower gastric emptying as compared to men. That is why women are more prone for developing symptoms like nausea, bloating, gas and abdominal fullness. Certain conditions like gastric surgeries, medications and hypothyroidism can also be responsible for slower gastric emptying.

Gastritis: Gastritis means inflammation of stomach. Older women tend to use pain killers more often and over the counter than men and are more prone for developing gastritis. Pain killers if used for longer duration can cause ulcers in the stomach lining and can cause blood in vomiting.

Constipation: Women have slower emptying from the large intestine than the men. Anal canal and the anal sphincter both are small shorter in women and women can tolerate less volume in the rectal area as compared to men. So, constipation is more common in females and worsens with older age and obesity.

Irritable Bowel Syndrome (IBS): IBS is 2 to 6 times more common among females as compared to males. Symptoms of IBS include diarrhea, constipation or both with abdominal pain and bloating. These symptoms worsen with emotional stress.

Inflammatory Bowel disease (IBD): IBD includes Crohn's disease and Ulcerative colitis. Symptoms are blood in stool, Malabsorption, anemia and weight loss. IBD is two times more common among females than males.

Gallstones: Women have slower gallbladder emptying than men and are two times more prone for developing gallstones. The symptoms include right upper abdominal pain after taking meals, nausea, vomiting and bloating.

Malnutrition: Malnutrition include both under nutrition and over nutrition. Nutritional status is an important indicator of wellbeing of elderly. Malnutrition affects elderly and even can precipitate preterm death. Malnutrition among elderly is multifactorial and can be divided into 3 categories as shown in table 1.

Table1. Factors responsible for malnutrition among elderly

Physiological factors	Social Factors	Psychological factors
Gastrointestinal Malabsorption Poor diet Poor appetite Oral problems: Loss of denture, dysphagia Loss of taste and smell Physical disability to feed self	Isolation Poverty Loneliness Inability to shop and cook food	Dementia Depression Anxiety Confusion

4) ***Genitourinary system***: In older women gynecological problems can affect the general wellbeing, self esteem, quality of life, longevity and confidence. Geriatric gynecologic disorders generally result due

to variations in neuro-endocrine or musculoskeletal function. The female genitourinary tract is primarily dependent on circulating estrogens: the changes in genital tissues associated with aging reflect the progressive decline in estrogen. During the reproductive years, estrogen causes the epithelium to differentiate and deposit glycogen in the vulva and vagina. Estrogen also promotes vaginal lubrication and tissue vascularity.

Progressive atrophy of the genital reproductive system results from the decreased levels of estrogen. Postmenopausal vulvar changes include thinning and graying of pubic hairs, thinning and pallor of soft tissue of vulva. Decrease in size of labia minora and presence of petechiae among sexually active older woman are common. Vaginal walls become thin and less secretary. Due to loss of subcutaneous tissue; atrophy, shortening and narrowing of vaginal is also common among elderly women. The cervix and uterine endometrium becomes atrophic and os becomes stenosed.

Vulvar Inflammations and Infections: Due to combined effects of aging and estrogen deficiency the vulva and vagina becomes inflamed. Fissures, ulcerations, or hypertrophic lesions on the vulva or perineum are also common.

Pruritus: Pruritus is a very common vulvar symptom and is most often related to either estrogen deficiency or a fungal infection. Patients with diabetes mellitus or those taking antibiotics or corticosteroids are particularly more prone to infection with yeast of the Candida species.

Another cause of vulvar pruritus is Lichen sclerosis. The white lesion of Lichen sclerosis resembles skin change associated with severe estrogen deficiency; appearance is often depicted as "cigarette paper" or "papyrus" skin. Atrophy may progress to vaginal stenosis and loss of labia minora. In addition to vulvar tissue, the perianal area may also involve.

***Burning and Irritation*:** Estrogen deficiency in older women can produce atrophic tissue that subsequently tingles or burns when it comes in contact with urine. Paget disease causes the vulva to have a velvety red or eczematous appearance.

Vulvar and Vaginal Discharge: Atrophic vaginitis is believed to be the most common cause of vaginal discharge in elderly women not being treated with corticosteroids or antibiotics. However, this discharge does not have a foul odor, as does discharge associated with some bacterial infections. In elderly women thinning of vaginal mucosa makes it easier for bacteria to enter the sub epithelial tissues and cause infections as compared to younger women.

Vulvar Swelling: Vulvar swelling or areas of thickening raise the possibility of malignancy. Swelling on either side of the posterior vulva located at approximately 4 & 8 o'clock usually reflects inflammation of the Bartholin's glands. Any enlargement of these glands should be taken seriously and evaluated carefully because adenocarcinoma of Bartholin's glands is very aggressive. Invasive carcinoma of the vulva is primarily a disease that occurs in older women; peak incidence is 85 years. The anterior part of vulva is the most common site of involvement with invasive carcinoma: around the clitoris, the vestibule, or along the labia. The malignancy often resembles either a flat infiltrative or ulcerative lesion. Associated symptoms include pruritus, mass, irritation, and bloody vaginal discharge.

***Uterine Prolapse*:** The precise cause of uterine prolapse is not known but may be the result of a number of insults and likely contributors include damage during child birth, striated muscle weakness or neuromuscular disease, trauma, aging, and loss of tissue elasticity and turgor produced by estrogen deficiency. Symptoms of pelvic prolapse include increased pelvic pressure exacerbated by activities increasing abdominal pressure (like, sitting, straining or bending over, coughing, laughing, jogging, sneezing, squatting etc) backache; urinary incontinence; and seeing a mass protruding from the vagina. Prolapse is graded by the extent of descent: Grade 1 prolapse is some movement of the organ from its usual position; grade 2 prolapse implies the organ is near the vaginal introitus; grade 3 prolapse is when the organ is at or bulging from the vaginal introitus. In addition to the presence of a protruding mass and feeling a sense of pressure, the prolapse can affect the bladder and rectum, leading to incomplete emptying. Cervical protrusion can also predispose to mucosal dryness and infection and later can develop ulcers (Decubitous ulcer).

Urinary incontinence is another important gynecologic concern for elderly women. Approximately 30% of community living older women and over 50% of institutionalized elderly women have involuntary urinary leakage.

Postmenopausal Bleeding: Any vaginal bleeding in a postmenopausal woman is abnormal and requires that the source of bleeding be identified. Approximately one third of cases are caused by a premalignant or malignant cervical or endometrial lesion. The most common malignancies causing vaginal bleeding are vulvar, cervical, endometrial and hormone secreting ovarian tumors. Some benign causes of postmenopausal bleeding are atrophic vaginitis, urethral caruncle, endometrial polyps, trauma and endometrial hyperplasia. Estrogen replacement therapy can also produce vaginal bleeding, but any bleeding in older women needs to be fully investigated to rule out other causes.

5) ***Musculoskeletal System***: After the age of 30, the bone density begins to decrease in both male and females. After menopause loss of bone density accelerates and makes the bones more fragile and more prone to fracture. Even trivial fall can cause bone breakage among elderly women.

   Osteoarthritis*:* As the age increases the joints and their cartilages becomes stiffer. The cartilage and the connective tissue becomes thinner which makes the joint less stable and more susceptible to damage. In old age the joints don't slide over one another smoothly and causes rubbing of the surfaces. This is known as osteoarthritis; which is one of the common causes of morbidity among elderly. As the ligaments and tendons also become rigid and brittle; the range of motion also decreases among elderly. Loss of muscular tissue is known as sarcopenia and starts at the age of 30 and progresses throughout the life. This loss of muscle tissue leads to stress on certain weigh bearing joints like knees and make them prone for early onset arthritis.

   Osteoporosis: Osteoporosis is a degenerative process in which bone mineral density decrease and makes bones prone for fracture. After menopause bone becomes more osteoporotic and brittle. So the elderly women are more prone for fracture as compared to elderly men.

6) ***Dental Problems***: Gum and dental diseases are associated with multiple health conditions that affects woman. Gum diseases are generally bacterial infections and have the potential to spread to other parts of the body like heart and lungs. After menopause

women faces oral symptoms like red gums, burning sensation in mouth, altered taste sensation and dry mouth. Denture loss can occur due to osteoporosis in elderly women. Gum diseases are generally pain less; so the women don't bother about it until reaches an advanced stage.

Mouth cancer: Nearly half of deaths from oral cancer between 2012 and 2014 were in those aged over 70, while risk increases drastically from around the age of 65-69 in women.

Some other common diseases in older women are:

Cancer: As the age increase the risk for some cancers also increases like breast, endometrium, ovary and lung. Some cancers don't have warning signs and very difficult to diagnose at an early age e.g. Ovarian cancer.

Sensory decline: In the process of ageing the sensory organs like eyes, ears, tongue, nose and skin declines in their senses respectively. Some age related disease like cataract, glaucoma; macular degeneration and dry eyes are also common for elderly women.

Obesity*:* Obesity is more common in older females as compared to males and obesity is main predisposing factor for many non communicable diseases like heart disease, stroke, and osteoarthritis. Obesity in females is more common due to their habit of eating leftover food in the home as well as lack of awareness about physical activity. According to CDC the Percent of non institutionalized persons with obesity (2011-2014) are as follows:

- Men between the ages 65and 74: 36.2%
- Men ages 75 and above: 26.8%
- Women between ages 65 and 74: 40.7%
- Women ages 75 and above: 30.5%

Hypertension**:** Hypertension is also more common in older women as compared to man. This may be due to obesity as well as stressful life of older women. According to CDC the percent of non institutionalized persons with hypertension (2011-2014) are as follows:

- Men between the ages 65and 74: 63.5%
- Men ages 75 and above: 72.3%
- Women between ages 65 and 74: 64.3%
- Women ages 75 and above: 79.9%

(B) Psychological problems:

Mental Changes: loss of memory is common problem in old age and known as senile dementia. Intellectual performance decreases due to senile dementia.

Diminished sexual activity: Lack of interest in sexual activity leads to stress and emotional outbursts.

Isolation: After the death of the spouse elderly feels lonely and tend to feel isolated. Lack of care by younger generation and social maladjustment along with generation gap further leads to isolation.

Depression: Depression is also very common among elderly. There are several symptoms of depression among older women like lack of interest in household activities and sadness. They become more irritable and emotionally disturbed and start crying without any reason. Other symptoms include confusion, disorientation, lack of appetite and sleep and persistent body aches. In extreme cases of depression the elderly can even think of suicide.

(C) Social Problems:

These are poverty (due to retirement, lack of income, more expenditure due to ill health, etc.), isolation (due to death of family members/spouse), maladjustment with younger generation; Idleness and boredom are main social problems. India is a country where a widow has to struggle a lot to prove her importance or identity in society. There are some occasions in the family where a widow cannot participate in worship and other rituals. As the females lives longer then the male; nearly all the older women faces this situation in her life.

Other major problems are:

- Generation gap.
- Loss senses.
- Financial insecurity.
- Feeling unwanted and solicit
- Difficulty with everyday tasks and mobility.
- Finding the right care provision.
- Access to healthcare services.
- End of life preparations.

Prevention of diseases among elderly woman:

The prevention should start early in the childhood and adolescence and continued during old age. The objective of the prevention is to reduce the age relate disability and to improve the quality of life of the elderly.

(1) Primary Prevention:

Health Promotion: Includes actions to remain healthy in old age.

(A) Periodic checkup of blood pressure and blood sugar levels and blood cholesterol levels.

(B) Periodic eye checkup and hearing testing and dental examination.

(C) Weight or obesity control: Women should know about healthy weight guidelines or weight-management strategies.

(D) Regular physical exercise: exercise in older women can help in strengthening of muscles, tendons and ligaments which increases self confidence, independence and reduces incidence of fall. Regular exercise can also prevent the osteoporosis. Due to moderate physical exercise incidence of heart diseases, diabetes and hypertension also decreases. Yoga and meditation further can improve general wellbeing.

(E) Bone mineral density scan: After 65 years of age; every woman should go for bone mineral density scan on a periodic basis.

(F) Well balanced diet including salad and fruits

(G) Interest creation in reading, writing and listening music. These activities can keep engage the elderly and prevent boredom.

(H) Avoid over the counter medication and self medication.

(I) Take plenty of water to keep kidney healthy.

Specific Protection:

This includes specific vaccination against many diseases like influenza, pneumococcal pneumonia, tetanus, hepatitis B, hepatitis A and rabies.

(2) Secondary Prevention:

Early diagnosis and treatment: Timely detection of many diseases in elderly women can improve quality of life.

(A) Clinical breast examination: women should be educated regarding self palpation of breast and should report any change in consistency or texture.

(B) Pap smear examination: This should be done for all women who attained menopause and complaints of vaginal bleeding to rule out the possibility of cancer cervix.

(C) Elderly are educated for danger signs of cancer

(3) Tertiary Prevention:

Rehabilitation: This includes the training of the elderly to cope up with the disability and build self confidence.

(A) Cataract surgeries and spectacles use

(B) Hearing aids and dental implants

(C) Physiotherapy, psychological therapy and social therapy

(D) Social security in forms of monetary support by the government

Conclusion:

Elderly female population is increasing more than the male and their life expectancy is also more than the male. Due to this reason elderly women are facing more morbid conditions than the men. Social structure also makes elderly women vulnerable to many psychological and social issues. Women presents with different symptoms then the men for many common diseases. Woman need to be more proactive for her symptoms to get medical relief on time and should take part in screening programs for early detection for her illness.

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CHAPTER 4

GENDER DISCRIMINATION

Ms. Sangeeta Dhar

Incident 1: Assistant Professor Rajat Sharma (imaginary name) came back from college to find his pregnant wife and mother weeping uncontrollably. The answer given by them to explain the situation was more shocking than the scenario. Professor was being told by them that they need to go for abortion right away because they had found out "it's a 'girl". Rajat knew that his family never wanted a girl although he always secretly nurtured a dream of being father of a baby girl. But he wanted to know what is wrong with his wife, who is carrying the baby inside her? Justification was simple; she didn't want be mother of a burden, that way her value in the family and even in the society will retrograde. Unable to find any alternative and to keep his baby alive, overnight he changed job and city to keep his wife away from his family. At least now she will not get any partners to fulfil the evil deed – 'Abortion' rather murder of an unborn baby.

Incident 2: In a hospital two ladies had delivered babies at same time. They both blessed with healthy babies. One couple blessed with boy another couple blessed with girl. The second couple was sad because it's a girl, which means an expense or investment without return. But what is wrong with the first couple? They are happy but not merry. In their family not a single girl child was born in last three generation. All boys, everyone is settled but still no one feels it's a complete family. No unconditional attachment, no selfless dependence were ever there, the festivals like Rakhi or Bhai Dooz seems to be incomplete. A girl can also earn but can a boy able to give care and develop attachment like a mother, daughter or sister?

Incident 3: Story of a self claimed modern family, they never underwent "sex determination test" before birth. According to them both genders are equally important. It is a six persons' happy family, grandparents, parents, one boy and one girl. Girl is elder. They have sent both babies to same school. Girl was academically good, always stands among first few in class where boy is above mediocre. Girl did really well in 12th standard board exam but somehow for few numbers she missed the seat in

Government Engineering College but was easily getting seat in reputed private engineering college, was denied to take admission saying they can't afford to spend money because they need to save money for future. The well known silent truth was her brother who is presently studying in 9th standard wants to become an engineer and accepted fact is he will never able to acquire a seat in Government College, so he will be needing lots of money for his engineering study.

Incident 4: It is been always said to Tanu (imaginary name) that she has to be polite, obedient and good home maker to get a good house after marriage. Tanu's father was also worried because she is having a dusty complexion. To marry off this kind of girl in Indian society needs lots of money because she is having less quality and value in 'bride market'. Tanu's parents started collecting money from her childhood even cutting cost from her education. They wanted their girl to settle with a well establish groom. After completing graduation they have arranged marriage for Tanu with a well settle NRI. Gave off almost everything as dowry thinking happily they got a good 'damad' (son-in-law). Even relatives started envy them for this. What a good luck this girl is having!! Tanu was very happy but her dreamland shuttered just after 6 months of marriage when she came to know her husband always liked and loved another lady from his office, who is dusty too but well educated, well established and smart. He never thought of cheating his office colleague, his love of life but at the same time he didn't want to betray his parents who setup his marriage in exchange of money. He literally doesn't care about Tanu. She is free to stay or go back; it's her choice now.

(All incidences are imaginary) In above incidents a particular pattern of gender discrimination is noted may be before, at the time of birth or later. Every time a selection was done and privilege was given to a particular gender 'male'. In all the cases the selection was bias and depending on presumed future. These incidents are far more common than everyone could even notice every time. This is now became a part of main stream culture like an adulteration. But exceptions are also noted. In second incident people acting differently because people feel the necessary of presence of a girl in society when they suffer the absence of them. According to a study conducted in rural part of Uttar Pradesh, 94% of females still do not enjoy equal rights as males. Eighty eight percent married woman wished to conceive boy.

Gender Discrimination is also known as ‘Sexism’. The word sexism probably first coined by Pauline M. Leet during a "Student-Faculty Forum" at Franklin and Marshall College on November 18, 1965. Cambridge dictionary define the word ‘sexism’ as ‘(actions based on) the belief that the members of one sex are less intelligent, able, skilful, etc. than the members of the other sex, especially that women are less able than men.’

Ancient world and gender discrimination

Gender discrimination or Sexism exist from ancient era and prevalent in history. Although the status of woman in pre historic era is not documented but it is presumed that this concept developed over time specially along in increasing of social problems. Let us consider the history of gender discrimination in India. The status of woman in pre vedic era to mid vedic era were equal to men. There were notable woman personalities were their like Khana, Chitrangada, Sanghyamitra. That time female even can choose their husband and had full authority to take their own decision regarding education, marriage or bachelorhood. From the mid of vedic era this concept were started concentrating to effluents families. With time the problem of aggression started and situation deterred in medilivian era. Females were started considering more vulnerable than man and eventually this concept took its branch and adulated main stream of culture. Now with time the concept of vulnerability has gone beyond concept of sexism and become a conflict between civilized and uncivilized activities.

Reasons of gender discrimination

Gender discrimination or sexism is an quite old mind set, which developed though course of history. There are many incidents which are directly or indirectly contribute to develop this outlook. Other than ancient incidents presently it is related and considered as social practice. Depending on nature of this problem the reasons can be categorized under five major aspects.

Social Reasons

Sexism now has become an established part of society. Each social problem is having a deep flaw in society. Present believing, mind set, work, and living- everything directly or indirectly related to society. Thinking gets its shape of believing only after getting approval from society, may be directly or indirectly. In developing countries and even in developed countries male are considered as heir and main important part of society where females are considered as second part of society who is

shadow of 1st part. The family survives by surname and surnames are survived by male. Female are liable to change their surname after their marriage although it is not compulsory by law but defined by ages long practice. In India girls are considered as “Paraya Dhan” means the girl child is belong to others family. The biological parents here just get the status of care taker who gets liability without anything. In such situation and social practice female gets neglected by their own parents where as she will be not getting full authority even after her marriage because in in-laws place she will be a new comer and dependent. This practice is old enough and rarely being question by society and people.

It is considered that responsibility of parents mainly born by male child. The parents and more of the females in developing country accepted this. They also think their first and foremost preference should be at in-law’s place. This mind set is developed depending on grooming from childhood. This outlook is not only uprooting females from their own origin but also in subconscious state creating a fear in parents’ mind for girl child. Even after globalization and so much advancement most of the females her selves think so. Now considering this practice in other words, it become implicit believing that if you want someone own you need to have a boy which is a complete misleading point of view.

Economic reasons

It is believed that the earning capacity of men is more than that of female. Male are considered to be fit in all kinds of job where female considered as weaker race. This concept is true and at same time bias for the same reason. Let us consider two jobs. 1st job requires physical strength and stamina like security duty or heavy weight lifting job. In this kind of job profile a male is more preferable. Now consider 2nd kind of job where taking care of elderly, children are required. Here it is accepted that these jobs can be better done by a female, because she is more reliable and more sensible. Work and job is not specific and limited area, it is a vast arena of activities where some parts like doctor, engineering, teacher, politician etc can be equivalently intuitive by both male and female. Whereas in some part which require more physical strength male are preferred but in some other areas where delicate details work is needed like embroidery, some handicraft, tea leave selection female are preferred for same reason. In both the cases its physical attributes which make a particular gender more preferable for particular work. Every work is necessary and superior in its own way so preferring one gender over other definitely an indication of biasness.

Physical reasons

In history it has been noted that whenever civilization forgets its civics the weaker are targeted. Here 'weaker' means a simple measurement of physical strength and judging everything depending on that. It is as simple as an average woman and children cannot win physical combat with male like the same way if a man and animal fight the man cannot win. Anything which is established depending on physical strength always put the question on civilization. History of wars shows it's the most inhuman activity where anyone can be targeted. Along with the brutality and sexual cruelty against female and children same kind of activities also noted against male although in lesser number. It is accepted that physical strength is needed for some good work also and obviously for natural traits male gets some advantage but at the same time female are carrier and bearer of future generation which male cannot do. A society is a complete establishment which requires everything to be in working condition.

Psychological reasons

Psychological setup comes from situation and environment. In this male dominated society a pre-perceived idea is taken in favour of man. But if we question this belief, we will find there are some reasons behind this.
"Why you like to have boy rather than girl?"

Answer of this question can be explained like these. They can provide more financial support, they are powerful, we need not be worried that much while bringing them up etc. If we extract the essence of that answers the simplified answer will be "to get something in return" or in other words 'investing something for long term gain". So here is the catch. Is it good enough reasons to get a boy? Or is not these reasons are selfish enough and at the same time bias? Now a day every aspects of these can be equally fulfilled by females also. The most important part and condition to get a healthy family is totally ignored that one family will never be able to sustain without emotional attachments, values and selfless contribution. It is being demonstrated, tested and cited many times that woman are more caring and emotional. Even our common believe said so. It is depends on culture, environment, bringing up and inculcated believing by which girls get manifested to be believe themselves as secondary part in their own family thus inhibit them to express their biological quality of nurturing to same. According to Wester *et al.,* (2002) "Empirical evidence suggests that girls are socialized to be emotional, non aggressive, nurturing, and obedient, whereas boys are socialized to be unemotional, aggressive,

achievement oriented, and self-reliant. Peers continue this process as children develop and mature in effect constraining how, where, why, and with whom certain emotions are expressed."

Effects of gender discrimination

Gender discrimination not only affects directly but it also having some indirect affects. Direct effects include female feticide, discriminated behaviour against particular gender, not giving facilities, properties, non consideration as heir and many more. Whereas indirect effect includes disturbed of sex ratio, increase of crime against female, psychological affects etc.

Foeticide

Sex selective abortion or feticide is the process of terminating pregnancy depending on sex preference of descendant. The sex selective abortion is mainly related to Female foeticide. It is prevalent in areas of world where cultural value prefer male over female. It is mostly seen in countries like China, India, Bangladesh, Pakistan, Caucasus, and Western Balkans. This horrific picture of sex selective abortion in China is documented and portrayed in documentaries like The Dying Rooms (1995) and It's a Girl: The Three Deadliest Words in the World (2012).

Deterrence from getting proper education

Still millions of girls are deprive of getting proper education. Around 516 million (2/3 rd of whole illiterate population) females are illiterate. 31 million girls of primary school age including 17 million who never being to school are school are beyond reach of further education. Giving education to girl in some places in some places still considered to be unnecessary. This directly affects the woman earning power, well being, self dependency at the same time indirectly hamper the family well being, structure, decrease overall quality of population of a country.

Practice of dowry

The result of expression of extreme gender inequity is dowry. This is an evil practice in South Asian country specially in India, where bride's family liable to give money to groom's family for marriage. Unfortunately even after accepting the demand of groom's family brides are subjected to physical and mental torture for further demand even get murdered for money. India's National Crime Bureau reported that there were approximately 8,233 dowry murders in 2012.

Increase of crime against females and abrogation of human rights

The concept of sexism portrayed an undignified image of female which led to more crime and sometime gender specific crime against female. There are some practices and incident where females are subjected not only to violation but their human rights are also desecrated. The summery of some of them are presented in Fig 1.

Violence against women throughout the life cycle

Phase	*Type of violence*
Pre-birth	Sex-selective abortion; effects of battering during pregnancy on birth outcomes
Infancy	Female infanticide; physical, sexual and psychological abuse
Girlhood	Child marriage; female genital mutilation; physical, sexual and psychological abuse; incest; child prostitution and pornography
Adolescence and adulthood	Dating and courtship violence (e.g. acid throwing and date rape); economically coerced sex (e.g. school girls having sex with "sugar daddies" in return for school fees); incest; sexual abuse in the workplace; rape; sexual harassment; forced prostitution and pornography; trafficking in women; partner violence; marital rape; dowry abuse and murders; partner homicide; psychological abuse; abuse of women with disabilities; forced pregnancy
Elderly	Forced "suicide" or homicide of widows for economic reasons; sexual, physical and psychological abuse

Fig. 1: WHO Typology of violence against woman throughout whole life cycle

Discriminative property rights

Right on property and ownership of assets are influenced by social and cultural practice, races, political, legal and sexual factors. In most of the countries even in developed countries like UK male child considered as first line of heir. Women who get reduced level of advantages over property matters are suffered from vulnerability, lack of economic opportunity, security at home, maintaining basic survival needs and other development and wellbeing related problems which ultimately affects their status in society. Unequal distribution of productive and non productive resources causes restriction of earning ability, instability and insecurity for the deprived ones. Although in many countries the equal

property rights are established by laws but still poorly implemented and practices. In India as per The Hindu Succession (Amendment) Act, 2005 daughters are equal share holder in parents' property but still as poorly practiced and implemented in northern and southern India.

Discriminative food sharing, care and medical facilities

Gender discrimination affects basic facilities and rights. It been studied and established by many times that preferred gender i.e. male always enjoys privileged facilities of heath and food inside and outside house hold. It is noted that on an average working day females runs short of 20% calorie intake against prescribed calorie intake. Average man get surplus of 800 calories where an average women has deficit of 100 calories. In rural area girl child get breast feed for shorter period of time than boys. Even in present day practice in food sharing male are allocated with best quality of food available where as female are allocated after allocating food to male. Even in health care female are neglected. Due to patritial structure of society women specific health issues are over looked. The concept of resource allocation from finite resource leads to negligence towards woman health care. Lack of awareness, house hold treatment, stress and less nutritional value make woman more vulnerable towards health issues. Reduced social status and less economical viability affects the medical advantages. According to UNFPA, although many countries has obligation to protect the right of women's health but unfortunately they do not do so. In today's context maternal mortality is a major problem and also an issue of violating human rights.

Wedge differences

A practice of 'adjusted' and 'un adjusted' pay gap in noted all over world. Although it is reported from all corners of world but largely found in developing countries. Adjusted wedge difference is stand for prefixed pay gap. For example employee who suppose to take maternity or child care leave are paid less than who don't require these leaves. A 'un adjusted' pay gap stands for a difference in pay depending on presumed work capacity of particular gender. For the year 2013, India was reported to have gender pay gap 24.81%.

Tendency of lowering exposure for females

Practice of Gender discrimination makes one gender (female) more vulnerable. This situation put question on female safety. Indirectly this

increase the tendency of barring exposure of females. Even some times they are destitute of basic education. Long distance from school, early pregnancy and marriage, vulnerability to HIV epidemic, stereotype in school, unwillingness of spending money on girls' education affects the ratio of literacy among female. 41 million among 71 million children who never been into preschool are girl.

Disturb gender ratio

According to Fisher's theory the sex ratio of male and female off springs produced by sexual reproduction is 1:1 and natural selection favours this ratio in course of evolution. Sexism indirectly disturbs the ratio and makes species more vulnerable through the course of evolution. Human the finest race, evolved from evolution now disturbing this natural concept through artificial means. Sex discriminatory abortions, unprivileged status of female indirectly affect the whole species balance.

Eradicating gender discrimination

Gender/Sex discrimination or sexism is deep rooted in human society. This problem cannot be eradicate by simply following directed measures or predefined formulae. Only multi dimensional reasonable approaches can bring the change. This change cannot be expected in one day, it will take years to change this mindset. Awareness, education, empowerment, legal rights, equal participation- are some key features of this movement.

Awareness

Awareness is the main influencer of the concept of sexism. The skeptical concept of gender selection can only be drive away with awareness. People especially female should be aware of their social status. The awareness related to work capacity, attachment, family bonding and value of women should be well spread. Positive and logical examples should be laid in front of general population. Digital media, street drama, printed media, social networks etc can largely influence the population.

Education

Education is the most powerful tool for regaining gender equity. Table 1 showed the positive effect of education on community. It not only elaborated the scope of empowerment but also contributes toward maternal-child health, and persuades overall family's wellbeing.

Table 1: Positive effect of women education

Woman have primary education • 2/3 rd less maternal death • 15% fewer child deaths • 1.7 million children would be saved from stunting from malnutrition • There would be 14% fewer child marriages • Empowerment rises to 50% if they have a primary education instead of 37% • In Pakistan, women with a primary education earn 51% what men earn • In Jordan, women with a primary education earn 53% what men earn • In Brazil, only 37% of women with less than primary education are in work. This rises to 50% if they have a primary education Woman have secondary education • Child deaths would be cut in half, saving 3 million lives • 12 million children would be saved from stunting from malnutrition • There would be two-thirds fewer child marriages • Empowerment rises to 60% if they have a primary education instead of 37% • In Pakistan, women With a secondary education, they earn 70% what men earn • In Jordan With a secondary education, they earn 67% what men earn • In Brazil, only 37% of women with less than primary education are in work. This rises to 60% with a secondary education *Source: UNICEF, Education for All Global Monitoring Report, 2013

Fight against superstition

There are many superstition related to females. In some part of India females are not allowed to participate in religious function, marriages etc. At the same time some religious rituals like bratas, are only concentrated to and considered to be perform by females only but in reality ritual practices can be performed by both the genders depending on their faith and believing. All these believing indirectly restricted the right and thought of female. This concept should be clarified.

Empowerment

Empowering woman can increase the status of woman in society. Earning power not only give the monetary security also increase the exposure. As a whole it improves the status of her and contributed to the overall upliftment of family. This also can act as stepping stone on future. 'Empowering woman' is considered and accepted as a vital step towards global poverty eradication. 48% of total population is female so no civilization can work along eliminating half of its population. Rio Declaration (1992) and Beijing Declaration (1995) recognize empowering woman is essential for sustainable development and key eliminate for removing gender discrimination.

Legal rights

Securing woman rights through law is essential. The necessity of relevant laws acknowledged all over world. There are few international conventions to secure the women's rights. In India there are laws related to woman general rights, against violence and against discrimination. But due to lack of awareness and poor implementation benefits are not enjoyed by all. Moreover a few more legal amendment is need to further ensuring woman rights and keeping laws compatible to present scenario. All laws which are relevant are summarized in Table 2.

Table 2: Constitutional Provisions and criminal Offence against woman

Constitutional Provisions and Privileges (i) Equality before law for women (Article 14) (ii) The State not to discriminate against any citizen on grounds only of religion, race, caste, sex, place of birth or any of them (Article 15 (i)) (iii) The State to make any special provision in favour of women and children (Article 15 (3)) (iv) Equality of opportunity for all citizens in matters relating to employment or appointment to any office under the State (Article 16) (v) The State to direct its policy towards securing for men and women equally the right to an adequate means of livelihood (Article 39(a)); and equal pay for equal work for both men and women (Article 39(d)) (vi) To promote justice, on a basis of equal opportunity and to provide free legal aid by suitable legislation or scheme or in any other way to ensure that opportunities for securing justice are not denied to any citizen by reason of economic or other disabilities (Article 39 a) (vii) The State to make provision for securing just and humane conditions of work and for maternity relief (Article 42) (viii) The State to promote with special care the educational and economic interests of the weaker sections of the people and to protect them from social injustice and all forms of exploitation (Article 46) (ix) The State to raise the level of nutrition and the standard of living of its people (Article 47) (x) To promote harmony and the spirit of common brotherhood amongst all the people of India and to renounce practices derogatory to the dignity of women (Article 51(A) (e)) (xi) Not less than one-third (including the number of seats reserved for women belonging to the Scheduled Castes and the Scheduled Tribes) of the total number of seats to be filled by direct election in every Panchayat to be reserved for women and such seats to be allotted by rotation to different constituencies in a Panchayat (Article 243 D(3)) (xii) Not less than one- third of the total number of offices of Chairpersons in the Panchayats at each level to be reserved for women (Article 243 D (4)) (ix) Not less than one-third (including the number of seats reserved for women belonging to the Scheduled Castes and the Scheduled Tribes) of the total number of seats to be filled by direct election in every Municipality to be reserved for women and such seats to be allotted by rotation to different constituencies in a Municipality (Article 243 T (3)) (x) Reservation of offices of Chairpersons in Municipalities for the Scheduled Castes, the Scheduled Tribes and women in such manner as the legislature of

a State may by law provide (Article 243 T (4))
Offences against women and children in the Indian Penal Code The Indian Penal Code, 1860 • Abandoning of child under 12 years of age • Adultery • Assault or criminal force to a woman with intent to outrage her modesty • Buying minor for purpose of prostitution • Causing death of quick unborn child by act amounting to culpable homicide • Causing miscarriage or miscarriage without the woman's consent • Cohabitation caused by a man deceitfully inducing a belief of lawful marriage • Concealment of birth by secret disposal of dead body • Concealment of former marriage • Death caused by act done with intent to cause miscarriage • Dowry death • Enticing, detaining or taking away with criminal intent a married woman • Fraudulent marriage ceremony without lawful marriage • Husband or relative of a husband of a woman subjecting her to cruelty • Importation of girl from foreign country • Intercourse by man with his wife during separation • Intercourse by a member of management or staff of a hospital with any woman in that hospital • Intercourse by public servant with a woman in his custody • Intercourse by superintendent of jail, remand home, etc • Kidnapping, abducting or inducing woman to compel her marriage • Marriage ceremony fraudulently gone through without lawful marriage • Marrying again during lifetime of spouse • Preventing a child from being born alive or causing its death after birth • Procreation of minor girl • Rape • Selling minor for purpose of prostitution • Word, gesture or act intended to insult the modesty of a woman

Medical facility

Female medical facilities should be increased. Woman specific treatment centre and awareness should be created. The recruitment of female doctors should be encouraged because sometime female do not feel comfortable discussing their personal problems with male doctors. More campaign should be conducted to create awareness among female and their family. Maternal heath, nutritional value, child care provision, dealing with emergencies should be discussed and published via governmental and non governmental media.

Conclusion

"All is in our mind" this is the most important reason of gender discrimination. Human beings are considered as most advance race of nature. To sustain as a race it is important to have both the genders. Nature is having own way of selecting a species as fittest. The interference not only disturbs the balance inside the species but also make it more vulnerable in terms of existence. Leaving a part behind cannot lead to sustainable future. The education acts as tools and adaptability to change. To understand the need of equity a free, analytical and adaptable mind is the ultimate need which can only be created by awareness and education. Every gender should be treated with equal respect. Until unless both the genders started act as complimentary to each the development will be hampered. Gender discrimination problem is not limited to culture, race, tradition or need, it is the global problem and it is necessary to eradicate it not only for the sake of female wellbeing but for achieving sustainability in human race.

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CHATPER 5

PRE- MENSTRUAL AND MENSTRUAL CYCLE EXPERIENCES AND DISORDERS

Dr. Poonam Khanna, Dr. Priyanka Sharma, Dr. Mamta Bansal

Menstruation is supposed to be invisible and silent.

The majority of menstruating girls and women are actually silent and become invisible. Millions of girls and women are subjected to constraints and restrictions in their routine life during their menstrual periods. According to estimates about 52% of women worldwide are in reproductive age and most of them thus are menstruating monthly. The majority of them have no access of safe sanitary products, private and hygiene space to wash and change pads. Besides, they face many health related problems and diseases due to poor hygiene during cycles, lack of facilities, unaffordability to purchase sanitary products. These situations sometimes compel younger girls to abscond from schools causing negative impacts on their mental and physical health.

Understanding menstrual cycle

The menstruation is a normal series of changes in reproductive cycle in female body from menarche (onset of menstrual bleeding) to menopause (periods stops completely). This is a monthly menstrual bleeding from vagina that a woman has from early teen years around age 12 till menopause around age 50. During menstrual cycle body goes through to prepare for a possible pregnancy. About once a month, the uterus grows a new, thickened lining (endometrium) that can hold a fertilized egg. When there is no fertilized egg to start a pregnancy, the uterus then sheds its lining. Menstrual cycles average approximately 28 days in length, but any cycle length between three and five weeks apart can be normal. In the first year of “reproductive life,” menstrual cycles are usually irregular and unpredictable. However, it is still possible to become pregnant during this time. Cycles in younger women tend to be longer and those in older or perimenopausal women are shorter. Normal menstrual flow lasts between

three and six days and involves between eight and sixteen soaked regular-sized pads or tampons. Most women experience cramps before their flow is about to start. Women commonly experience some changes in how they feel across the menstrual cycle. For example, bloating and water retention are most intense on the day flow starts. Women may also note that their breasts become mildly tender. Some women become more sensitive to emotional issues (either happy or sad) in the days before flow. All of these experiences are normal.

Phases of menstrual cycle:

The day count for menstrual cycle begins on the first day of menstruation when blood starts to come out of the vagina. The average length of menstrual cycle among women has been assumed to be 28 days. The entire duration of a menstrual cycle can be divided into four main phases (Fig. 1):

1. Menstrual phase (From day 1 to 5)
2. Follicular phase (From day 1 to 13)
3. Ovulation phase (Day 14)
4. Luteal phase (From day 15 to 28)

Menstrual phase (day 1-5):

Menstrual phase begins on the first day of menstruation and lasts till the 5th day of the menstrual cycle. The following events occur during this phase:

- The uterus sheds its inner lining of soft tissue and blood vessels which exits the body from the vagina in the form of menstrual fluid.
- Most of blood loss happens during the first three days. Blood loss of 10 ml to 80 ml is considered normal.
- During this period girls may have cramping pain in pelvis, legs and back. These cramps help in shedding endometrium by contraction of uterine and abdominal muscles.

Follicular phase (day 1-13):

This phase also starts on the first day of menstruation, however it lasts till the 13th day of the menstrual cycle. The following occasions occur in the course of this section:

- The pituitary gland secretes a hormone that stimulates the egg cells inside the ovaries to grow.
- One of those egg cells starts maturing in a sac-like-structure known as follicle. It takes 13 days for the egg cellular to reach adulthood.

- While the egg cellular matures, its follicle secretes a hormone that stimulates the uterus to broaden a lining of blood vessels and smooth tissue referred to as endometrium.

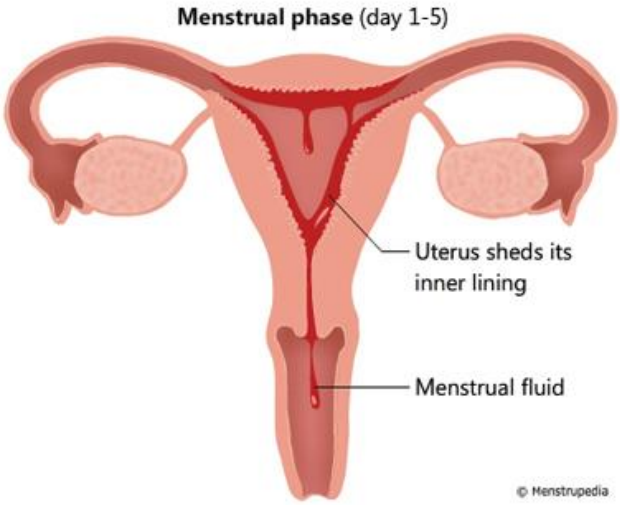
Ovulation phase (day 14):

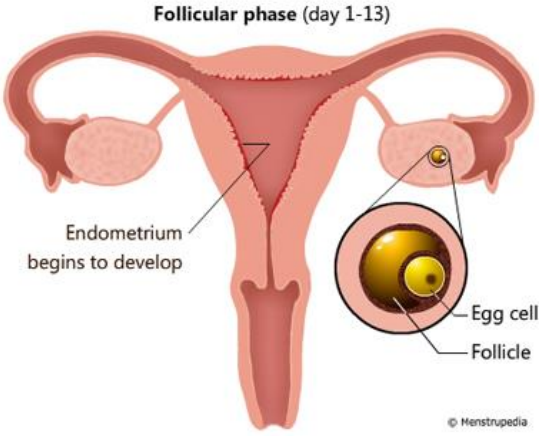
On the 14th day of the cycle, the pituitary gland secretes a hormone that causes the ovary to release the matured egg cell. The released egg cell is swept into the fallopian tube by the cilia of the fimbriae. Fimbriae are finger like projections located at the end of the fallopian tube close to the ovaries and cilia are slender hair like projections on each Fimbria.

Luteal phase (day 15-28):

This phase begins on the 15th day and lasts till the end of the cycle. The following events occur during this phase:

- The egg cell released during the ovulation phase stays in the fallopian tube for 24 hours.
- If a sperm cell does not impregnate the egg cell within that time, the egg cell disintegrates.
- The hormone that causes the uterus to retain its endometrium gets used up by the end of the menstrual cycle. This causes the menstrual phase of the next cycle to begin.




Follicular phase (day 1-13)
Endometrium
begins to develop
Egg cell
Follicle
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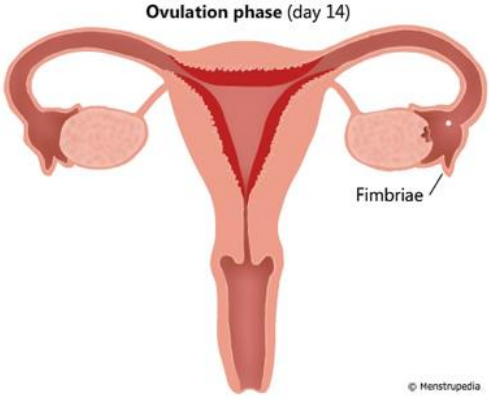
Ovulation phase (day 14)
Fimbriae
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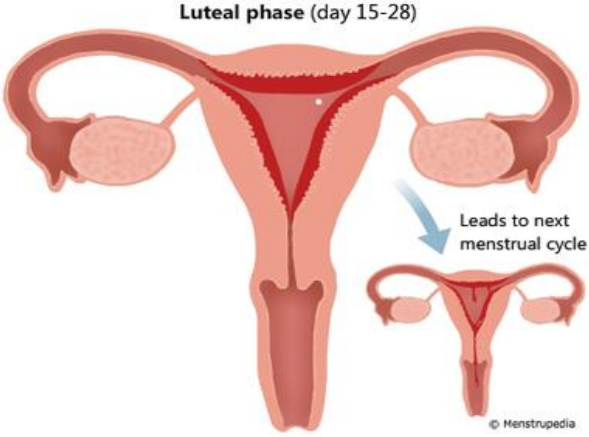
Luteal phase (day 15-28)
Leads to next
menstrual cycle
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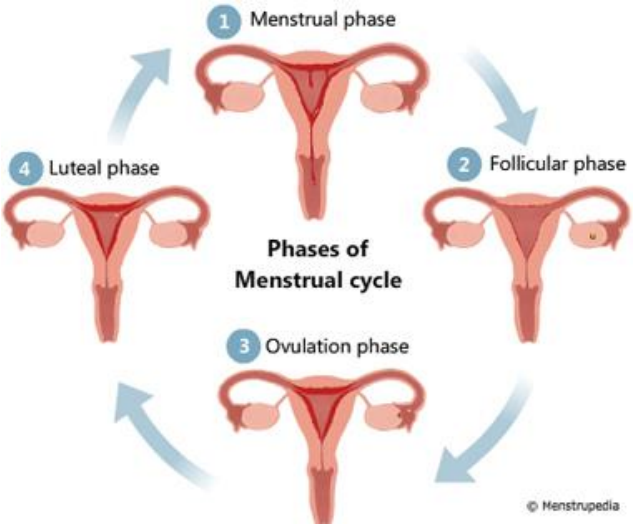

Fig. 1: Phases of Menstrual Cycle.

Beliefs, myths and taboos

Menstruation is a normal process of reproduction in women's life in which the thick lining of endometrium remove through the vagina in the form of blood. This process indicates the onset of puberty among girls. This so-called normal phenomenon among girls always surrounded by multiple taboos and myths in the most part of the world. These unwritten rules and practices exclude women and girls from many rituals, social and cultural events of life. They are also associated it with evil spirits, superstitions, shame and embarrassment due to 'impurity' during menstruation. In Tanzania, it is belief that if a menstrual cloth is seen by others, the owner of the cloth may be cursed. In other cultures, during menstrual cycle girls should not take bathe otherwise they will become infertile or touch a plant or it will die. The awareness and education of parents regarding menstruation is almost everywhere paradoxical. In much of Asia and Sub-Saharan Africa, girls' understanding and awareness regarding menstruation in still very near to the ground.

Beliefs, myths and taboos associated with Indian culture:

Many parts of Indian culture, women are prohibited to do work in kitchen as they are treated as impure, unhygienic and unclean. It is believed that the specific smell emitted by menstrual women contaminates and spoils the food. Sometimes, they are not allowed to touch preserved food like

pickles. They are also not allowed to enter in 'puja room', exclude them to attend ceremony, offering prayers and touching holy books. Women and girls must be 'Purified' after 5-7 days of menstrual cycle to perform daily chores and return to her family. Although, there is no scientific tests showing menstruation is related with spoilage of any food.

There are also strict restrictions for women and girls who are menstruating regarding eating of certain food. They need to avoid eating sour and cold fruits and food like tamarind, pickles, curd, ice creams, cold shakes and milk. It is believed that these products will hamper the menstrual flow. Physical exercises and activities are also avoided by menstruating women but in real exercises help women and girls in relieving cramps and pains. It also helps in managing premenstrual syndrome, relieve bloating and make women feel much happier by release of hormone serotonin. In some regions of India, women bury their clothes or pad used during periods to avoid evil spirits as it is also believed that this cloth can be used to do 'Black magic' by any malevolent person. Moreover, the secretions from women body during menstruation and child birth are treated as contaminating natural resources mainly water. Moreover, water is considered as most easily contaminated and most common medium for purification. For this reason, probably, they are not allowed to take bath for first few days of their menstrual period. However, it seems to be illogical and there is no scientific proof to it.

Millennium Development Goal and women rights

Millennium Development Goal 3 is to promote gender equality in all levels of education and empower women. It has only one target i.e., to eliminate gender disparity in primary and secondary education by 2005, and in all levels of education by 2015. It is important to promote the total and equitable participation of both men and women in efforts aimed at improving poverty reduction, food security, and sustainability of rural development. Without gender equality and the economic and social improvement for rural women, food security cannot be achieved. On the

Fig. 2: Symbol of Millennium Development Goal 3

other hand, taboo of menstruation among adolescent girls push them out of school temporarily or permanently. This monthly period issue also creates impediments for female teachers. In such cases, they leave early from schools or report themselves unwell.

The probable reasons in both the cases remain same like gender-unfriendly school culture and infrastructure, lack of clean, safe and private space in schools to change sanitary cloth/pad, lack of adequate menstrual protection alternatives and appropriate sanitary products. These lack of facilities for adolescent girls and women undermine the right of privacy, resulting in basics fundamental infringement of the 'human rights', with no equal opportunity.

This stigma around menstruation also abuses and violates the right of human dignity, right to non-discrimination, integrity, privacy and right to freedom from inhumane. Due to these many reasons which acts as obstacle, Millennium Development Goal 3 (promote gender equality and empower women) seems very difficult to achieve either (Fig. 2).

Common menstrual disorders:

1. Abnormal uterine bleeding
   a. Heavy menstrual bleeding or menorrhagia
   b. No bleeding or amenorrhea
   c. Bleeding between periods or metrorrhagia
2. Dysmenorrhea: painful menstrual bleeding
3. PMS (Premenstrual syndrome)
4. PMDD (Premenstrual dysphonic disorder)
5. Menstrual Cramps

1. Abnormal uterine bleeding:

a. <u>Heavy menstrual bleeding or menorrhagia</u>

Heavy bleeding interferes with the normal routine activities in women's life. Bleeding is treated heavy when one or more sanitary pads or tampons are soaked every hour in a day or bleeding for longer than a week. Normal blood loss during periods is about five tablespoons, but during heavy flow, it can be increased upto 10-25 times. This heavy bleeding can be more common during menarche or menopause. Women or girl may suffer from anaemia, tiredness, fatigue, low-energy or shortness of breath.

Causes of heavy menstrual bleeding:

The causes of menurrhagia are sometime unknown. But there are many factors which can be responsible for heavy bleeding.

Following are the few common factors:

- Imbalance of hormones in the body: The imbalance of reproductive hormone i.e. estrogen and progesterone are either produced too much or not enough by the body, that cause heavy menstrual flow. Hormone imbalance may occur due to various conditions like hypo or hyper thyroidism, PCOD (Polycystic Ovarian Disease), obesity, stress in daily life, insulin resistance, use of tobacco etc.
- Ovaries malfunctioning: If the ovaries are anovulatory i.e., egg is not released, body is not able to produce reproductive hormones, as it would during the normal periods. It leads to imbalance of hormones and in turns responsible for menorrhagia.
- Uterine fibroids and polyps: These are non-cancerous or benign tumors in the uterus or on the lining of uterus which is responsible for causing prolonged and heavy bleeding during menstrual cycle.
- IUD (Intrauterine device): It is a nonhormonal intrauterine device use by women for birth control. Menorrhagia is a side-effect of using this device in many cases.
- Miscarriage or other complications: Miscarriage also leads to heavy, prolonged and late periods. Abnormal position of placenta may also responsible for heavy bleeding.
- Adenomyosis: In this conditions glands from the lining of uterus i.e., endometrium become embedded in the uterine muscles, causing painful periods and heavy bleeding.
- Cancer: Cancers like cervical or uterine is responsible for excessive and painful bleeding.
- Medical complications or excessive medication: There are number of medical complications like diabetes, thyroid problems, liver or kidney disease, associated with heavy flow. Some medicines like hormonal (estrogen/ progestins/ steroids), anticoagulants (Lovenox), anti-inflammatory can responsible for heavy or prolonged menstrual bleeding.

b. No bleeding or amenorrhea

It is the absence of menstruation or no menstrual periods at all. This condition is common during pregnancy, after menopause or before puberty. But if a woman faces this problem during normal life time, they need to consult doctors. Amenorrhea can be primary or secondary.

Primary amenorrhea means if a girl turns 16 and yet not menstruated. It may be due to many reasons like endocrine/hypothalamus disruption, genetic abnormalities, ovaries disorders, eating disorders, under weight, excessive medications and exercises.

Secondary amenorrhea means a woman with regular periods, suddenly stops for three months or longer. The reasons may be disruption in estrogen level, thyroid level, pituitary gland, stress, depression, weight loss, excessive exercise or prolonged illness. Sometimes this condition occurs due to cysts in ovaries or ovaries are surgically removed.

c. Bleeding between periods or metrorrhagia

This is abnormal vaginal bleeding between periods also known as intermenstrual bleeding, spotting or metrorrhagia. There are many possible causes to this problems like sudden change in medication, hormone imbalance, stress, a growth in uterus, miscarriage and cancer.

2 **Dysmenorrhea: painful menstrual bleeding:** Most girls and women experience menstrual cramps and painful bleeding before and during their periods. Sometimes it is treated as normal but in other cases if cramps are persistent and heavily painful, called dysmenorrhea, which actually needs health care professionals.

These cramps during bleeding are due to uterine contractions triggered by hormone prostaglandins, released from uterine lining cells. This hormone also travels in blood stream and cause contractions in the intestine which results in diarrhoea, low blood pressure, nausea or sometimes faintness and sweaty.

3. **PMS (Premenstrual syndrome):** Premenstrual syndrome describes the physical and psychological symptoms correlated with the menstruation. This syndrome is sufficient enough to disrupt and interrupt woman's lifestyles and daily routine. About 30-40% of women suffers with this trouble and more than 150 symptoms are documented. The most common symptoms is 'depression' which develops 5-7 days before the onset of period and disappear once the periods starts.

   The causes of PMS is still not clear but seems to be caused by imbalance of estrogen and progesterone hormones, which may in turn influence serotonin, a chemical of brain and has effect on mood and anxiety. Physical symptoms can be tiredness, swollen breasts, bloating, constipation, headaches and shortness of breath Emotional symptoms include stress, lack of concentration, anxiety, anger, mood swings, crying and depression.

4. **PMDD (Premenstrual dysphonic disorder):** It is the most severe form of PMS. There are more than 100 symptoms that categorize PMDD but no two women demonstrate a similar range of symptoms. Symptoms among women sometimes also vary from month to month.

   The causes and symptoms of PMDD remains the same as for PMS. Hormonal shift or imbalance during menstrual cycle results in headache, fatigue, body aches, cramps, mood swings, abdominal bloating etc.

   Data reveals that 3-8% women are affected from PMDD during their child bearing period. PMS and PMDD are diagnosed by looking at the symptoms of the patients for consequently for three months. This problem can be severe in case of medical impediments of a woman like asthma and migraine issues.

5. **Menstrual cramps:** During or before menstruation, it is quite common to feel discomfort, pains and cramps in abdomen, thighs and in lower back. This is because of contraction of womb muscles which is required to shed endometrium lining of uterus. Some adolescent girls and women many experience nausea, vomiting, diarrhoea, stress or headaches.

There are many factors which are associated with severe pain like

- Heavy blood flow
- Overproduction of prostaglandins
- Growth in the womb or abnormal uterine tissue growth
- Use of birth control
- After a first child
- Underweight of a girl of women

Management for healthy menstruation:

- Disposable sanitary pads or towels are the most hygiene and safest methods of managing menstruation.
- Reusable clothes and pads can prove to be sustainable method, but more hygiene is required in washing and drying under sunlight and proper stored in clean dry area.

  But out of shame girls and women usually hide their sanitary clothes out of sight. It means they hide and keep in a damp and unhygienic place.
- Menstrual cup can prove be an appropriate technology to manage menstruation. It is made up of medical silicon rubber which is reusable and inserted in to the vagina to collect menstrual blood.

  This technology may establish practical, sustainable and cost-effective alternative.
- There should be sufficient, clean and safe sanitary facilities, access to adequate water for washing hands and sanitary clothes, private space to change sanitary materials in schools and in workplaces.
- There should be safely disposal of used material or safe dry places if reusable.
- Proper awareness, education and information in schools regarding managing menstruation hygiene, combat myths and beliefs related to it also empowers adolescents and women with factual information of their bodies and care about it.
- In schools, teachers should also be trained in teaching menstrual hygiene.

Good Hygiene practice

There are plenty of hygiene and health issues apart from social issues related to menstrual cycle. Most of the women in developing countries do not have access to clean and cheap sanitary products; they are resort to use an old cloth, which is frequently reused. Other atrocious products uses by poor women are ashes, newspaper, dried leaves, husk sand to abet absorption. Washing facilities are also lacking in many areas. So, inadequate hygiene, poor protection and washing affect have significant implications on their physical, emotional and mental health.

They become susceptible to infections, as the risk of all infections including sexually transmitted infections is higher than normal during the periods. A specific odour of menstrual blood creates stigma in girls and women that force them to miss their schools and jobs during those days. These 5-6 days away from office become a huge economic loss to working women (Table 1).

Table 1: Potential health infections due to poor practices during menstruation.

S. No.	General practices adopt by girls and women	Health related issues
1.	Unclean and unhygienic sanitary pad, clothes or other material	Local and uterine infections
2.	Irregular changing pads	Skin irritation and chances increase for infections in broken skin
3.	Insertion of unhygienic material for blood absorption in vagina	Bacteria can easily travel to the cervix and the uterine cavity
4.	Use of tampons during light blood loss or during non-menstruation	Vaginal irritation and/or unusual vaginal discharge
5.	Unprotected sex	Sexually transmitted diseases including HIV or Hepatitis B
6.	Unsafe disposal of sanitary pads	Risk of spreading infections including HIV or Hepatitis B
7.	Improper hand-wash during menstruation	Facilitate spread of infections

Relief from cramps and pains:

Some remedies are very supportive for mild to temporary cramps during menstrual cycle. Following are the few tips to lessen the pains and fast relief during periods.

1. Hot message at lower abdomen by using hot water bottle, heat pad or hot towel will help in relieving pain. A study on 18-30 years women with regular menstrual cycle demonstrated that heating pads at 104°F (40°C) is as effective as ibuprofen.
2. Message from essential oils i.e., naturally occurring, plant derivatives, volatile organic compounds; prepared from flowers, bark, stem, root, seeds or other parts of the plants, for about 15-20 minutes can help in reducing pain significantly and immediately. Essential oil like lavender oil, rose oil etc. can be diluted with a vegetable oil, almond oil or mild oil like coconut oil with the concentration one drop of essential oil with one spoon of other oil.
3. There are some foods which cause bloating and retention of water during menstrual cycle. It is good idea to avoid these foods during periods like alcohols, fatty foods, carbonated beverages, caffeine, salty foods and tobacco products. Instead try caffeine- free drinks like ginger or mint teas, lemon flavoured hot water or fruits like strawberries or raspberries.
4. Rest when needed and listen favourite music.
5. If pain persists, take mild pain killer like ibuprofen, acetaminophen or aspirin.
6. Make exercise a part of your life. Regular exercise helps in preventing cramps and lowering menstrual pain..

How to make your own heating pad:

1. Cut and sew together two pieces of fabric, leaving a hole at the top.
2. Fill with uncooked rice and sew up the hole.
3. Microwave for a few minutes to the desired temperature. Don’t overheat!
4. Let cool, if necessary. Or wrap your homemade pad in a towel to reduce heat transfer. Reuse as necessary.

Dealing with taboos, beliefs and myths:

These beliefs and myths not only affect reproductive health but also physical, emotional and mental health of adolescents and women. Adolescents especially from developing world have very less or limited knowledge about menstrual health and hygiene. In addition, their mothers, sisters, sister-in laws or other women in family and neighbour are not open enough to discuss this menstrual issue with them. Moreover, they themselves not very aware of the biological changes and safe, clean and hygiene practices needed during menstrual cycle. Rather they pass various other myths, restrictions and taboos in young brain which further affect their health in many ways.

There can be various steps to deal with these menstrual related beliefs and myths as follows:

- The foremost step to combat this problem is raising the awareness among adolescent girls and women related to personal hygiene during menstruation cycle.
- Campaigns on health and hygiene during menstruation would prove a worth.
- Community based health programs is also required.
- Teachers of schools can also be trained and taught regarding menstruation.
- Empowering and educating women can play an important in improving health and hygiene and overcoming the taboos in families and in turn in society.
- Cheap, clean and hygienic sanitary pads and napkins should make available in rural and slum areas.
- Adequate facilities for washing and sanitation should be made available to both urban and rural areas.
- Awareness should be increased among male partners so that they can facilitate in combating deep rooted beliefs and myths. Moreover, they can mentally support their women in families, friends and employees.
- Health workers, ASHA (Accredited Social Health Activists), Anganwadi workers may also be sensitized regarding menstruation and biological changes in body. They can further disseminate this information in the society and will helps in combating social myths and taboos.

- There should be more adolescent friendly health services clinics in region with trained professional to tackle this menstruation issue.

Conclusion:

Menstruation is a normal biological phenomenon in female body which requires proper knowledge and awareness to tackle with. To manage menstruation multi-sectoral and strategic approaches should be followed. Adolescent girls and women should understand that they have the power of procreation only because of this virtue. We need to link physical infrastructure and water and sanitation projects to health education and reproductive health programs and address the issue in more holistic ways.

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CHATPER 6

POLYCYSTIC OVARIAN SYNDROME (PCOS)

Dr. Nidhi Srivastava and Dr. Mamta Bansal

PCOS, also known as Stein-Leventhal syndrome, is a complex condition that affects about 68 percent of female or 1 in 10 females of reproductive age globally. It is characterised by elevated levels of the male hormone (androgen and testosterone), other hormonal imbalance, irregular, missed or extended menstrual cycles, cysts in the ovaries, weight gain and problems related to metabolism. These problems affect their overall health and standard fitness. Studies conducted by Deswal *etal.*, in 2020, in 19 countries reveals the prevalence of 2.2%, 3%, and 4% PCOD cases in Southern China, Iran, and the USA respectively whereas the range of 5-10% prevalence in Spain, Beijing, Brazil, Palestine, , Greek, Sri Lanka and the UK. The higher percentage of PCOS cases i.e., 15%–20% also observed in Denmark, Australia and Turkey. PCOS is becoming more and more common**.** In India also the cases of PCOS is increasing. Based on the Rotterdam criteria, PCOS is determined by the presence of at least one of the following three symptoms: oligo- or amenorrhea, scientific or biochemical hyperandrogenism, and the presence of any one of these. Because PCOS is becoming more common, early detection and treatment are necessary in order to reduce symptoms and complication that arises in women due to the syndrome.

Polycystic Ovary/Ovarian Syndrome (PCOS) and Polycystic Ovary/Ovarian Disease (PCOD)

It is a problem associated with reproductive health which is caused by an imbalance of reproductive hormones. The imbalances in the hormones can create severe reproductive health related problems in the ovaries. The major function of the ovaries is to make and released the egg every month for the healthy and regular menstrual cycle. But girls or women with PCOS or PCOD, the egg may not develop appropriately as it should and/or it is not able to release properly during the process of ovulation as it should be. Due to this, it leads to irregular, missed or prolonged menstrual cycles which further can lead to major reproductive health issues like Infertility among women and/or development of small follicles in one or both ovaries called cysts.

Difference between PCOS and PCOD

The terms PCOS (Polycystic ovarian syndrome) and PCOD are frequently misunderstood (Polycystic ovarian disease). These two medical conditions both involve the ovaries. While PCOS is a problem of the endocrine system, PCOD is a syndrome brought on by a hormonal imbalance. Although neither disease has a known exact origin, hormone imbalance and heredity are thought to play a significant part. In simple language, increased quantities of androgen are produced by PCOS-affected girls' ovaries, which interfere with the formation and release of eggs. Some of these eggs develop cysts, which may lead to reproductive issues. Additionally, the hormonal imbalance in girls with PCOD causes the ovaries to produce mature eggs, which subsequently develop into cysts similar to those seen in PCOS. Women therefore experience infertility troubles as well as a number of other health problems, including those that stated above.

Who can get PCOS?

PCOS can happen at any age after puberty. It may develop at the time of puberty or first menstrual cycle or sometimes it develops later for example, in response to substantial weight gain. About 5% and 10% of women between the age of 15 and 44, or during the childbearing age can get PCOS. Mostly PCOS affected women see the doctor when they face problem of getting pregnant and/or infertility in their 20s and 30s. The threat of PCOS is higher if a girl or a woman is obese or if any of family members a sister, mother or aunt is already suffered with PCOS. PCOS can affect any female around the world of all races, culture and ethnicities.

Signs and Symptoms

If a woman is obese, the signs and symptoms are typically more severe. If a woman is suffering with PCOS, she experiences at least two of these following symptoms (Fig. 1, Table 1).

- **Irregular menstrual periods:** The most common sign are infrequent, irregular, missed or prolonged menstrual periods. Female suffering from PCOS may have less than nine menstrual periods a year or more than 35 days between periods and/or abnormally heavy flow during periods.

- **Excess androgen level:** Physical manifestations of elevated levels of the male hormone androgen may include excessive body and facial hair (hirsutism), as well as occasionally severe acne and male-pattern baldness.

- **Polycystic ovaries:** The ovaries swell and develop follicles around the eggs. The ovaries may consequently stop working consistently.

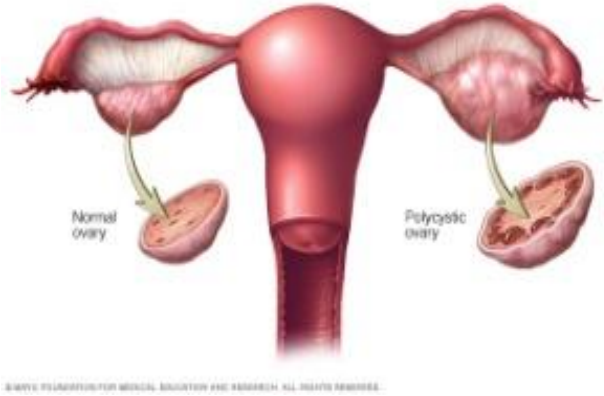

Fig. 1: Normal and cystic ovaries

Table 1: Polycystic ovary syndrome and clinical characteristics

S.no.	clinical characteristics	Types of clinical characteristics	Impact
1.	Menstrual abnormalities	Oligomenorrhea,	Infrequent menstruation at intervals >35 days
		Amenorrhea	Absence of menstruation
		Hypermenorrhea	Heavy and prolonged menstrual periods
2.	Clinical/ Biochemical hyperandrogenism	Hirsutism	Thinning and diffuse hair loss, Male pattern baldness
		Acne	Rashes or spots on the skin
		Serum androgen level	Elevated testosterone level in serum
3.	Polycystic ovaries	Numerous small cysts in a “string-of-pearls” appearance	Presence of ≥12 follicles of 2-9 mm diameter increased ovarian volume >10 ml in either ovary

	Infertility	Primary infertility Secondary infertility	Failure to achieve a live birth
	Endometrial cancer	Endometrial hyperplasia	Endometrial biopsy
	Metabolism disorder	Digestion problem and increase in cholesterol level and blood insulin level	Abdominal obesity >35 inches, TG ≥150 mg/dl, HDL-C: <50 mg/dl, BP ≥130/85 mmHg
	Obesity	Defined by body mass index	BMI ≥30 kg/m^2
	Type 2 diabetes (DM2)	Characterized by high blood sugar, insulin resistance, and relative lack of insulin	Fasting plasma glucose level, oral glucose tolerance test, fasting glucose ≥110 mg/dl
	CVD (Cardiovascular disease)	Group of disorders of heart and blood vessels	Arrhythmia, stroke, atherosclerosis
	Anxiety	BAI (Body adiposity index)	0-21: Low anxiety 22-35: Moderate anxiety >36: Severe anxiety
	Depression	BDI (Body density index)	0-9: Minimal depression 10-18: Mild depression 19-29: Moderate depression 30-63: Severe depression

Causes of PCOS

The exact cause of PCOS isn't known. Factors that might play a role include (Fig. 2):

- Excess insulin level in blood: The hormone insulin, which is made in the pancreas, enables body cells to utilise sugar or carbohydrates. Blood sugar levels may increase and the body may produce more insulin if cells become less responsive to the effects of insulin. Extra insulin may boost the production of androgen, making ovulation challenging.

- Low-grade inflammation in blood: This term refers to describe the substance produce by white blood cells to fight with infections and illness. According to research, polycystic ovaries have a kind of low-grade inflammation that is stimulated to generate more androgen hormone, which can cause problems to heart and blood vessels.
- Excess androgen level: The poly cystic ovaries produce abnormally high concentration of male hormone androgen, which result in hirsutism, baldness and acne.
- Heredity: According to research, few genes may responsible to cause PCOS.

Fig. 2: Cycle showing causes of PCOS

PCOS and health complications:

Researchers have found connections between PCOS and a number of health issues and complications, including:

- Diabetes or gestational diabetes: Before the age of 40, more than half of women with PCOS develop diabetes, prediabetes (glucose intolerance), or both.
- Hypertension or hypertension brought on by pregnancy: Compared to women of the same age without PCOS, women with PCOS have a higher risk of high blood pressure. Heart disease and stroke are primarily caused by high blood pressure.
- High Cholesterol: LDL (Low-density Lipoprotein) or bad cholesterol levels are frequently higher and HDL (High-density Lipoprotein) or good cholesterol levels are lower in women with PCOS. The risk of heart disease and stroke increases if woman have high cholesterol.
- Sleep apnea: This is the stage of repeated sleep disruption due to force breathing. Obesity and overweight are common in PCOS patients, which can result in sleep apnea. The risk of diabetes and heart disease is increased by sleep apnea.
- Stress and depression: Among women with PCOS, anxiety and depression are prevalent.

- Cancer of the endometrium: Endometrial cancer is more likely to develop in women with PCOS due to problems with ovulation, obesity, insulin resistance, and diabetes (lining of the uterus or womb).
- Eating disorders: women or girl with PCOS start eating more or less which further interferes with metabolism, hormonal imbalance and obesity.
- Infertility: the severe cause of PCOS lately is infertility.
- Miscarriage or premature birth: In severe cases miscarriage or premature birth also takes place.
- Non alcoholic -steatohepatitis — a severe liver inflammation caused by fat accumulation in the liver
- Metabolic syndrome — a cluster of conditions including high blood pressure, high blood sugar, and abnormal cholesterol or triglyceride levels that significantly increase risk of cardiovascular disease.

Researchers do not know if PCOS causes some of these problems, if these problems cause PCOS, or if there are other conditions that cause PCOS and other health problems.

Diagnosis of PCOS

PCOS cannot be detected with a single test. The doctor would interview you about your medical history, perform a physical exam, and run various tests to help diagnose PCOS and rule out alternative explanations of the symptoms:

- A medical exam: Blood pressure, body mass index (BMI), and waist size will all be measured by the physician. Additionally, they will check the skin for acne, skin discolouration, or hair growth on the face, chest, or back. Your doctor may examine you for any symptoms of hair loss or other health issues (such as an enlarged thyroid gland).
- Pelvic exam: A pelvic exam may be performed by your doctor to look for signs of excess male hormones (such as an enlarged clitoris) and to determine whether your ovaries are big or swollen.
- Pelvic sonography (sonogram): This examination checks the endometrium and your ovaries for cysts using sound waves (lining of the uterus or womb).
- Blood exams: Your androgen hormone levels, sometimes known as "man hormones," are measured through blood testing. In addition, your doctor will look for additional hormones linked to other prevalent health issues that may be misdiagnosed for PCOS, like thyroid illness. Additionally, your doctor might check your blood sugar and cholesterol levels.

A patient may be identified as having PCOS if she exhibits at least two of the following symptoms after all other illnesses have been ruled out:

- Period irregularities, including overly frequent, infrequent, or nonexistent menstrual cycles
- Androgen excess
- Hirsutism, which is an excessive amount of body and facial hair development
- Acne
- Hair thinning on the scalp and elevated androgen levels in the blood
- Numerous cysts on either one or both ovaries

Treatment of PCOS

PCOS cannot be cured, however its symptoms can be controlled. To develop a healthy treatment plan, doctors need to know about clear symptoms, family history and likelihood of developing long-term health issues like diabetes and heart disease. Doctors generally recommend a combination of treatments as below and in table 2:

• Home remedies for symptom relief
• Medicines

Home remedies

Losing weight: it is one home remedy for PCOS symptoms that you might try. PCOS-related symptoms can be reduced with a healthy diet and frequent exercise. Your blood glucose levels may drop, your body's ability to use insulin will likely improve, and your hormone levels may return to normal with weight loss (Table 3). Your menstrual cycle can become more regular and your chances of getting pregnant can increase with even a 10% body weight drop (for instance, a 150-pound woman shedding 15 pounds).

Healthy diet: Healthy diet surely manages PCOS among girls and women. This topic explained in detail below under heading "**Healthy diet plan in PCOS".**

Medicines

The different types of medicines that are able to treat PCOS and its symptoms include:

- Hormonal birth control medicines: it may include pill, patch, vaginal ring and/or hormone intrauterine device (IUD). For women who don't wish to get pregnant, hormonal birth control can:
    - Make their menstrual cycle more regular
    - Lower the risk of endometrial cancer
    - Help improve acne and reduce extra hair on the face and body

- Anti-androgen medicines: These medicines block the effect of androgens and can help reduce scalp hair loss, facial and body hair growth and acne. They are not recommended by the Food and Drug Administration (FDA) to treat PCOS symptoms. These medicines can also cause problems during pregnancy.
- Metformin: Metformin is often used to treat type 2 diabetes and may help women with PCOS symptoms. It is also not approved by the FDA to treat PCOS symptoms. Metformin improves insulin's ability to lower your blood sugar and can lower both insulin and androgen levels. After a few months of the use of the medicine, may help restart ovulation, but it usually has little effect on acne and extra hair on the face or body. Recent research shows that metformin may have other positive effects, including lowering body mass and improving cholesterol levels.

Table 2: Treatment approaches to treat PCOS and its side effects

S. No.	Category of PCOS	Side effects of taking drugs
	Medical treatment for irregular menstruation	
1.	Oral contraceptive pill	May increase insulin resistance and weight
2.	Combined oral contraceptives	Mood changes, bloating, acne, hair loss
3.	Progestins and progesterone	Increase risk of heart disease
	Medical therapy for insulin resistance and diabetes	
4.	Insulin-sensitizing drugs	Nausea, abdominal bloating, vomiting, and loss of appetite
	Medical treatment for fertility	
5.	Ovulation induction	Multiple births, ovarian cancer
6.	Gonadotrophins	Multiple pregnancies
7.	Assisted reproductive technology	Cost and failure
	Medical therapy for acne, hirsutism and hair loss	

8.	Antiandrogen	Birth defects, weight gain, depression
9.	Sebum-reducing cream	Dry skin and eczema
	Medical therapy for obesity	
10.	Lipase inhibitors	Risk for heart disease
11.	Central nervous system stimulants/anorexiants	Dizziness, diarrhea, anxiety, hair loss
	Medical therapy for depression	
12.	Antidepressants	Fatigue, weight gain, tremors, bladder problems
13.	Antianxiety drugs	Confusion, stomach upset, dizziness

Table 3: Lifestyle management to treat PCOS

S.No.	Lifestyle management	What patient can do?	Side effects
1.	Diet	Wholegrain cereals, low glycemic index foods, less Na and sugar intake	No Side effects
2.	Physical activity	Walk, running, aerobics	No Side effects
3.	Yoga	Morning and evening	No Side effects
4.	Meditation	Morning and evening	No Side effects
5.	Natural supplements , Herbs	Licorice root/Maca/Vitex/Chasteberry/inositol	No Side effects
6.	Stress free life	By adopting healthy life style	No Side effects

PCOS/PCOD Diet Plan for Weight Loss

<u>After waking up:</u>

Pick any option you like.

- 10 soaked almonds, warm water, or any herbal tea of your choosing without sugar.
- 10 soaked almonds plus 1 glass of green juice (made with bottle gourd, mint, cucumber, ginger, and lemon).

Breakfast:

Pick one of the following.

- One cup of oatmeal with skim milk and your preferred fruit on top
- One wheat bread toast with pieces of apples or bananas

Millet porridge, 1 cup

- Smoothie with Chia or Flax Seeds with Apple and Banana
- Two Idlis and sambar
- 1 cup of either vegetable oats or broken wheat upma

Snack in mid morning (11am)

1 cup green tea, guava or few berries

Lunch:

Any one of the following will opt for lunch.

Avoid all white rice recipes.

- 1 bowl of curd or yoghurt and 2 whole-wheat rotis (pulkas) with dal and vegetable salad
- 1 cup of curd or yoghurt and 1/2 cup of your preferred vegetables, except pota toes.
- 1/2 cup of brown rice with chicken breast, a vegetable salad, and curd;
- Chicken soup and vegetable salad
- 2 Chapatis, Veggie Salad, Chicken, Paneer, and Egg Curry

Evening Snack:

- Two fiber-rich biscuits and coffee or tea without sugar
- Fruit salad (without the banana, if you had one in your morning smoothie), along with a little amount of walnuts or any other type of nut
- A glass of buttermilk and a single little bowl of sprouts

Dinner:

Choose one of the following for dinner.

- Fruits of your choice • Grilled chicken or fish with sautéed vegetables (no potato) • One chapati with raita,

A sample PCOS Indian diet plan for vegetarians and non-vegetarians is provided here. The menu and the dishes of your choice can be combined.

What to Eat When Following a PCOS Diet Plan

It includes the following food groups in the normal diet for better PCOS or PCOD management. It not only helps in naturally reducing weight but also regulate blood insulin levels.

Fig 3: Green vegetables

1. **Leafy green vegetables:** All types of green leafy vegetables are low in calories and abundant in vitamins K, C, B group, and E as well as minerals including calcium, iron, potassium, and magnesium. Most significantly, they include B vitamins including B2, B3, B5, and B6 which aid in fat and sugar metabolism, hence enhancing hormonal balance and maybe assisting with PCOS-related infertility.

2. **Bright coloured vegetables:** These are full of potent antioxidants that aid in bringing PCOS women's oxidative stress levels back to normal. The healthiest vegetables to include in the diet on a regular basis are tomatoes, carrots, sweet potatoes, red and yellow bell peppers, eggplant, bitter gourd etc.

3. **Fruits and nuts:** Fruit is healthy, particularly if you have a sweet craving. But be sure to incorporate foods like lemons, apricots, grapefruit, strawberries, cantaloupes, guavas, pear, watermelons, blueberries, apples, nectarines, kiwi fruit, and others that have a low Glycemic index. Include some nuts and seeds in addition to the fruits to obtain a protein boost that will help you handle the abrupt rise in your blood sugar levels.

4. **Healthy Fats:** It is myth that fats are unhealthy and not good for weight loss. In fact, there are good fats and bad fats. Including the good fats promote hormonal balance and help with weight management. You can include all kinds of seeds, nuts, avocado, olive oil, tuna and salmon. They not only contain healthy fats, but also B vitamins and Vitamin D, both are recommended for treating PCOS.

5. Natural and organic meat: Protein promotes fullness, which makes you feel fuller for longer, which aids with weight reduction
   but eat right amount of protein without consuming additional calories.
   For this, use lean and organic meat in PCOS diet.

6. **Low-Glycemic Index Carbohydrates:** Carbs aren't always harmful to weight loss. For weight loss, one should prefer complex carbs over simple carbs. Low-glycemic-index foods including legumes, beans, lentils, and non-starchy vegetables take longer to digest and do not result in a quick jump in blood sugar levels. Additionally, these foods help to stay fuller for longer.
7. **Aloevera:** By improving ovarian steroid level and changing important steroidogenic action, aloevera gel formulation protects against the PCOS phenotype. Gel formulation exerts a protective effect against the PCOS patient by restoring the ovarian steroid status and altering key steroidogenic activity.

What not to eat when following a PCOS diet plan

In case of PCOS or PCOD, one should absolutely stay away from these food groups as they worsen PCOS/PCOD symptoms and insulin resistance.

1. **Foods High in Glycemia:** Stay away from packaged foods, white rice, white pasta, cakes, cereals, dried fruit, candies, pies, sodas, ice creams and bread.
2. **Dairy Products:** Milk consumption on a regular basis may raises levels of testosterone, which worsens PCOS symptoms. Therefore, try to avoid dairy products like milk, yoghurt, cheese, butter, and anything else that uses milk as a byproduct.
3. **Soy Goods:** Because soy has been associated to delayed ovulation in some women, it is not recommended to consume soy-based products if you have PCOS, particularly if you're trying to get pregnant.
4. **Bad Fats:** As mentioned earlier, it is recommended to eat good fats and avoid bad fats at any costs as they inhibit weight loss process. Try to avoid foods that are high in saturated, trans and hydrogenated fats.
5. **Caffeine:** Coffee is incredibly effective in helping people lose weight when drank in moderation (once or twice day) without milk or sugar. However, excessive drinking (three or more cups per day) may affect a woman's ability to conceive. It increases estradiol, a hormone that affects ovulation and menstrual cycle of women.
6. **Alcohol:** It makes perfect sense. Skip the alcohol. I'm done now.
7. **Packaged or processed foods:** Last but not least, stay away from processed and packaged foods because they are full of chemicals, tastes, additives, and preservatives that disrupt hormones and make the body more resistant to insulin.

Additional Helpful Advice for Managing PCOS Naturally

There is currently no specific or permanent treatment for PCOS/PCOD. However, you can naturally reduce the symptoms of PCOS by living a healthy lifestyle and adopting appropriate eating habits. The only thing you should do is alter your lifestyle to better meet the issue in order to increase your insulin sensitivity. Here are a few beneficial health tips for managing PCOS/PCOD and increasing insulin sensitivity.

- Avoid refined foods and replace them with whole grains.
- Avoid fruit juices and replace them with whole fruits.
- Avoid sugar completely and replace it with dates or honey.
- Eat foods with good carbs that are low in glycemic value, as they will not suddenly raise the blood sugar levels.
- Eat fibre rich foods like legumes and vegetables
- Get your protein right by including Lentils, Meat, Eggs etc.
- Eat just 2 to 3 times per day. Don't eat too frequently. It gives your body enough time to digest the meal that you've already eaten.
- Limit salt intake and instead make use of lemon juice, vinegar, spices etc.
- Say no to processed and packaged foods
- Munch on healthy snacks like nuts and seeds
- Include healthy fats like Tuna, Salmon, Avocado in your diet
- Get enough sleep for at least 6 to 8 hours
- Exercise daily for at least 30 minutes
- Track your menstrual cycle regularly using a mobile app and consult the doctor in case if the gap is more than 40 days.

PCOS and Pregnancy

Can someone with PCOS still become pregnant?

Yes. Even if a woman has PCOS, she can still become pregnant. One of the most prevalent but manageable reasons of infertility in women is PCOS. The hormonal imbalance in PCOS patients prevents the development and release of eggs from the ovaries (ovulation). Doctor can discuss methods with the patient that will increase the probability of getting pregnant and help you ovulate. To determine which days of the menstrual cycle are the most fertile, utilise the ovulation calculator.

How is PCOS impacted during pregnancy?

Both the mother and the foetus may experience issues during pregnancy as a result of PCOS. Women with PCOS are more likely to have Preeclampsia, gestational diabetes, miscarriage, and caesarean section (C-section). Additionally, a newborn baby is more likely to be overweight (macrosomia) and may admit in a neonatal intensive care unit (NICU).

How can I avoid PCOS-related issues when pregnant?

Pregnancy-related issues can be decreased by:

- Getting to a healthy weight prior to becoming pregnant: To determine the healthy weight before becoming pregnant and how much weight to gain while pregnant.
- Achieving healthy blood sugar levels prior to conception: It can be achieved by combining healthy eating practises, consistent exercise, weight loss, and medications like metformin.
- Consuming folic acid: as the dose prescribed by the doctor.

Alternative for PCOS pregnancy treatment

If a woman has PCOS, she has number of choices to increase the chances of becoming pregnant:

- **Weight reduction:** shredding weight via good, nutritious food and regular exercise can help the menstrual cycle become more regular hence improving the fertility.
- **Medicine:** There are many medicines which helps in producing eggs in the ovaries, so doctor may recommend medication that helps in ovulation.
- In vitro conception (IVF). If medication does not work, IVF may be an option. In IVF, egg is fertilised in a lab with the partner's sperm before being implanted and growing inside the uterus. IVF has higher pregnancy rates than medication alone
- **Surgery:** Surgery is a possibility as well, but typically only after all other remedies have failed. Women with PCOS have enlarged ovaries in their cortex, which is likely to contribute to the inhibition of spontaneous ovulation. A few holes are drilled into the surface of their ovary during ovarian drilling surgery using lasers or a small, heated needle. Ovulation is usually restored after surgery, but only for 6 to 8 months.

Latest research, international and national status

Latest research on PCOS

New approaches to treating PCOS are still being explored by researchers. Some recent research concentrates on:

- Heredity and PCOS
- Environmental exposure and the risk of PCOS
- Variations in PCOS symptoms by race and ethnicity
- Restarting ovulation with vitamins and medications
- Obesity and its link to PCOS
- PCOS women and linking health concerns of their infants

International Status

A metabolic, endocrine, and reproductive condition called PCOS affects 5 to 10 percent of women in developed countries. This frequency is the derived from a research, which was primarily carried out in the U.S. and Europe countries. The American College of Nurse-Midwives discovered that nearly 6.8 million women in the USA, or 5 to 30 percent of all women, have PCOS. The total yearly cost to treat women with PCOS between the ages of 14 and 44 years in the USA was more than US$430 million in 2006. PCOS and related problems are also having a great economic burden. At the global, regional, and national levels between 2007 and 2017, there were 1.55 million incident PCOS cases and 0.43 million associated DALYs (Disability-Adjusted Life Years) among women of reproductive age.

National Status

The World Health Organization (WHO) has underlined the importance of utilising the local, plant-based, indigenous medical systems that are locally available in developing nations. According to estimates, 5 percent to 10 percent of women in reproductive age have PCOS, and this number is rising quickly as a result of westernised diets and lifestyles. However, new research from developing nations like China and India suggests that PCOS prevalence rates are comparable. In nations where type 2 diabetes and obesity diagnoses are more prevalent, the estimation of PCOS prevalence rates appears high. Even though women with PCOS might range from being slim to being fat, but more often about 30-75 % of PCOS sufferers struggle with being overweight or obese . Both metropolitan areas of developing countries like India and China as well as most of the industrialised world have been touched by these parallel patterns between

metabolic health disturbances and PCOS. *Pterocarpus marsupium* was recently discovered to have potential reproductive benefits in female albino rats with PCOS induced by testosterone propionate and may be employed as an alternative therapy.

Fig. 4: *Pterocarpus marsupium* Plant

Conclusion

Since Polycystic Ovary Syndrome (PCOS) adversely impacts women's health and poses a major burden on healthcare resources, it is a serious clinical and public health concern. While selecting synthetic medicines to cure PCOS cause negative effects on the health, weight gain, lethargy, nausea, oedema, diarrhoea, sinusitis, hypoglycaemia, renal toxicity and high expense on fertility treatments must be taken into account. In order to treat PCOS and prevent infertility, the current research emphasizes the need for novel medications derived from various plant sources that are affordable, safe and highly effective. There is no recognised cause for PCOS. But it is common and treatable.

Diagnosis of disease on time and right treatment along with nutritious diet and weight loss help to reduce the chance of developing long-term problems such as type-2 diabetes and coronary heart disease. There are various synthetic medications available for the effective treatment and control of PCOS, but due to their adverse effects and high cost, patients are increasingly turning to plant-based therapies. In addition to seeking medical help, it is strongly encouraged to follow a healthy PCOS/PCOD diet and engage in regular exercise to lower blood sugar levels and regulate the menstrual periods.

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CHATPER 7

CANCERS AMONG WOMEN

Dr. Ravneet Kaur, Dr. Poonam Khanna

In India, the health systems with regard to women's health are largely focused on reproductive matters, particularly maternal health and contraception. Although these are crucial areas, however, there has been a neglect of women's health in other stages of life. Due to demographic transition and changing lifestyles in developing countries, the burden of non-communicable diseases (NCDs) is increasing. According to WHO Global Burden of Disease 2016 update, NCDs cause 65% of all female deaths. Majority of deaths are due to cardio vascular diseases, cancers and diabetes. As million individuals diagnosed each year, cancer is an important public health concern in India. It is projected that by the year 2050, there will be 17% increase in deaths due to cancer among women.

What is cancer?

- Cancer may be considered as a group of diseases with following features:
- Abnormal growth of cells
- Ability to invade adjacent tissues and even distant organs
- If the tumor progresses beyond the stage where it spreads to distant organs and is not amenable to removal, it eventually leads to death of the patient.

Cancer can involve any site of the body or any type of cells.
The term "primary tumour" is used to denote cancer in the organ of origin, while "secondary tumour" denotes cancer that has spread to regional lymph nodes and distant organs. Although signs and symptoms of cancer are different, depending on the organ of origin of cancer, however, there are certain warning signals that indicate that cancer should be suspected. These symptoms can be summarized in the acronym *CAUTION.*
C – Change in bowel or bladder habit
A – A sore that does not heal
U – Unusual bleeding or discharge

T–Thickening or lump
I – Indigestion or difficulty in swallowing
O – Obvious change in wart or mole
N – Nagging cough or hoarseness of voice

Initially, when cancer cells multiply and reach a critical size, the cancer appears as a lump or ulcer localized to organ of origin. If it is not diagnosed and treated at an early stage, symptoms and signs of invasion and distant metastasis become evident. The most common cancers among men in India are: Cancer of the oral cavity, lung, pharynx, stomach and oesophagus. In men, most of the cancers are related to tobacco consumption. Among women, breast cancer is now the leading cancer, and cancer of the cervix is the second-most common cancer in Indian women. Cancer of oral cavity is also common in India. In India, due to lack of systematic screening and delay in diagnosis, most of the cancers are diagnosed at an advanced stage, and the outcome is poor. The cancer mortality in India is high and less than 35% of the patients survive for five years or more after the diagnosis of cancer.

Risk factors for cancers

A number of factors are associated with risk of cancers. Although some risk factors are specific to some cancers, however, there are common risk factors also. The risk factors for cancers are as follows:

Tobacco: Tobacco is one of the most important single cause of cancers worldwide. It causes cancer of lung, nasal cavity, larynx, oral cavity, stomach, pancreas, cervix, kidney and urinary bladder.
Chewing of tobacco products is particularly prevalent in south-east Asia and is major cause of oral and pharyngeal cancer.

Alcohol: Alcohol has been shown cause cancer of stomach, oral cavity and oesophagus. Along with tobacco, it increases the risk of cancer to many-folds for various cancers.

Ionizing radiations: Ionizing radiations cause several cancers including acute lymphocytic leukaemia, acute and chronic myeloid leukaemia and cancer of breast, bone, lung and brain.

Obesity and physical inactivity: Obesity is associated with an increased risk of many cancers. Physical activity and avoidance of obesity has a protective effect on cancers of large intestine, gall bladder, breast, endometrium, kidney and oesophagus.

It has been suggested that physical activity should be an integral part of all cancer prevention programs.

Genetic factors: In some cancers, familial susceptibility is seen. If parents or siblings have suffered from cancer, the risk of an individual getting the cancer is high due to certain genes. Genetic susceptibility is noted in cases of cancer of breast, colon, prostate and lung.

Biological factors: Many organisms (viruses, bacteria and parasites) have been associated with cancers. Table 1 shows biological agents and human cancers caused by them.

BREAST CANCER

It is the most common type of cancer among Indian women. Earlier, cervical cancer was the most common type of cancer among women in India. However, due to rapid urbanization and change in lifestyles, breast cancer has emerged as the most common cancer.

Risk factors for breast cancer

- ***Genetic:*** Inherited changes (mutations) to certain genes, such as BRCA1 (Breast Cancer gene 1) and BRCA2 (Breast Cancer gene 2). Women who have inherited these genetic changes are at higher risk of breast and ovarian cancer.
- ***Early menstrual period:*** Women who start their periods before age 12 are exposed to hormones longer, raising the risk for breast cancer by a small amount.
- ***Late or no pregnancy:*** Having the first pregnancy after age 30 and never having a full-term pregnancy can raise breast cancer risk.
- ***Late menopause* (after age 55):** Like starting one's period early, being exposed to estrogen hormones for a longer time later in life also raises the risk of breast cancer.
- ***Not being physically active:*** Women who are not physically active have a higher risk of getting breast cancer.
- ***Family history of breast cancer:*** A woman's risk for breast cancer is higher if she has a mother, sister, or daughter (first-degree relative) or multiple family members on either her mother's or father's side of the family who has had breast cancer.
- *Age*: As the age increases, the chances of developing breast cancer

increases. The usually has a peak, and keeps on changing over decades. It is seen that the peak is at around 40 to 60 years, and then decreases after 60 years of age.

- ***Stress*:** Stress is a general risk factor that increases chances of developing any cancer.
- ***Obesity*:** Obese individuals have a higher incidence of developing breast cancer. This effect may be related to the production of estrogen by the fat cells.
- **Diet:** A healthy balanced low fat high vegetable diet and regular fruit intake will decrease chances of developing breast cancer.
- ***Radiation Exposure:*** An exposure to radiation in the second or third decades of life will increase the chances of developing breast cancer in the future.
- ***Hormone Replacement Therapy (HRT):*** This has been realized as an important risk factor. Hormone replacement therapy consists of giving a combination of estrogen and progesterone to relieve menopausal symptoms. If HRT is indicated, it must be taken with a clear idea of risk of breast cancer and awareness about screening.
- ***Oral Contraceptives:*** There is a slightly higher risk of developing breast cancer in women taking oral contraceptives.
- ***Alcohol:*** Intake of alcohol may increase the risk of developing breast cancer.

Signs and symptoms of breast cancer

In early stages breast cancer may appear as a painless lump (Fig. 1).
However, the other symptoms are:

- ***Change in appearance:*** Any change in the appearance or symmetry of breasts.
- ***Lump:*** Any lump, if noticed, needs immediate doctor's consultation. Lumps may be benign or malignant i.e. cancerous.
- ***Colour and texture:*** Discolouration of skin over the breast or changes in texture, making it thick, may be due to infection/ inflammatory condition and very rarely due a so called 'inflammatory breast cancer'.
- ***Discharge from the nipple:*** Discharge from the nipple may also occur in benign conditions, but a blood stained discharge from the nipple is not a good sign, and may be an indicator of an underlying cancer.

- ***Excoriations:*** Excoriations are seen as erosions occurring in nipple and are commonly seen during lactation. If they occur in older age group, they must be brought to a doctor's attention.
- ***Nipple:*** Nipple may be a source of symptoms for some women. Like for example, a nipple discharge, which is new and not milky, needs to be evaluated by a doctor. Bleeding or moist red area around the nipple, which doesn't heal easily, may be a sign of underlying cancer.
- ***Skin:*** During regular examination, the skin over the breast must be paid special attention to. In some advanced cancers, the skin over the breast becomes like the 'peel' of the fruit orange.

Fig 1: Different forms of breast cancer

Diagnosis of breast cancer

It consists of three main aspects, called as the 'triple investigation'. The latest recommendations are clinical breast examination, followed by ultrasound of breast, and mammography.

Clinical Examination: Clinical Examination by an experienced medical professional is a must. Palpation of the breast will give an idea of abnormality. If there is even a slightest doubt, it must be investigated further.

- *Mammography:* Mammography will give more details about the tumour in the breast, and in more than 85% to 90% of cases, features on mammography will point towards a breast cancer.
- *FNAC (Fine needle Aspiration Cytology):* In this procedure, a fine needle attached to a syringe is inserted into the tumour. This will result in some cells from the tumour coming into the needle. Immediately, the material is collected on a slide, and is viewed under the microscope.
- *Core Biopsy:* In this procedure, a special instrument, called as the 'core biopsy gun' is used. The process gives bits of tissues from the tumour.
- *Other* tests include the hormonal receptor status.

Treatment

There are various modalities for treatment. The treating doctor decides on the basis of stage of cancer, type of cancer etc. The available modalities are Surgery, and Chemotherapy.

Early detection of breast cancer

Screening enables us to detect a cancer much before it produces symptoms. This cancer will be in its initial stages, before it spreads to other areas of the body.

Thus, adequate treatment of this initial stage will lead to a longer life. Self-breast examination has proved to be important method by which women can themselves palpate their breasts and can notice if there is any abnormality or lump (Fig. 2). Creating awareness among women regarding breast cancer is of utmost importance. Most of the women do not seek help as breast cancer in early stages is painless.

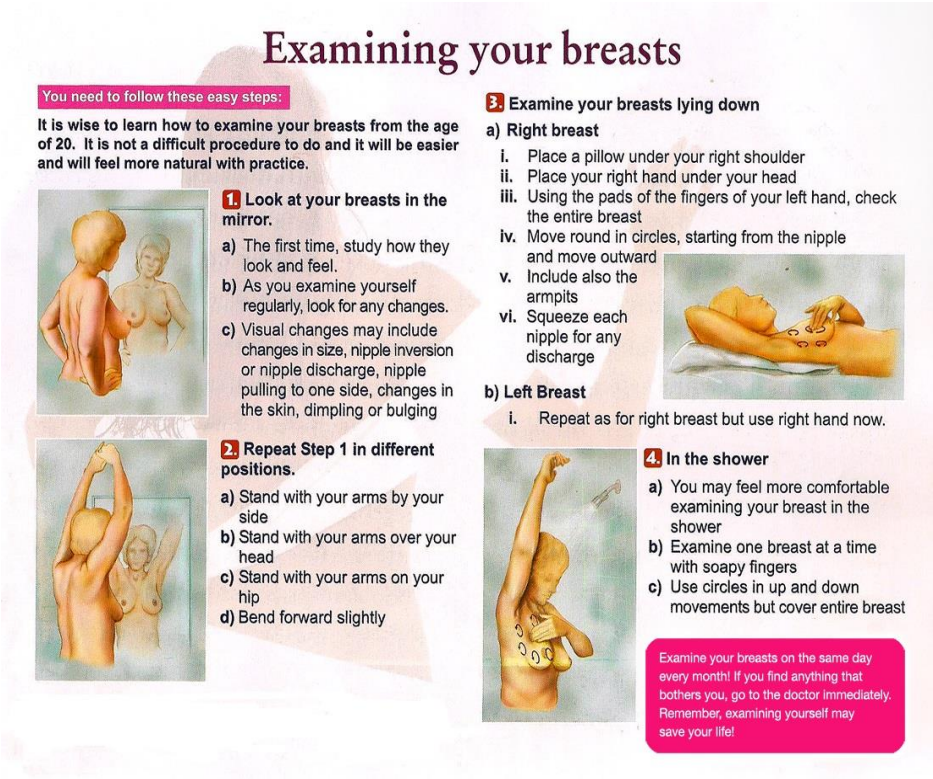

Fig 2: Breast self-examination

CERVICAL CANCER

Cervical cancer is cancer that starts in the cervix, which is the narrow lower part of the uterus, or womb. It is the passageway that connects the uterus to the vagina. It is the second most common cancer among women to India, next to breast cancer. Earlier cervical cancer used to be the most common cancer among Indian women.

Cancer of the cervix mostly occurs among middle-aged women, between 35 and 55 years of age. Human Papilloma Virus (HPV) is found in about 99% of cervical cancers. More than 70 percent of cervical cancer cases are caused by two types of the virus, HPV-16 and HPV-18.

Besides cancer, changes to cells of the cervix can also cause precancerous conditions. This means that these cells are not yet cancer, but there is a chance that these abnormal cells might become cancerous if not treated. Precancerous cervical cell changes and early cancers of the cervix generally do not cause symptoms.

However, cervical cancer has a long latent phase i.e. precancerous lesions persist for a long time before progressing to cancer, sometimes even up to ten years (Fig. 3). For this reason, regular screening through Pap and HPV tests can help catch precancerous cell changes early and prevent the development of cervical cancer. Most women with precancerous changes of the cervix can be successfully treated, and progression to cancer is prevented.

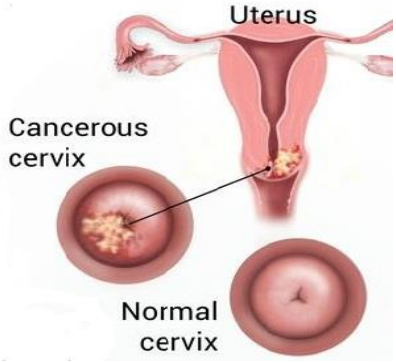

Fig 3: Diagram showing normal cervix and cancer of cervix

Risk factors for cervical cancer:

- Early marriage and child birth at a younger age
- Repeated child birth
- Sexual activity at young age, including adolescence
- Many sexual partners
- Other sexually transmitted infections (STIs**)** such as chlamydia, gonorrhoea, syphilis and HIV/AIDS.
- Smoking.

Symptoms of Cervical Cancer

Possible symptoms of more advanced disease may include abnormal or irregular vaginal bleeding, pain during sex, or vaginal discharge. A healthcare provider should be consulted if there is:

- Abnormal bleeding, such as-
- Bleeding between regular menstrual periods
- Bleeding after sexual intercourse
- Bleeding after a pelvic examination
- Bleeding after menopause

Pelvic pain not related to menstrual cycle

Heavy or unusual discharge that may be watery, thick, and possibly have a foul smell

Pain during urination

Diagnosis

If a Pap test result is abnormal, more tests are required to rule out or diagnose a precancerous condition or cervical cancer. These tests include:

- Another Pap test after a few months (usually 6 months) if there are mild changes.
- HPV test.
- Colposcopy (a procedure that uses a light magnifying instrument, called a colposcope, to examine to examine the vulva, vagina and cervix) and biopsy
- Endocervical curettage (a procedure that uses a special tool, called a curette, to remove cells from the endocervical canal, usually done at the same time as a colposcopy)
- Biopsy

Treatment

It depends on the stage of the diseases, age of the patient, general health condition etc. Surgery, radiotherapy and chemotherapy are various modalities of treatment.

Prevention of cervical cancer

Cervical cancer can be prevented by following measures:

- **Safe sex.** Using a condom, having fewer sexual partners and delaying intercourse may reduce risk of cervical cancer.
- **Routine Pap tests.** Pap tests can detect precancerous conditions of the cervix, so they can be monitored or treated in order to prevent cervical cancer. Most medical organizations suggest women begin routine Pap tests at age 21 and repeat them every few years.
- **Vaccination against HPV.** Vaccination is available for girls and women ages 9 to 26. The vaccine is most effective if given to girls before they become sexually active. However, this vaccine has yet not been included in the National Immunization Program of India due to conflicting results of some trials.
- **Early detection:** As per the latest recommendations, women aged 30-49 years should undergo visual inspection with acetic acid (VIA). It should be done once every five years and awareness should be integrated along with screening.

ORAL CANCER

Oral cancer accounts for over 30% of all cancers in the country. Each year approximately one million people in India are diagnosed with oral cancers and half of them die a painful death within twelve months of diagnosis due to late presentation. Important risk factors for oral cancer include:

- Tobacco
- Betel quid
- Alcohol
- Infection with Human papilloma virus and candida

Individuals who smoke as well as drink alcohol have higher risk of developing oral cancer.

In Indian population, these risk factors are widely prevalent (Fig. 4).

Oral cancers are also preceded by pre cancerous lesions.

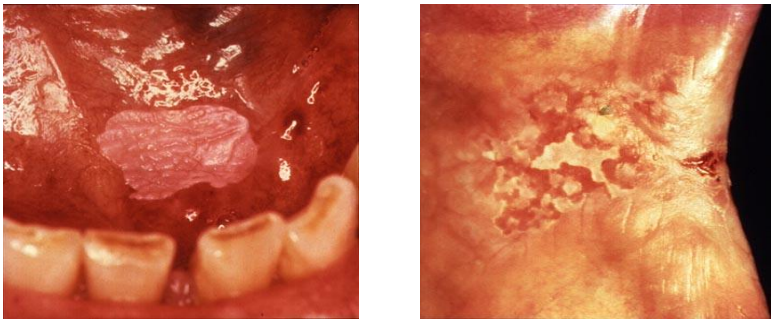

Fig. 4 Precancerous lesions of mouth

Early detection: Oral visual examination should be done by trained workers by systematic visual inspection of oral cavity, tongue, palate and floor of mouth under adequate light with disposable instruments. It can detect potentially malignant disease (PMD) or oral cancers. In persons with associated risk factors, oral visual examination should be done every three years.

Principles of prevention of cancer

Primary prevention – It implies prevention of occurrence of cancer. It includes the following measures

- ***Control of tobacco and alcohol*** – It is an important intervention for prevention of all tobacco and alcohol. Legislative measures to prohibit smoking and awareness programs can control the use of tobacco products. It has been estimated that nearly one million cases of cancer can be prevented by control of tobacco.
- ***Personal hygiene*** – Improvement in personal hygiene can lead to decline in incidence of certain cancers e.g. cancer cervix.
- ***Radiation*** **–** Efforts should be made to reduce the amount of radiation exposure to minimum so as to minimize the harmful effects. This will significantly reduce the incidence of skin cancers.
- ***Occupational exposure*** **–** Measures to protect workers from harmful effects of occupational carcinogenic agents should be enforced strictly in industries.

- ***Immunization*** – For protection of cancer of liver and cervix, immunization is an important preventive measure.
- ***Awareness and Education***–IEC (Information, Education and Communication) activities should generate awareness about warning signals of cancer. The aim of cancer education is to motivate people to seek early diagnosis and early treatment.

Secondary prevention – It involves early detection and treatment of cancers. Since prognosis of cancer largely depends on stage of diagnosis of cancer, early detection is an important measure.

- ***Cancer registration*** – It is of two types – Population-based cancer registries and Hospital based cancer registries.
  Early detection of cases and Treatment –Facilities for early detection and treatment should be available at all levels of health care.
- ***Cancer screening*** – Screening for cervical cancer by Pap smear and VIA (Visual Inspection with Acetic Acid) is recommended for early detection. Screening for breast cancer has also shown favourable effect on reduction in mortality.

Tertiary Prevention –In India, many patients are diagnosed at an advanced stage where distant metastasis has already occurred and disease prognosis is poor. In such a situation, provision of palliative care and hospice is pertinent in our country.

Conclusion

Cancer among women is an important public health problem in India. Most of the cancers are potentially preventable and have common risk factors. Certain occupational exposures pre dispose to higher risk of cancers. Prevention and control of cancers include reducing population exposure to risk factors through effective multi-sectoral policies, regulations and programmes. Facilities for early detection of cancers and treatment should be available at all levels of health facilities. Health promotion and awareness generation are important measures for prevention of all types of cancers.

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CHAPTER 8

HIV/AIDS: THE STORY UNFOLDED

Dr. Suninder Kaur Sodhi, Dr. Punita Sharma

76.1 million people have become infected with HIV and 35 million have died from AIDS related illnesses since the beginning of the epidemic till 2016. 1.8 million new infections with HIV were seen in 2016. An estimated 17.8 million women living with HIV (aged 15 and older), constitutes 51% of all adults as of 2016. More than 17 million children have lost either one or both parents to AIDS.

Millions died, many innocent victims, as the AIDS virus spread across the globe over decades. Still more remained undiagnosed for years, choosing to remain undercover, fear of stigma and discrimination pushed them away from tests and treatment. Stories of discrimination abound, AIDS is not just any physical malady, its social and cultural dimensions leave people helpless, lonely and isolated, even criticized, boycotted and disdain. The dooms day comes before the disease elopes them, as many committed suicides. Each person dying from AIDS or AIDS related death was not just a figure; each had a story to tell, volumes to convey. It is a reflection of humanity, it questions the very fabric of our culture which we take pride in, it questions the very dynamics of sustainability of life. As the numbers of people infected with the AIDS virus swelled, gays, lesbians, homosexuals, transgender, etc. were identified, they no longer were lost in the main crowd, but were segregated, fought for their identity and rights. Without any cure in sight, and discrimination thronging, people were paralyzed, in rage and powerless. Suffering and death increased with every passing day. Concerted efforts were needed by all of us, as citizens, activists, NGOs, health professionals, social workers, Academicians, Anthropologists, Sociologists and Psychologists, political people, government and private sector, parents and families. When parents die and children are orphaned, their lives automatically become difficult. When the productive work force dies, and children are taking care of the aged and vice versa, life comes to a standstill. The way this pandemic has caused ripples in our lives and affected us over the years goes beyond

being just a health malady, its tentacles go far and beyond, touching emotional, economic, personal, social, and cultural aspects of our lives. How do we deal with death, and the situation that comes after one is gone? How to we come out of remorse and walk the struggling path, how to we prevent innocent victims succumbing to this disease? Which door do we knock for help? How do we save ourselves from the stigma, humiliation and outcast?

Defining AIDS

Acquired Immuno Deficiency Syndrome, as it came to be known as, this disease breaks down the body's immune system. The body is unable to fight off infections called 'Opportunistic Infections' (OIs). Slowly the AIDS virus, called HIV eats away the immune system, the army of the body which fights off infections, thereby making the person weak and diseases such as pneumonia, OIs, TB (Tuberculosis) etc. grip him, loss of weight, re-erupting fever, weakness, diarrhoea are some of the symptoms and either the person dies with AIDS related diseases, or the CD4+ (cluster of differentiation 4) counts falls very low and the patient dies because of AIDS. Spreading by unsafe sexual contact with an HIV positive person, or infected blood transfusion, even sharing infected needles and syringes can cause AIDS. HIV can also pass on from an infected parent to the fetus, before or during birth and afterwards during lactation. The HIV can remain in the window period for about 6 months, and during that duration tests do not show the presence of HIV in the blood.

The story of AIDS

During the summer of 1981, a lung infection, sparked the attention of doctors and scientists in CDC (Centre for disease control) Atlanta, USA. It was pointing out to immunosuppression, usually caused in cancer patients and is very rare. On 5th June 5 gay men of Los Angeles were found to have a rare lung infection. This case published in the Morbidity and Mortality report (MMWR) by CDC of which 2 had died by the time the report was published. Shortly thereafter cases of KS (Kaposi Sarcoma),a type of cancer and other OIs were observed. KS, a rare cancer was observed in 41 Gay men of New York and California, reported by 'The New York Times' on July 3rd. Later that year, 159 cases of KS and OIs had been reported in the United States mostly from California or New York, with the earliest cases retrospectively identified in 1978. By the end of 1981, 270 cases of this new disease was seen in Gay men of which 121 had died. Within 6

months, it was recognized that a new, highly concentrated life threatening illness was occurring in the United States, which would reach epidemic proportions. The co-occurrence of KS and OIs suggested that the outburst was of immunosuppression and KS or OIs were a consequence of the immunosuppression. It was the tip of the iceberg being seen as a spectrum of illnesses. During April 1982, CDC announced that tens of thousands of people might be infected with this disease, sounding that it may be an epidemic. By September the same year $5 million was apportioned by the government to CDC for surveillance and $10 million to the National Institutes of Health (NIH) for AIDS research. In 1982, it was found that that the new syndrome was caused by a sexually transmissible infectious agent. In December, the same year, infants were identified having the AIDS virus and cases of transfusion also showed similar results. This proved that the infecting agent could be passed on from mother to the infant, through blood transfusions, through the sexual route and from homosexual to heterosexual population. Hence, initially thought as a '***Gay Plague***', receiving little attention from mainstream population and media, this disease spread to all populations very rapidly and caused panic. (Source CDC 1982)

On 24th September 1982, the U.S. Centre for Disease Control, Atlanta, uses the term AIDS, for the first time to and releases its definition as *a disease at least moderately predictive of defect in cell-mediated immunity, occurring in a person with no known case for diminished resistance to that disease*.

The AIDS Journey in India

Suniti Solomon a microbiologist and physician, teacher, counsellor, and activist, was teaching microbiology in Madras Medical College where one of her students Nirmala took up research on people with HIV in 1986 as her Ph.D topic on the insistence of Dr. Suniti. They tested 100 people, including those who had undergone blood transfusion, truck drivers, sexual workers etc. and one day 6 samples collected from FSWs lodged at the government home on Kutchery Road in Mylapore, Chennai tested positive for HIV. The 1st person detected was a 13 year old girl who was abducted and forced into prostitution. The samples were sent to John Hopkins University for confirmation. The stories of innocent victims and stigmatization begin then. This 13 year old girl did not consent for 3 days

after being kidnaped but was starved for 3 days when she finally gave in. After about 6 months she escaped and Dr. Suniti took care of her. Dr. Suniti was trained in clinical pathology in Britain, the US and Australia, but returned to India to work in a government hospital as early as in 1973. She remembers how when someone is found having AIDS, the society looks down on the person as being 'immoral'. She established the YR Gaitonde Center for AIDS Research and Education in Chennai. It was the 1st Voluntary Counselling and testing centre (VCTC) in the country and was known to accept patients who were cast aside by other hospitals. It was later called YRG Care. The Hindu, Quartz India online newspaper, Times of India)

According to the United Nations, in 2013, 2.1 million Indians—the third largest in the world—were living with HIV. More than 130,000 people died from AIDS-related illness in 2013 alone (UNAIDS).

The AIDS Timeline

(Source : https://www.hiv.gov/hiv-basics/overview/history/hiv-and-aids-timeline)

1981-Ist cases identified among Gay men in NY and CA, USA

1982-AIDS term coined, found in children and heterosexual population as well

1983- WHO holds its 1st meeting announces International Surveillance. A New York doctor is threatened with eviction from his building for treating AIDS patients and in response Gay Men's Health Crisis (GMHC) and Lambda Legal file the first AIDS discrimination lawsuit.

1984-In June, Dr. Gallo From National Institute of Health USA and Professor Luc Montagnier, from the Pasteur Institute in France, hold a joint press conference to announce that Dr. Montagnier's Lymphadenopathy Associated Virus (LAV) and Dr. Gallo's HTLV-III virus are almost certainly identical and are the likely cause of AIDS. SFO officials close bathhouses to avoid high risk behaviours.

1985-The FDA licenses ELIZA to test blood for the AIDS virus. Ryan White a haemophilic teenager from Indiana publicly speaks out on Stigma and discrimination, when refused from Middle school as he got AIDS though blood transfusion

1986- The term HIV was coined as the Virus causing AIDS. AIDS activist Cleve Jones creates the first panel of the AIDS Memorial Quilt.

1987- On February 4th, Emmy-award winning pianist, Liberace, dies of AIDS-related illness. On 19th March FDA approves the first antiretroviral drug, zidovudine (AZT). In October, the AIDS Memorial Quilt is displayed for the first time on the National Mall in Washington, DC and draws ½ a million visitors. On May 31, President Reagan makes his first public speech on AIDS.

1988- WHO declares Dec 1 as world's AIDS Day, EGPAF started,

1989- Photographer Robert Mapplethorpe dies of AIDS-related illness on March 9. There are 100,000 AIDS cases reported in the USA.

1990- CDC reports AIDS virus can spread through dental procedure. Pop artist Keith Haring dies of AIDS-related illness on February 16. On April 8th Ryan White dies at the age of 18.

1991- Caucus launches the Red Ribbon Project for AIDS awareness and compassion. On Nov 7th, Magic Johnson, basketball player, announces that he is HIV positive. On Nov 24th, Freddie Mercury, lead singer/ songwriter of the rock band Queen, dies of bronchial pneumonia resulting from AIDS

1992- AIDS is the no. 1 cause of deaths of men between ages 25 to 44 in USA. Florida teenager Ricky Ray dies of AIDS-related illness on December 13th (The 15-year-old haemophiliac and his two younger brothers sparked a national conversation on AIDS after their court battle to attend school led to boycotts by local residents and the torching of their home)

1993-President Clinton establishes the White House Office of National AIDS Policy (ONAP) On Jan 6th, Rudolf Nureyev, a ballet dancer dies of AIDS-related illness and on Feb 3rd, tennis star Arthur Ashe dies. Based on a true story, the film "Philadelphia" starring Tom Hanks as a lawyer with AIDS, airs. Angels in America, Tony Kushner's play about AIDS, wins awards. CDC declares that people with CD4 counts below 200 have AIDS and adds more conditions to the list of clinical indicators of AIDS.

1994- Randy Shilts, a U.S. journalist who covered the AIDS epidemic and who authored And the Band Played On: Politics, People and the AIDS Epidemic, dies on Feb 17th, of AIDS-related illness at age 42. On 11th Nov Pedro Zamora, a gay man living with HIV who appears on the cast of MTV's popular show, "The Real World." dies at age 22. On 23rd Dec FDA approves an oral HIV test.

1995- Greg Louganis, Olympic gold medal diver, discloses that he is HIV-positive. FDA approves the HAART treatment. Rapper Eazy-E (Eric Lynn Wright), dies on March 26 from an AIDS-related illness one month after being diagnosed. 500,000 cases of AIDS reported so far in the USA.

1996- UNAIDS begins operations. FDA declares Nevrapine as another drug for AIDS,. Urine tests for AIDS detection. The AIDS Memorial Quilt is displayed in its entirety for the last time. AIDS no more the leading cause of death in USA in age group 2 25-44 and AIDS cases begin to drop. HIV/AIDS researcher Dr. David Ho advocates for a new strategy for treating HIV – "hit early, hit hard, named *Man of the Year* by TIME Magazine. HAART treatment is effective.

1997- FDA declares Combivir, a combination of two antiretroviral drugs in one tablet. UNAIDS declare 16,000 people are newly infected with the virus each day.

1998- CDC states that 49% of U.S. AIDS-related deaths are African Americans. President Clinton declares AIDS to be a "severe and ongoing health crisis" in African American and Hispanic communities in the United States and announces a special package of initiatives.

1999- WHO announces that HIV/AIDS has become the fourth biggest killer worldwide and the number one killer in Africa. WHO estimates that 33 million people are living with HIV worldwide, 14 million have died of AIDS. In March, VaxGen, a San Francisco-based biotechnology company, begins conducting the first human vaccine trials in Thailand.

2000- G8 Summit (Canada, USA, UK, Italy, Japan, Germany, France and Russia) release a statement acknowledging the need for additional HIV/AIDS resources. Reversing the spread of HIV/AIDS, malaria, and TB is one of the Millennium Development Goals (MDG) declared by the UN

2001- 7th Feb is the first annual National Black HIV/AIDS Awareness Day in the U.S.A and May 18 is the first annual observance of HIV Vaccine Awareness Day. To reduce drug prices in developing countries, generic medications are announced by WTO in the Doha Declaration affirming the right to buy or manufacture them.

2002- UNAIDS states that average life expectancy in sub-Saharan Africa falls from 62 years to 47 years as a result of AIDS. U.S. National Intelligence Council releases Next Wave of the Epidemic, a report focusing on HIV in India, China, Russia, Nigeria, and Ethiopia.

2003- President George W. Bush announces the creation of the United States President's Emergency Plan for AIDS Relief (PEPFAR). On march 31st the Bill and Melinda Gates Foundation give a $60 million grant to the International Partnership for Microbicides for research and development for prevention of HIV transmission.

2004- US Congress authorizes $350 for PEPFAR. The end of 2004 has an estimate of 700,000 people infected with HIV.

2005- G8 summit focuses on Africa in HIV/AIDS Scenario. Nelson Mandela's son Makgatho, dies of AIDS-related causes.

2006- University of Illinois at Chicago study indicates that medical circumcision of men reduces their risk of acquiring HIV during heterosexual intercourse by 53%.

2007- CDC reports over 565,000 people have died of AIDS in the U.S. since 1981.

2008- President Bush signs legislation reauthorizing PEPFAR for an additional five years for up to $48 billion lifts blanket ban on HIV positive travellers to the USA. September 18 is the first observance of National HIV/AIDS and Aging Awareness Day. National Gay Men's HIV/AIDS Awareness Day is first recognized on 27th Sept.

2009- President Barack Obama calls for the development of the first National HIV/AIDS Strategy for the United States, announces that his administration will officially lift the HIV travel and immigration ban in January 2010 by removing the final regulatory barriers to entry.

2010- AIDS Action merges with the National AIDS Fund to form AIDS United.

2011- Actress Elizabeth Taylor dies on March 23. She advocated on behalf of people living with HIV and AIDS, and was the founding national chairman of AMFAR (American Foundation for AIDS Research)

2012- The FDA approves the use of Truvada for pre-exposure prophylaxis (PrEP). people not infected but at risk for infection, can now take this medication to reduce their risk.

2013- 10th anniversary of U.S. President's Emergency Plan for AIDS Relief (PEPFAR). With its support more than 1million infants are born HIV free since 2003. NIH-funded scientists announce "the Mississippi Baby," an HIV infected child shows no detectable levels of virus or signs of disease, even without antiretroviral therapy.

2014- NIH announce that the “Mississippi baby” now has detectable levels of HIV after more than two years of showing no evidence of the virus.

2016- CDC reports that an estimated 50% of young Americans who are living with HIV do not know they are infected.

The National AIDS Control Program

AIDS control program was implemented in India in a phased manner. It began with ‘awareness generation’ and setting up of Sentinel surveillance sites in 1992. It was the initiation of prevention intervention. The National AIDS Control Program began in 1992. The Government of India took timely proactive measures to frame effective policies and ensure proper implementation of the program with robust monitoring.

Over the past decades, as a result of these focused efforts, the disease once considered deadly is now like any other chronic manageable disease. This has been possible due to the political commitment of the highest level, involvement of the community, civil society, support from developing partners and stakeholders. Initial Interventions began with the 1st phase of NACP (National AIDS Control Programme)(1992-1998).

1097 is the toll free number for AIDS helpline services

Every 5 years, more NGOs were roped in and program strengthened to make inroads in the most remote areas and not to leave any person indulging in risk behaviours from enrolling in the program and receiving services. During the 2nd Phase State level AIDS Control societies were set up (SACS). It is considered a revolutionary phase, as the program was decentralized. The VTCT (Voluntary Counselling and testing) program began in NACP II with the aim that no child should be born HIV positive. A landmark services to provide free ART (Anti Retroviral Therapy) services for people living with HIV/AIDS (PLHIV) began, spread and gained strength during these 5 years. To address the vulnerability of the youth was also the prime focus. Appropriate age related Intervention began with adolescent educational programs in schools and red ribbon clubs in colleges. While NACP II (1999-2006) saw limited coverage of services, there was decentralization of resources to the States. Unprecedented up scaling of services was done in Phase III.

NACP phase III (2007-2012) focused on reaching the district levels and massive scale up of quality assurance mechanism. The launch of strategic Information management services (SIMS) was in Phase III of the program. To reach the unreached, several flagships of services was undertaken such as the red ribbon express project, the largest mass mobilization in the world, to fight AIDS and to launch North East Multimedia campaign.

The Red Ribbon Express train has travelled more than 27,000 kilometres (16,777miles) across India carrying message of AIDS awareness to more than 50 000 towns and villages.

Consolidation of gains made in the previous phases and focusing on emerging vulnerabilities is being done in the current NACP IV phase. The thrust has been on building partnerships with other health programs and non-health departments. NACO (National AIDS Control Organization) has signed 50 MoUs (Memorandum of Understanding) with key departments and ministries. A balance with growing treatment needs, Integration and mainstreaming is focused upon and a drop of 60% in new infection rate is the accomplishment and success story of efforts on all fronts and collaboration between the NGOs and various departments, government and private sector. The target of achieving 75% reduction in new infection rate till 2020 and ending the AIDS epidemic by 2030 is the aim of the program. NACP is considered one of the most successful public health programs of the country. The 6th millennium development goal of halting and reversing the HIV epidemic in the country is being achieved.

Global HIV Statistics

- As of June 2017 20.9 million people on ART treatment worldwide.
- Around 36.7 million PLHIV in 2016.
- 1.8 million newly infected cases in 2016.
- 1 million people died from AIDS-related illnesses in 2016..

Resources and Helpline

NACO (National AIDS Control Organization) launched a toll free number **1097** which answers questions and misconceptions regarding HIV/AIDs and imparts information about the disease, which began in 1995 but by 2014, has been implemented successfully throughout the country

and declared as a helpline number. Other foundation also has helpline phone numbers 23543333 and 23542222, which are manned 9am to 5pm Monday through Friday, launched by actress Shabana Azmi.

Red Ribbon Express (RRE) is a train, started by NACO, which goes through out the length and breadth of the country, crossing many cities and states and imparting awareness along the way. At every station it stops for 2-3 days. It goes every alternate year. Launched on Dec 1st 2007, i.e. on the world AIDS day, it is a seven-coach train with three exhibition coaches, one each for counselling and medical services, an auditorium cum conference coach and a pantry car.

Test and treat Policy

Working towards eradication of AIDS the Government of India has launched the test and treat policy, to bring all PLHIV under the ambit of ART services, regardless of CD4+ counts.

NACO and the success story of its NACP

NACP is a 100% centrally supported scheme by NACO, India's voice against AIDS. The Ministry of Health and Family Welfare launched NACO in 1992. A steady decline of the HIV adult prevalence over the years reveals the success of NACO and NACP to combat the AIDS epidemic (Fig. 1).

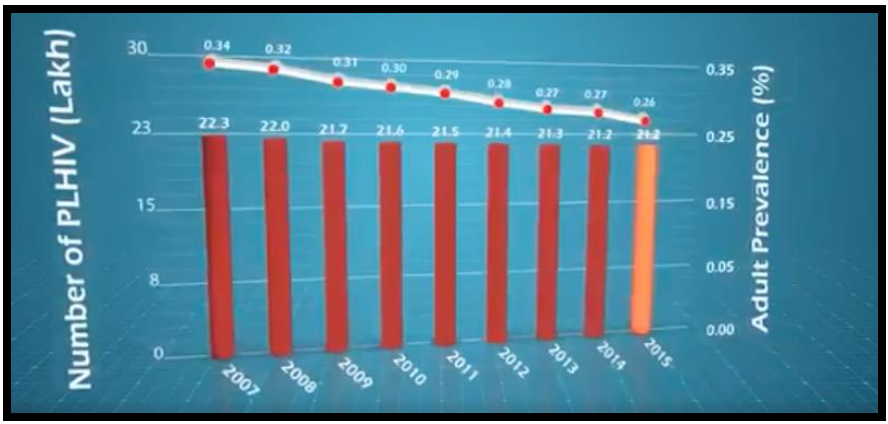

Fig 1: AIDS prevalence over the Years (Source NACO)

The magnanimous success of the program is endorsed by the fact that the HIV infections from 2000-2015 have dropped by 67% against a global average of 35%. There has been a sharp decline in AIDS related deaths from 2007-2015 by 44% against the global average of 41% decline (Fig. 2).

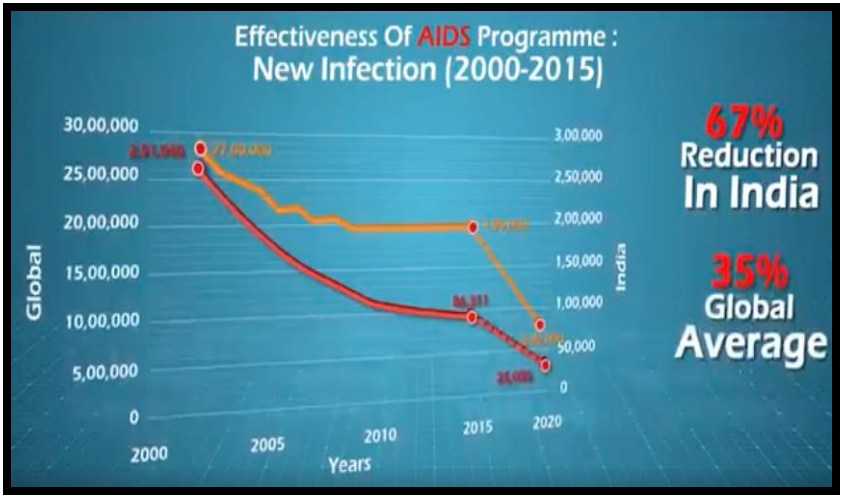

Fig 2: Effectiveness of AIDS Program

Woman and AIDS

Gender inequality continues to fuel the 35-year-old AIDS epidemic in majority of nations worldwide. Women were initially seen either as partners of Gay men, or drug abusers, or as FSWs (Female Sex Workers). As it was discovered that MTCT transmission (Mother to Child Transmission) is also possible, women became even more guilty. Medical care, treatment and benefits were long neglected.

Today, more than 80% of Infections are seen among women in sub Saharan Africa and a leading cause of death among 15-49 age group. Policy makers have not focused so much in women oriented interventions in the way the statistics of cases are revealing. 80% of money should have m been earmarked for women tests and treatment and awareness, which is not the case. Power relations play a prominent role in policy drafting and its implementation. Girl power and women empowerment has started to make inroads but there is still a long way to go before we see equality in terms of opportunity and social equity in terms of treatment and protection. Sexual and domestic violence needs to be curbed to lift women from the cycles of violence and disease, no control on self or decision making in important matters etc. Women's biological susceptibility is greater than men as she is at the receiving end when it comes to vaginal intercourse and has more than twice the possibility of infection when exposed to HIV (Gupta 1996; Mehta 2000; Pilot 2001; Ida 2002; Higgins *et. al.*, 2010; CSIS website).

Stigma

The Indian parliament has recently in April 2017 passed the HIV/AIDS bill. It enacted the HIV/AIDS act 2017, which addresses stigma, and discrimination faced by PLHIV (People living with HIV) and ensures stigma free services to all PLHIV. This ensures equal rights while seeking treatment, education and job by people living with HIV. In spite of the law being passed, stigma still endures and many people do not get themselves tested for the fear of being stigmatized and isolated. Children have not been allowed to attend school, people have lost jobs, even told to vacate their rented property and leave the vicinity. People with HIV try to keep their condition secret to avoid such incidences.

Neelimi Mohanty who is Project director of HIV children's home in Hyderabad run by the NGO ' Desire Society' in India, opines that little have been changed after 2 orphan siblings of parents who died of AIDS in Kerala State were expelled from school in 2000. Bency and Benson's case caught the headlines, but change is minimal thereafter. People still have to live with hostility; School officials and parents need to be convinced, as they dislike HIV positive children being taught in the same classroom. Anjali Gopalan, founder of the Naz Foundation though welcomed the law, says that people find filmiest of reasons for refusing treatment and it will take decades for things to improve. People do not want to rent the house where previously an HIV positive person was staying; attitude will change but will take a long time and effort from all walks of life. Stigma also needs to be understood it's in cultural context. Transgender and other neglected groups are also stigmatized. Deep rooted traditional beliefs, moralistic values and tag attached has made it even more difficult for people to come out of their hideouts and get tested.

Anthropology and AIDS

Sharing experiences and learning from failures was the road that was treaded over decades in our fight against AIDS. India falters while world fights because the population remains under covers for fear of being ostracized and stigmatized leading to further spread of the HIV Infection. This silent killer engulfed thousands of us before we actually fought back, mustered the courage to come out in the open, get tested, declare our status, ask for help and begin treatment. Life is complicated, both by cultural beliefs and by inequalities among people. Not only were women considered immoral and stigmatized, gay, transgender, and other groups

were equally blamed and hence remained silent and in confinement to avoid the pain, violence and isolation as a result of being identified as AIDS victim. The first wave of the epidemic caught the attention of a few, considered as a disease of homosexuals. Only when heterosexual population was found to have the AIDS virus did any serious research and concern cascade from the doctors and scientists, to politicians, health professionals and policy makers, and finally the NGOs and academicians also played a key role in imparting positive education and breaking the ice. Fear, stigma abounds and people commit suicide much before the disease finally grips them. Cases of suicide started increasing, just as the cases of rare ailments such as KS and PCP and even tuberculosis related to AIDS started rising.

This invoked discussions, became a barrier to anthropologists, sociologists, psychologists, health workers, social workers NGOs, wanted to know the source of the stigma, extent of fear, ways to combat discrimination so as to address the ailment, provide quality treatment and care, and also address logistic and economic barriers to access services. There are structural factors shaping the course of the epidemic in different settings that needs to be understood for the success of any intervention program. Research on the synergistic effects of HIV/AIDS, social exclusion, and related social problems in various pockets needs to be undertaken. Prevention policies for HIV/AIDS requires dealing comprehensively with diverse dimensions that determine differential vulnerabilities to the epidemic, thus requiring substantial social transformations. Stigma possesses several challenges, is not easily defined, and has many angles to it. Stigma highlights powerful social inequalities and violates human rights. Stigma and discrimination needs to be understood in its social setting and addressed properly in our fight against AIDS as it is at the heart of the AIDS pandemic.

AIDS sans aid makes the syndrome worse. Without informed choice, timely tests and treatment, AIDS can spread in communities and engulf people of all age groups silently and rather quickly. Though not contagious, it spreads through fluid contact, sharing needles, infected blood transmission, through mother's milk to the baby, or during or prior to birth, unsafe sex with an infected person, can lead to HIV passing on from the infected person to the uninfected person. In this era of Technology, media and communication play a key role in spreading

awareness and also spreading stigma. One should be very cautious in imparting information and understand how the media news can impact the masses. Communication involves dialogue. Participants of the dialogue, who play the role as news sources, not only exchange knowledge but also critically analyze their beliefs, attitude and behaviour patterns. It is important to understand the cultural contexts of the situations, the power relations that come into play and the beliefs associated with it. TV and Radio account for main sources of AIDS awareness besides the little discussion by teachers, as AIDS awareness is now part of high school curriculum.

Multidisciplinary research is the need for understanding complex behaviour related to pandemics and epidemics. From 1981 till 2017, HIV/AIDS has claimed more than 35million lives. It is never too late to begin prevention efforts and parliamentarians can earmark funds which can go a long way in reaching out to the needy, to the hideouts, make inroads in the remote pockets of the country and help those who need it before the disease keeps spreading in geometric progressions. 30% of the people do not know of their HIV status (avert.com) that means that the source is not blocked and the high-risk behaviour people still can spread the disease in geometric progressions. Information and education is the key to control the further spread of the disease. Curtail myths and misconceptions and spread knowledge and awareness. Enforce laws against discrimination and stigma, favouring equality and human rights so that barriers can be broken and AIDS control can be enforced with full zeal.

Conclusion

With more concentrated efforts and partnerships, the AIDS virus detected in 1981 in USA and 1986 in India, the journey which began in 1992 has to go a long way, and the governments strive for an AIDS free India by 2030. The only milestone achieved in the AIDS trend is that more and more people are being linked to life saving treatment. Executive Director of UNAIDS states that the target of 15million people or ART in 2015 has been met. People irrespective of their CD4 count have access to ART and other forms of treatment that can increase their lifespan. Commitment from all departments and people from all walks of life is needed to combat this pandemonium rocking the world. Stigma and discrimination are part of complex systems of beliefs about disease and illness and are often

grounded in social inequalities. The issue needs to be handled with sensitivity. Proper counselling is required; support systems need to be strengthened with focus on AIDS prevention and care. Cost of medication is another issue that requires immediate attention. Government and politicians should work todays the welfare of its people by addressing this issue with urgency. Lack of access to drugs compromises treatment. Public health experts have also debated poor patient adherence to treatment as insurmountable barriers to effective AIDS control, which also needs to be addressed case-to-case basis with sensitivity. The underlying cause could be personal, many and sensitive issues. People need to be understood in their emotional, social and cultural context.

Those denying the ailment or hushing off treatment often do not want their family and loved ones being ashamed of being associated with them. Though AIDS therapy can rejuvenate life, bring back energy and can spark a "virtuous social cycle." Anthropologists and other social scientists noted that the introduction of therapy and by accessing comprehensive AIDS care might alter profoundly the social interpretations of the disease. The social construction of stigma reveals that stigma aggravated by an undefined etiology and the lack of an effective treatment generates new cultural models of the etiology and expected course of this newfound disease. These models change with time because diseases have a social course i.e. pathology is embedded in social experience. The social experience of AIDS is affected profoundly by the advent of effective therapy.

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CHAPTER 9

MENTAL HEALTH: DEPRESSION AND SUICIDE AMONG WOMEN

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Depression is a brain disorder that occurs when the brain circuits i.e., responsible for regulating mood, sleep, thinking, appetite and behaviour go awry and other hormones and chemicals used by the nerve cells and neurons, fluctuate out of balance. Depression is feelings of sadness and despair, of helplessness and hopelessness.

- It is predicted that by the year 2025 depression will be the second-leading cause of disability globally.
- Women are more prone to depression than men.

Depression is more than just feeling low, low in self-esteem, low in self worth or low in confidence. It is a sombre situation, an illness where the person looses interest and vigour in day-to-day activities. This serious illness is due to changes in brain chemistry. There are many factors contributing to the onset of depression, including genetics, stress, hurt and grief, changes in hormone levels, certain medical conditions, difficult life circumstances, heart break, sudden loss of a loved one, domestic violence, imbalance in family setup and lack of love, protection or security at the home front or school or work area etc.

Women face various challenges in life when she is expected to do more, give more and expect less. There are also phases in life when the women have no say in her health status, decision making, in her life per se. persistently being sad and discontent, having mood swings and being anxious, are some of the symptoms women feel in their life. Lack of women empowerment is a major cause for the rise in depression and suicidal tendencies among women.

The result is loss of interest or pleasure in activities, including sex. Physical symptoms such as headaches, digestive disorders, and chronic pain also occur which do not respond to medication or treatment many a times.

Depression: Psychological distress

Depression affects each one of us differently. The signs and symptoms depend upon the stage of ailment. The severity and frequency of symptoms, and how long they last, vary from individual to individual and also the stage of a particular illness. The cause of depression if diagnosed and eliminated can lead to sudden improvement in health condition and depression can vanish. However, curing depression is a slow process and needs very careful steps both medical as well as psychological, especially if the person is not on 1st stage of depression as it can also lead to fatal consequences as the depressed person can give up on life, in her low self esteem or self worth feeling.

The mental state a person is at any given time in to be very carefully understood and slowly and steadily elevated. Stress, tension, sadness, loneliness, even angry or guilty feelings, hopelessness, helplessness, lost feeling, hurt and emotional pain are also psychological distress symptoms and if these feelings prolong or keep returning again, making it hard to stay positive and enjoy life, it is called depression. The deeper, more intense and prolonged the feeling, it is called severe depression or clinical depression. If the feeling is mild, it is called mild depression or dysthymia.

Depression: causes and symptoms

The feeling of emptiness, loneliness, that no one understands, and that there's nothing left in life to see or know are all symptoms of being depressed. It starts in the mind, in our thinking, of course there are external factors, which trigger depression, there are people or circumstances that lead to such depressing feelings, but the thought happens and channelizes in the brain. Every minute the brain is processing complex thoughts and emotions. The chemicals, hormones, neurotransmitters are working to bring out various moods, pleasure and pain etc. We can also inherit some traits or tendencies. However it might not have the same influence or result as it had from the parents you inherited these traits. Nurturing, environmental factors, positive or negative vibrations surrounding you have an impact on the output of the inherited behavioural trait.

Some of the causes of Depression are:

- Stress
- Age
- Gender
- Cultural norms
- Nutrition
- Surroundings/Social and Physical environment
- Society and its rules
- Substance abuse
- Medications
- Family history
- Physical illness
- Your temperament

Researchers proved that some cases of depression occur as a result of influence of several genes and their interaction with environmental factors. Researchers also believe that depression also linked to many gene variants that act together in some way, with either a child's environment or his/her unique development. Experts believe that genes affect the risk of depression by shaping a person's response to stress.

Even if someone has a genetic tendency towards depression but lives in a positive environment, which is supporting and accepting, he or she may never develop full-blown depression. On the other hand, if the environment is hyper critical and demanding, people may develop depressive symptoms. Eliminating domestic violence, physical, and sexual abuse, and living in a happy, accommodative and cordial environment can prevent depression.

Types of Depression unique to women

Clinical depression, also called major depression is a mental health disorder characterized by persistently depressed mood leading to loss of interest in activities and causing significant impairment in daily life. It is a severe case of depression. Also called unipolar depression at times, it causes sadness and a feeling of loneliness or emptiness and an individual looses interest in daily chores.

National Institute of Mental Health (NIMH) says that certain types of depression are unique to women such as:

1. **Premenstrual Dysphoric Disorder** (PMDD)- most women feel some sort of irritability or pain during pre menstrual period days or during the cycle. These can become signs of depression if the symptoms are not mild and lead to either anger, sadness, depressed mood, severe muscle pain, breast tenderness, appetite changes, suicidal thoughts, bloating etc.

2. **Perinatal Depression**- Women have a lot of body changes during pregnancy, tastes buds change, there's anxiety, fatigue, there is morning sickness, and there are mood swings, and feelings of worry. Sometimes these feelings are short-lived and go in a week or two when the woman adjusts to her new body changes, however if these prolong or worsen, then these are no more 'baby blues', but are symptoms of Perinatal depression. This depression can occur both during pregnancy and after delivery (Peripartum and post partum depression). The feeling of exhaustiveness, lack of sleep, energy draining activities including lactation can be signs of depression if they are prolonged and extreme feelings.

3. **Peri-menopausal Depression**- It is a state where sometimes a woman may experiencing abnormal periods, get mood swings, and hot flashes and might also have difficulty sleeping. This can bring in depression in some women. Experiencing irritability, anxiety, stress, sadness and lack of enjoyment can be symptoms of Peri menopausal depression.

Other clinical/severe depressions are;

- Persistent Depressive Disorder
- Bipolar Disorder.
- Seasonal Affective Disorder (SAD)
- Psychotic Depression.
- 'Situational' Depression.

Frequent complaints of sickness such as stomach pain or headaches, eating either too much or too little, concentration problems, drop in grades, feeling of guilt or worthlessness, lack of energy, prolonged sad mood, and loss of interest in favourite activities, loss of interest in sex, fatigue,

memory difficulties, loss of appetite, sleep problems etc. are some of the common signs of depression.

Mania and Depression

People with **bipolar disorder** have rapid mood swings and are more likely to be anxious or irritable and when their mood disturbance is continuous, it's a mixture of mania and depression.

Manic Depression symptoms are racing thoughts and speech, lot of energy, feeling less need for sleep, exaggerated optimism and daring behaviour, Self importance feel, explosive anger, aggressive behaviour, extreme happiness or silliness, talking a lot and very fast, delusions, impatience and poor judgment etc.

Causes of Mania and Depression

Manic-depressive illness or manic depression is another term for bi-polar depression. Bi-polar depressive people go through periods of depression or mania and may also experience frequent mood swings. The condition varies from person to person. *Imbalance in the brain chemicals i.e. neurotransmitters* that normally affect mood is the cause for *bipolar disorders*. Genes can also play a role in causing bipolar disorder. It can be *heredity*. Sometimes, *environmental factors* play a large part in triggering such reactions like the impact kids have on parents frequent arguments or separation, divorce or death. Serious money problems or even sibling jealousy can lead to such disorders.

There are roller-coaster mood swings in bi-polar depression, with lowest and highest points and life feels being out of control. When the high point or mania comes, there are energetic and fidgety feelings and when people feel mania, they become manic. If people have both high and low feelings at the same time, it's called mixed state.

One of the hardest parts of having Bipolar disorder is having meltdowns or rages as they can come suddenly without any warning. People may feel normal a minute ago, and suddenly become angry and say mean things, screaming and having violent thoughts. It feels as if the thinking part of the brain has shut down, and the meltdown can last few minutes or even hours. Cause of such behaviour is because the part of the brain in charge of emotions called the *limbic system* may be

overtly active in people with bi-polar disorder. The limbic system shuts off the prefrontal cortex i.e that part of the brain that solves problems and makes decisions. Hence these people are unable to handle their frustrations.

Treatment for bi-polar disorders

Proper treatment and therapy can help control bi-polar depression even if it doesn't go away totally. Taking medicines such as mood-stabilizers also help. They help make brain stay calm and not over react. Therapy includes counselling, which is important because it helps you learn how to cope with outbursts of anger and/or sadness.

Triggers of depression

There are many triggers of depression, which can also be due to some medication being taken, or situations such as abortion or miscarriage. Situational depression also needs to be handles with utmost sensitivity and understanding for example sudden loss of a loved one, death in the family, disruption of the family structure, divorce or separation of parents, bullying at school/play, college, stress at work, financial upheaval, domestic violence or child abuse, serious illnesses, and social problems etc. are triggers of depression. Problems like social isolation because of other mental illnesses or being out cast from a family, friends or social group can contribute to the risk of developing clinical depression. Sometimes depression co-exists with a major illness or may be triggered by some other disease or medical condition.

Neglect, not giving adequate attention and care, both in children and women, can also cause depression. Eating disorders, alcohol and drug abuse are both a cause and an outcome of depression.

Eating Disorders

There are many kinds of eating disorders. **Anorexia** is the most common where a people think and feel they are too fat no matter how thin they are. Hence they try to eat too little. Another eating disorder is called **Bulimia**, where people go on eating binges i.e. eating extreme amount of food in a very short time. This can make them sick and even guilty afterwards. Guilt feeling makes them go bathroom many times or keep doing exercises a lot. This can damage the body in ways that can last a lifetime. Eating disorders is more common in girls than boys.

Obesity and Depression

Sometimes obesity isn't about diet or genetic make-up, it could be a case of being clinically depressed and struggling. People sometimes eat not to fill up the stomach but the emptiness they feel inside. It is vice versa too: Obesity leads to depression and depression leads to obesity.

Obese people are more likely to feel isolated and obese, ignored and neglected or treated in a degrading manner leading to depression. Sometimes it is hard to figure out if depression leads to obesity or obesity to depression (Fig. 1). Many experts suspect that it is just a repetitive cycle of depression and overeating and unhappiness.

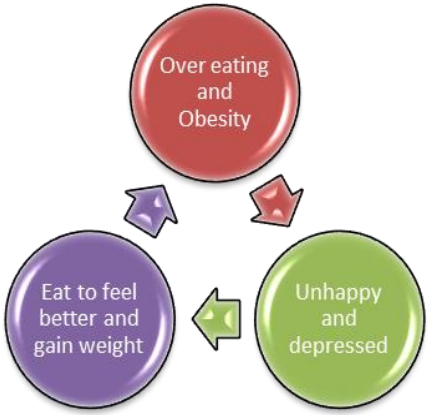

Fig.1: Relationship between depression and suicide

Statistics

- *350 million people worldwide suffer from depression and now it is a leading cause of disability (WHO Factsheet, updated February 2017)*
- *16 million adults had at least one major depressive incident in 2012 i.e. 6.9 percent of the population.*
- *Adolescent depression and related suicidal tendencies are becoming a serious problem, in almost all cultures and countries.*
- *Asia accounts for 60% of the world's suicidal cases, it means at least 60 million people are affected by either suicide or attempted suicide in Asia every year.*

Depression induced Suicide

Suicide is becoming a global public health problem and considered as top three causes of death among youth worldwide. Almost one million people die from suicide and 20 times more people attempt suicide; a global mortality rate of 16 per 100,000, or one death every 40 seconds and one attempt every 3 seconds.

> ❖ *Suicide is the main cause of mortality among girls 15 and 19 years, globally. In Asian and rural Latin American countries, poisoning with pesticides is a major suicidal approach among girls. (Ajdacic-Gross, 2008)*

In 1998, 1.8% of the total global burden of disease was due to suicide worldwide and it is projected to be 2.4% in countries with market and former socialist economies by 2020.

Suicide with depression

Some people think life is not worth living or that no one cares about them. They start feeling that they are a burden on this earth. Their life is meaningless. They think or talk about committing suicide. This is a sign of severe depression and needs treatment immediately. Meeting a counsellor, confiding to a friend or acquaintance, can help tremendously.

The depressed person should be given encouragement and shown self worth, and his belief of how beautiful life is and value of living should be inculcated. Initially a promise should be taken from the person that he/she would not think or committing suicide under any circumstance. The person's will power to live should be increased. This is a slow process, needs counselling, medication etc. Many other external factors and environment also impacts the mental state of an individual.

India statistics on Suicide

As per the report of National Crime Records Bureau in 1990, the suicide rates in India rose from 6.3 per 100,000 in 1980 to 8.9 per 100,000 in 1990, an increase of 41.3% during decade from 1980 to 1990. It shows the compound growth rate of 4.1% per year. The leading cause of depression and suicide by women s linked with sexual abuse. If sexual abuse decreases in females, the suicide attempt by females over lifetime would fall by 28% relative to 7% in men.

Role of Marriage in suicide: Indian Scenario

Marriage is not a strong protective factor for suicide attempts in developing countries (*World Report on Violence and Health*. Geneva: WHO; 2002.) Among those who attempt suicide, men are usually single and women, married. In 2009, 70.4% of all suicide victims in India were married and 21.9% were unmarried. Divorcees and individuals who were separated accounted for about 3.4%, while widows and widowers comprised 4.3% of the total suicide victims (National Crime Records Bureau 2009). The reason for more women attempting suicide is mainly domestic violence and sexual abuse within marriage.

Depression and Suicide: An Anthropological perspective

Having put forth the various causes, symptoms and factors related to depression and suicide, what is most striking and usually missed out is the socio- cultural scenario an individual is groomed in and how it affects his behaviour, thoughts feelings and lifestyle. Cultural context is one of the determinants that regulate a person's mood and mind and influences his or her activities and life. Societal rules are different for men and women in almost every region. There are a set of norms to follow and taboos to adhere to. For example in Asian cultures, women in their menopausal days are refrained from various activities including entering the temple for prayers in some places in India. There are many social customs and rituals that are being practiced for centuries and have come to harm more, such as the practice of giving gifts before, during and after marriage by the girl's parents to the boys family, such dowry practices, including domestic violence relating to dowry or son preference is socially and culturally condoned.

- In Brazil 48% of women, Egypt 61%, India 64%, Indonesia 11%, and in Philippines 28% of women had significant correlation between domestic violence and suicidal ideation (WHO Report, 2001).

A huge social, economic and emotional burden on the women to give birth to '*sons'* is also an underlying cause of depression and increasing suicides in countries such as China and India. The one child per family norm imposed by the Chinese Government have led to many married men especially in the rural areas, threaten to divorce or even divorce their wives

if a male child is not born. In Asia, methods of committing suicides is usually by poison consuming and in case of domestic violence, there are also cases of burning. Male dominated feudalistic-democratic society also leaves little scope for women to express themselves and their feelings, and have decision-making rights. Women do not have the required education, support and empowerment to speak for themselves, nor are allowed to vent out their grief, and if living in subjugated position and role, being abused sexually, domestic violence, or shamed, loses her self-esteem and gets depressed and even leads to suicidal attempts.

Even after such an extreme step is attempted, the family members do less in treating the person or finding the root cause of the problem and trying to eliminate or reduce it. Rather, the women is looked down upon even more and considered a coward who cannot face situations bravely or cope up with situations. The society creates a lot of psychological pressure to their females, with more expectations and females are not allowed to express or have any say in decision making activities, and do not have any economic support or independence either. Studies reveal that Depression related suicide (DRS) is overtly connected with the cultural pattern of some societies, for example: in Kaliai, West New Britain, Papua New Guinea. Even in West Bengal State of India, 15 suicides takes place every hour.

World Health Day 2017 with the Theme: Depression, Let's Talk

- More than 300 million people of all ages suffer from depression globally (an increase of more than 18% between 2005 and 2015)
- Depression is the one leading cause of ill-health, poor mental health and disability worldwide.
- Women are at high risk than men.
- Depression can also lead to suicide.
- Depression is treatable.
- 36% of Indian population is likely to suffer from depression at some point in their lives.

Women feel helpless, in male dominated feudalistic societies because, there is no alternative shelter for them except husband or a male-led family. Myths, prejudices and long-standing wrong beliefs, along with delusive superstition, folk tales and constant talks and stories about the bravery of suicides, modesty in avoiding shame, rape, abduction of widows or other similar tales may induce a female to kill by herself believing it is legitimate.

In many societies earlier and even in some remote areas now, wife burns herself on the funeral pyre of her husband, either as a ritual or to avoid sexual abuse by other male members. This is called the SATI practice and is very shocking. Social evils of society need to be combatted and status of women in society alleviated to bring about a decrease in depression, suicide and other mental health problems as well.

Stigma associated with mental illness

People fear asking for help or treatment from a parent, friends, relatives, counsellor, psychologist, social worker, health specialist or psychiatrist because they feel that they will be labelled as being abnormal or mad. Usually stigma delays onset of treatment and mostly those seeking treatment are at a severe stage of the illness**.** There are a lot of myths and misconceptions related to such feelings and people do not want to be labelled as *coward*, being unable to live up to expectations or unable to cope up with challenges in life. 'Failure' feeling also make people resist going for treatment or asking for help. Fear of Stigma and Discrimination make people remain secretive about their depressive feelings and suicidal urges and results in lack of timely help and therapy. People usually resort to temples rather than doctors for issues related to mental health because of the stigma. Statistics reveal that more than 50 million people in India suffer from a mental illness. In 2011, India recorded the highest rate of major depression in the world at 36 percent.

According to doctors, roughly 10 percent of India's population suffers from depression. Approximately, one in five women and one in 10 men are victims of a major depressive episode at some point in their life. This is also true because of the prevailing power dynamics in the Indian family structure where either the husband or the mother-in-law holds the authoritative position. The fear of being looked down upon is also a reason for shunning away any help or treatment.

Treatment for mental health problems:
Studies have shown pregnant women, and those who have children are less likely to attempt suicide. With better verbal and social skills than men, women may respond better to psychological and cognitive-behavioural therapies for depression. In the year 2000, the WHO launched the *multisite intervention study* on suicidal behaviours '***SUPRE-MISS'***.(http://www.who.int/mental_health/prevention/suicide/supremiss/en/) The study was done to gain more insights about suicidal behaviours and to gauge the effectiveness of interventions for suicide attempters in culturally diverse places around the world. (Suicides and the attempts to suicide remain under reported and the statistics are just the tip of the ice berg. Two large epidemiological verbal autopsy studies in rural Tamil Nadu revealed that the actual suicide rate is six to nine times the official rate. Early detection and proper treatment of a *primary psychiatric disorder* is vital. *Anti depressants* and behavioural interventions such as *dialectical behaviour therapy* (DBT) have been shown to have anti-suicidal effects**.**

Cognitive behaviour therapy or CBT is a psychotherapeutic treatment that helps people identifies their behavioural patterns that reinforce distorted thinking. It aims to help people understand the thoughts and feelings that influence behaviours. People on CBT realize and become aware that they are making negative interpretations and inferences. CBT is an effective way of treating many mental health conditions. CBT helps people with depression or anxiety disorders, phobias, addictions, and obsessive-compulsive disorder (OCD) panic disorder.

Rational Emotive Behaviour Therapy (REBT) is the 1st form of cognitive-behavioural therapy (CBT). It is a humanistic approach that focuses on our ability to create our own positive and negative emotions. The American psychologist 'Albert Ellis' developed REBT in 1955.

Role of NGOs
There is a need to spearhead projects that aim at decreasing suicides because of depression. Many NGOs and other sectors both private and government sector have rolled out projects in an attempt to address the issue but still lots of efforts are needed to bring out a positive change. Suicide attempts and depression are the outcome of many factors such as the stigma imposed by society isolating people. It can be due to an illness, a habit, or imposed by social and cultural norms. NGOs can help people

by maintaining confidentiality and secret so that the people asking for help may not fear being exposed and thereby isolated or stigmatized. Social workers, health workers, NGOs and Governments schemes can work in close co-ordination to improve the status of people, women in particular and help them with coping strategies and life skills to manage well and live a positive life.

Types of Psychotherapy

- **Behavioural Therapy:** Earlier treating the behaviour of a depressed person was a popular method. The basic idea being that what really matters, is how you act, what are your thoughts and beliefs doesn't matter. Therapists may have you practice ways to respond and focus on your behaviour instead of tour thinking using social **skills and training.**

- **Cognitive behavioural Therapy-** Now, cognitive therapy has been added to treatment. Cognitive means 'Thinking" i.e. attention to beliefs and thought process.

  Proper understanding of the underlying causes and factors and trying to eradicate them and bring out a positive thinking and outlook can help curb feelings of depression and suicide and help lead a happier and more productive life.

Life skills needed

There is a lack of sensitivity and understanding about how deep-rooted depression is in our society. There is a lot of stigma attached to mental health illnesses and unlike physical ailments, people do not immediately consult a doctor or ask from treatment and/or medication. Improper work life balance and fast paced society is among the major causes of rise in depression and suicidal tendencies. Coping up strategies and mechanisms are not arms length away or easily accessible and stigma attached also is a major hindrance in asking for help. Medication is required for chronic depression. However, early depression can be cured by understanding the sensitivity, counselling and providing, support systems, coping mechanisms, ensuring life style changes such as good sleep, good breakfast, yoga, positivity in life, teaching how to manage stress and failures in life, cope with anxiety etc. Living with bi-polar disorder can be challenging, we need to identify when our moods are changing and then find ways to calm it down and keep them under control, if there's extra

energy or anger feeling, sometimes a punching bag helps, for others counting till 10 or drinking a glass of water relieves extra energy, for others activities like strolling in grass, listening to relaxation music, doing chores like cleaning, or painting, writing or sculpturing helps. Each individual needs to identify what works best for him/her and can take help from family and friends, counsellor, teacher etc. Identifying triggers that cause such mood variations and then finding solutions to control them is among the best life skills for a balanced life.

There is a dire need to scale up mental health services in order to prevent further rise in depression and suicide. Suicide is a complex and multifaceted issue. Prevention efforts require coordination and collaboration among multiple sectors of society such as the health sector, education, labour, agriculture, business, justice, law, defence, politics, media etc. Suicide usually is more prevalent among low and medium income group societies in India and 30% of suicides take place by pesticide consumption. Suicide prevention efforts must be comprehensive and integrated to make an impact and reduce depression, anxiety and suicidal attempts.

Conclusion

A multi-sectoral public health approach is needed to combat depression and suicide. In India, ere is a tendency to stigmatize and discriminate people with mental illness. The breakthrough from this cocoon came when Bollywood famous actress started spoken about her depression and anxiety and thereafter the ice was broken and many dignitaries, actors and singers also spoke openly about their depression symptoms at some point in life. Suicide is a very grave issue with complex dimensions, but largely preventable through timely intervention. However, it is a difficult and diverse public health problem as it incorporates many factors, personal, traditional, societal and economic. Gender related vulnerability to psychopathology and to psychosocial stressors increase the burden of female suicidal behaviour. Women are more likely than men to attempt suicide, and hospitalized for suicide attempts. Suicide prevention programs should particularly focus on women and include woman specific approaches. More research on suicidal behaviour in women particularly in developing countries is needed. Reducing suicidal behaviour in women should be a public and social objective rather than a traditional exercise in the mental health sector. If public, private and government sector strives to reduce female suicide attempt, it can bring about a positive change in the society.

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National Institute of Mental Health
www.nimh.nih.gov

CHATPER 10

HEALTH PROMOTING BEHAVIOUR AMONG WOMEN

Dr. Suninder Kaur, Dr. Sushma Negi

A healthy lifestyle encompasses health promotion behaviours which act as determinants of disease prevention, health and fitness. Several factors are responsible for health conditions in people such as environmental, social, personal, biological, religious and economic factors. Health is also defined differently for men and women. Age also makes a significant contribution to our requirements of health. Besides socio-cultural norms even 'mass media and advertising' play an important role in defining the health system of a particular society.

Defining Health

Health is a combination of physical, emotional, mental and social wellbeing. It implies being in total fitness, both physically and mentally. A state of wellness means being healthy i.e. in total health and having a balanced life. Achieving this means making committed efforts and taking responsibility to eat well, exercise if and when required, be happy as much as you can (if possible at all times), reduce any kind of stress in life, developing health promoting habits. Health is critical to quality living and prevention is the key to wellness.

For example, prevent mouth, throat and lung cancer caused by excessive smoking; prevent accidents caused by drunk driving, prevent depression, stress and anxiety by remaining calm, at peace and happy. Good sleep, a good laugh and good company all help in increasing the overall health and quality of life.

Components of Health

Physical health is when your body functions well. The factors that help improve physical health are adequate rest and sleep, good nutritional diet and water intake, being physically active and having enough

energy for day to day activities. It encompasses getting regular medical checkups including dental checkups and undertaking any treatment if and when required. It also included getting proper vaccinations. Avoiding excessive intake of bad foods such as soda or junk food, alcohol and tobacco is also required.

Emotional/mental health means that you should feel good. It encompasses the ability to imbibe any emotional feeling positively and move on in life without major setbacks. The role of relationships, family setup, friends, acquaintances and social bonding, all play a role in shaping this emotional strength and feeling. To deal with frustrations without dwelling in and holding onto negative thoughts is very important for emotional wellbeing. To be able to manage stress and take responsibility for their actions is also vital. Another essential element is to be able to stand up to their beliefs and values.

Social health is how you get along with others. It includes co-operating and living in harmony during life's ups and downs. It means to increase friends and reduce foes. It involves communicating well, caring, sharing and respecting self and others.

Health Promotion

WHO (1986) defines health promotion as the process of enabling people to increase control over and to improve their health. Factors affecting health can be hereditary, environmental factors, technology, and media and also the values, beliefs and attitudes one adopts over the course of life especially during the formative years.

The genes inherited can strengthen the immune system and help protect from illness and disease better. Some genes can also pass on the ailment from parent to their offspring or increase the risk of getting certain infections. Genes also define and influence the personality, intellect to some extent though other outward factors also play a part in it.

The environment is the surrounding - physical, social, and institutional such as the neighbourhood, family, school, job etc. which has an effect on the overall health of an individual. Beside the physical condition that one lives in, the social and cultural setting of a person also influence his or her health behaviour.

Factors affecting health behaviour

Socio-cultural factors affecting health behaviour

Many sociocultural factors influence behaviour especially health-related behaviour. Society refers to the people you interact with. It connotes the people with a similar lifestyle and organization. Society is composed of groups of people, institutions that are cohesive, having a definite culture. Gender is a key factor associated with health-related behaviour. Other factors affecting health is race, location, disability, education etc. Health is determined to a large extent by the cultural and social confinement of an individual. Women especially have different customs in different cultures. Societal norms vary with each culture, place and region and also ethnicity.

For example, health and hygiene norms for women during her mensuration days differ in different societies. Some women are needed to stay in isolation during these days, not allowed to do kitchen work or cooking and not permitted to enter the religious institutions. Some women during these days are also supposed to wear old white clothes. In general, women are considered weaker than men and live in a very confined circle of house and family; their only contact aside from male family members is with other women whereas the menfolk have a much broader sphere. Over generations, spanning decades of progress have had little or no change in this realm and practice in many regions.

Another major custom is the discussion or choice over sex and sexuality. Though there have been women's movements and change or shifts in lifestyle patterns altering women's lives, nevertheless, many parts of the world still sees male dominance and miniscule levels of transformation in what is called 'taboo' or sexual repression.

Health is by and large linked to these customs and practices and with no informed choice, it becomes even more difficult to practice and promote safer, healthier habits and lifestyle among women. The role and status of women in the basic unit 'the family' needs to be re-evaluated. The lack of clear economic role for women also needs to be deliberated upon which relates to her health status. The level of respect, status and authority a woman gets in the family structure defines her will to control and improve her health.

Religious beliefs affecting health behaviour

Religion plays a major role in defining women's position in the family and the society. It influences the choices she can make, which weigh on her lifestyle and health status. The taboos limiting discussion on sex and sexuality and differing roles and opinions defined for men and women have, for centuries, made sex something that men sought and women feared. The ban on using contraceptives has been seen among both Muslim and Christian faiths though the trend seems losing ground in some countries as populations continue to spread at alarming rates uncontrollably. However religion is one institution that is most rigid and prohibited from questioning or opposing easily.

Severe punishments and large stirs, protests, demonstrations, and agitations occur or are predicted and so it leaves people less choice to counter and no option of forgoing such customary and religious practices. Among religions such as Hinduism, Judaism, Islam etc. women during their mensuration are considered 'impure' and thereby barred from performing certain tasks or visiting certain areas.

Menstruation Myths, misconceptions and taboos abound:

*The ancient Greeks believed that if a girl's menarche was late, blood would accumulate around her uterus would wander around her body. This could produce erratic behaviour, from violent swearing to suicidal depression. (*Janie Hampton is the founder of the World Menstrual Network, the author of books on health in Africa, and an international health planner. For more information see https://aeon.co/essays/throughout-history-and-still-today-women-are-shamed-for-menstruating)

Education factors affecting health behaviour

Health education is vital not just for students but people of all age groups. Providing complete and adequate information on health and related issues help people make healthy and right choices. The goal of health education is to provide tools that assist people make positive decisions and help live longer and have energetic, fruitful, happy and productive lives.

Health literacy refers to the capacity of an individual to imbibe knowledge and use it for promoting his or her health and fitness. To receive basic

health information and understand health issues and their priority is a key element to living a healthy life. Health literacy is also to be able to use these resources and services optimally there by maintaining good health.

Age related Health-promoting behaviours

Health-promoting behaviours of women vary with age. Women in the reproductive age have differing needs than youngsters. Similarly women in the menopause stage or post menopause stage have differing needs from those of younger or reproductive age. Midlife women are willing to make positive health behaviour change but require education about it and support.

Physical factors affecting health behaviour

Living in a safe environment is a boon whereas living under constant stress or scare is detrimental. Living in high polluted areas can shoot up the risk of respiratory infections. Living in areas where crime rate is high is living under constant pressure and scare. Living in a friendly environment increases positive vibrations. One should always strive to be as calm and as peaceful as possible for better health and good quality life.

Challenges of women in health promoting behaviours

A cross-sectional study among 1359 Islamic Iranian women of reproductive age selected by proportional random multistage cluster sampling in Tehran was conducted and it was observed that among the six dimensions of health-promoting behaviours, women scored highest in interpersonal relations followed by nutrition, then physical growth and then was stress management and health responsibility whereas they scored lowest in physical activity. The study highlighted that Social support was a major aspect influencing health-promoting behaviours and some socio-demographic characteristics predicted health-promoting behaviours.

Another study in Nova Scotia, Canada among mid-life women examined the connection between physical activity, self-esteem, health status and health-promoting lifestyle activities of women between age group 35-65 years. The results showed that it has a ripple effect on health behaviour practices and other lifestyle behaviours as well.

A study by scientists on Gender and determinants of health was conducted to see biological, social and economic determinants of health and its differing health outcomes for both males and females throwing light on

coping mechanisms with illness. Results revealed the food preferential of men over women in developing countries; for example in Zimbabwe husbands had total control over all decision making, women had significantly lower nutritional status than men, showing how power relation impacts health status. A research on 194 American Asian Indian women in California, on breast cancer knowledge, attitudes, and screening behaviours showed monthly breast self exam adherence was low (Sadler *et al.,* 2001). Majority of them reported that their breast cancer knowledge was inadequate.

Indian Scenario

Assessment of the degree of education, awareness about health behaviours and knowledge provided by various institutions and NGO have had to women both in rural and urban settings and also the extent to which ethnic subgroups are perceiving and acting on the proffered information is vital in creating effective public health interventions. A study conducted in 2007-08 among Chandigarh college students regarding health behaviour in 100 males and 100 females showed that women were more conscious about their health than men. Results also showed that Chandigarh college students had good health promoting lifestyle and sense of coherence.

The leading cause of cancer related death, in recent times in India, for women is breast cancer and cervical cancer (two of the most common cancers among women worldwide). A study in Karnataka showed that people were not aware that it can lead to death, many were not educated enough and have limited or no knowledge at all, besides there was a lot of stigma attached to it and hence reluctance to get tested and receive proper medication. Two Qualitative exploratory studies were done, one on breast cancer (59 in-depth interviews) and the other on cervical cancer (147 in-depth interviews) among patients and caregivers. Lots of stigma is attached to this disease, which limits screening and treatment, such as inevitability of disability and death with a cancer diagnosis.

Health promoting behaviours, perceived stress and interpersonal support

Unhealthy behaviour and its impact

Unhealthy behaviours increase ill health, which has a ripple effect; it not only affects health and wellness but also slows down other positive

behaviours and jeopardizes the positive flow of life. Ill-health affects all other lifestyle behaviours. If woman's health jeopardizes or is compromised to some extent, it affects all other household activities, affects other members in the house especially the children who both by observation, gestures and also by words and behaviours learn and imbibe so much. It affects their mood and attitudes as well. Hence, women's health is very crucial to maintain the family balance and to have a healthy happy household and life. Hence it affects negatively on other family members and domestic activities. Similar is the effect on jobs and absentees may increase. Work output is compromised and so is work quality.

Lifestyle and eating habits

Nutrition transition, considered as the main cause of obesity, began in the 1990s. The developed countries followed by the developing world changed their food consumption habits from home cooked fiber rich food to processed foods. Diabetes, joint pains, weight gain or obesity are modern diseases and have reached far and wide as a result of globalization. Changes in not only diet but also physical activity led to these maladies. Health promoting behaviours are the needed to address these conditions to promote positive and healthy lifestyle.

As more options of food choices at affordable prices flood the market, it becomes easier to buy these 'fast' foods and junk foods and no physical activity needed in preparation. Juices as a replacement to eating fruits have diminished the fibre intake and body also absorbs food rather than breaking it down into smaller particles and thus causing obesity. Urban areas have more obesity ration compared to rural areas as lesser physical activity and more access to processed food in cities than suburbs. Migration of people from towns and villages to the city compounds the problem. Similarly both men and women in jobs leave them with no time or energy to cook at home. It seems easier and worthwhile to spend time in entertainment and grab a meal from the fast food joint rather than using the little time after work to cook and eat. There is no link observed in education and obesity as learned people well aware of the disadvantages of eating junk food also find the option a preference over healthy, safe and nutritious home cooked food. Food safety is also an issue that is being overlooked besides the nutritious value. People have become a foodie and eat not only when hungry but lots of time for their good taste and tempting

presentations. Another issue that still remains neglected as a result of bad eating habits is unhealthy teeth and constant tooth decay and teeth treatment required these days. There is less research and even lesser hue and cry over the increase in dentition requirements, hygiene of teeth and increase in tooth replacements and artificial dentine and crowns as a result of modern lifestyle and dietary patterns. A reason could also be the increase in wealth of these dentists and hence there is no need to alarm and alert the society.

Obesity- a Major Health Behaviour Malady in Today's Era

Too much of weight gain or weight loss results in fatigue and imbalances in body fluids. Obesity has come to affect people of all ages. There is no single cause resulting in obesity and there is no single problem as a result of obesity. Obesity is a malady that does not occur overnight. It is a slow process involving not only dietary patterns but is an outcome of the 'lifestyle' in totality. To deal with obesity without taking into account the overall lifestyle will not cure the problem. It may diminish the problem for some time but it will re-occur and may recuperate with an even more psychologically upsetting feeling and setback. Obesity not only causes lethargy, it also brings with it a handful of psychological problems such as depression, lack of confidence, unable to face the world with an unfit body size, clothes of the wardrobe stop fitting and there is always a scare that people would stare and comment.

The feel bad feeling coupled with depression and loneliness results in more hogging of food and thereby increasing the problem in a geometric progression. The questions at the back of our minds are: What are adequate food and a healthy lifestyle? Where are we lacking and how do we achieve it? Though there are solutions but these are tough solutions, and so we must resonate time and again that 'Prevention is better than cure', follow a healthy lifestyle as changing to a healthy lifestyle later on would be all the more difficult. Changing lifestyles have stemmed global trends of increasing obesity.

Obesity is one of the most worrisome problems of industrialized economies. Till the 1900s only 10 % of the population lived in the cities and by 2000, the number rose to about 50%. Although science and technology increased, but the physical activity has decreased in villages. The today's villagers are not immune to the diseases of modern lifestyle,

however the various food choices available in the cities have not reached villages to such an extent and hence diseases like diabetes and obesity are at bay. Farming in the villages has seen a sea change from subsistence farming of multiple crops to a single high yielding cash crop. Foods rich in fat content are not only jeopardizing the developed world but have also made inroads in the developing world. In the earlier eras obesity used to mean a sign of richness and wealth and hence not much emphasis was given to curb it. It was considered that 'the poor' do not have access to the food choices as the rich and affluent do and neither do they have nutrition education. However, both underweight and overweight have similar effects as both decrease life-span, both shares high levels of sickness and disability, reduced productivity and quality output, lack of concentration and shortened lifespans.

Problems associated with obesity

There are both medical and psychological problems associated with obesity and all of them are chronic. Medical problems include hypertension, diabetes, various cancers, stroke, heart disease etc. Psychological ailments include depression and anxiety, loneliness, lack of confidence etc.

Diverse benefits of healthy lifestyle interventions

Modification in lifestyle, addressing nutritional and caloric intake of food and exercising for better blood circulation and energy expenditure can help maintain good health and reduce ailments such as obesity, depression and thereby increasing self esteem and have better health outcomes.

Having healthy lifestyle curriculum in school and other professional education and also integrating healthy lifestyle programs, workshops for people of all age groups can address problems before it harms health and wellbeing and girls /women should be especially focused upon because creating awareness in girls, their participation and immersion in such healthy lifestyle intervention programs, may ultimately lead to better health outcomes for household, workplace, entire family, and especially children i.e. future generations.

Promoting health behaviours

The process of enabling people to increase control over their health and its determinants, and thereby improve their health is *health promotion* (WHO definition)

In the USA two major agencies working towards health promotion are CDC (Centres for Disease Control and Prevention) and NIH (National Institutes of Health). They provide information about various diseases and health related goals such as exercise and nutrition. Research in preventive medicine for maintaining a healthy lifestyle is also being done in CDC and NIH for diseases such as cancer. Health behaviour related information is also provided by these agencies. A person's viewpoint and actions regarding their health, and well being is called *health behaviour*.

What is good health behaviour?

There are several things, which can help promote healthy behaviour and lifestyles such as:

1. Healthy nutrition
2. Proper and regular exercise
3. Yoga is also helpful
4. Avoid cigarettes smoking and tobacco products
5. Avoid going in high pollution areas
6. Avoid excessive alcohol consumption
7. Have a positive outlook towards life
8. Cut down on junk food, oily food, soda and sweeteners, and even processed food such as cheese pizza etc.
9. Food safety i.e. wash, clean, cook properly all food before consumption
10. Get adequate sleep
11. Avoid physical and mental stress
12. Take proper rest
13. Dental care
14. Eye checkup
15. General body checkups
16. Avoid substance abuse / Do not take illegal drugs
17. Take steps for stress management
18. Avoid depression and anxiety
19. Middle and old age women should test for osteoporosis
20. Cervical cancer and breast cancer can be avoided by timely and regular medical checkups

Engaging is health is critical for wellness. However, disease prevention is also equally important i.e. taking steps to thwart disease from occurring in the 1st place. A study was conducted in Maryland to find out the beliefs of 1040 physicians on 25 health promoting behaviours and it was found that by and large physicians felt all the behaviours were important barring vitamin supplement intake, barring from alcohol totally and annual exercise tests which 1/3rd of respondents did not feel as important.

Role of media and communication

Mass media helps in many ways. It helps spread people voice quickly to large areas. It echoes and resonates among all people quickly and helps brings result and change at a faster pace. Mass media helps health professionals and social workers expand their audience reach. The face-to-face channels of communication and working at the grass root level by heath workers, anthropologists, sociologists, psychologists, NGO members and other social workers and activists helps communicate and understand problems so as to address it at their level.

However, mass media is also crucial as it acts as extensions to these workers; and reach larger groups both in urban as well as in large, underserved rural areas. Use of mass media in developing and promoting effective opinion leadership thereby promoting changes in communication behaviour within social networks may in tum influence individual health behaviour. We need to make optimum use of mass media and other means of communication which can be crucial in our effort to improve women's health status.

Promoting health behaviours in medical education

It has been seen that obesity is becoming widespread globally. Changes in lifestyle is the major cause with lesser physical activity, more mental stress and changed eating habits fostering quick snacks, junk food etc. This is true for all sitting jobs and professionals such as computer engineers, office workers, administrative staff, teachers etc. Educating doctors and other health professionals warrants greater attention to promoting effective weight management through health behaviours related to eating and exercise. For those who are obese, interpersonal support is required to deal with stress, anxiety and depression which comes with obesity. Interpersonal support becomes a driving force by which exercising and good eating habits can be inculcated thereby increasing wellness and weight management

Disease Prevention

Prevention includes a wide array of activities, which are also called interventions. There are 3 levels of disease prevention, which can be employed to keep healthy and full of vigour. These are primary, secondary and tertiary disease prevention processes.

Primary preventions refer to the process entailed to stop disease from starting in your body. The few things that can help avoid diseases are things like food safety, proper hygiene, cleanliness in surroundings, avoiding sun to prevent skin cancers, washing and cooking food properly, avoiding smoke and tobacco to help prevent TB or lung cancers, avoiding excessive alcohol consumption to save the liver. Further one can use vaccinations to avoid flu and use condoms to avoid STIs (Sexual Transmitted Infections).

Secondary prevention, unlike primary prevention does not prevent disease or stops its onset or reduce disease incidence but is combating the disease after its onset, at the early clinical trial stage itself. There are certain screening processes, tests that can be performed to get hold of the disease at the early stage and cure it before it becomes fatal. AIDS detection tests, cancer tests, pap smear, mammography are some examples.

Tertiary disease prevention is not the screening stage but basically disease cure and infection containing methods by proper medication. There should be proper adherence to medication and it promotes recovery phase, tries to minimize suffering and impairments. For example: management of diabetes, rehabilitation, occupational therapy etc.

Mental Health Promotion

Mental health promotion is also an area widely neglected with very limited studies done and very few intervention programs to address mental health and promote mental health and well being. Women's mental health is neglected for generations and rural women all the more. Proper education and women empowerment is the call of the times. With no or very little say in any decision making process, with more duties and minimal rights given to them, women have no control over their physical as well as mental well being. Considered primarily as child bearers and expected to be son bearers, the women are rarely allowed to speak for themselves. Social, cultural and economic factors play a vital role in this regard.

Mental health problems include:

- Suicide
- Violence
- Depression
- Eating disorders
- Psychotic disorders such as hallucination
- Obsessive compulsive disorder (OCD)
- Stress and fear

India was amongst the first countries to develop a mental health programme in the early 1980s. NIMHANS (National Institute of mental Health and Neurosciences) India, reveals 17.3% people being mentally ill in India and 150 million people in need of immediate medical intervention (Times of India October 12, 2016).

Some of the reasons are:

- Economic Instability
- Stigma around tests and treatment
- Lack of education
- Powerless in decision making process
- No women empowerment
- Poor implementation of programs
- No prevention strategies
- Paucity of mental health professionals and proper treatment
- Stress factors relating to husband and mother in law behaviour

A study was conducted in rural Maharashtra among 32 women enrolled in Primary health care (PHC) project on mental health and inferences drawn were that the PHC programs could indirectly promote mental health by addressing its determinants i.e. by *decreasing discrimination, increasing social unity, generating income opportunities* etc.

Inference

Health system

Comprehensive modifications in the existing health system provides the opportunity to address health needs of women encompassing innovative approaches and focusing on bridging the gap of gender inequality and also empowering women as ‘drivers of change’.

The Individual

Individuals can and should play an active role in their health behaviours.

1. They should make informed decisions.
2. They should promote the health of others. They should access reliable health information, systems and services.
3. They should strive to remain as healthier as possible by basic health and hygiene techniques such as maintaining surroundings and self clean, taking regular baths, wearing clean clothes etc.
4. Food safety techniques should be employed such as cleaning/washing all food well and cooking well before eating.

The Family

Families can help ensure hygiene, cleanliness, food safety and also shape attitudes and behaviours in favour of a healthy lifestyle. Family is a great support system, it can help individuals cope up with their stress and achieve milestones. The values shaped within the family helps make better and stronger individuals. It is amongst the best mechanisms for emotional and mental development. Coping with distress and emotional upheavals can be very difficult but a strong and intact family system can help individuals overcome difficult situations with positivity as the family system provides a child the necessary skills for a healthy living.

Community

The community at large can help promote wellness by engaging in programs such as cleanliness, food safety, offering behaviour modification programs (such as tobacco-cessation classes, drug control, and AIDS control program etc.), imparting awareness and providing health services.

Risk Behaviours

Indulging in risk behaviours can jeopardize an individual's life. Risk behaviours threaten a person's health or the health of others. Cumulated risks could be related and may become detrimental as they increase in effect with every added risk behaviour.

Conclusion

Women both in urban and rural areas lack a voice. There is dire need of programs to uplift women from their position, empower them, provide necessary and timely education, and remove stigmas and discrimination so that women can come forward without any fear to improve their health, both physically and mentally. Adequate health professions, and other bodies, both government and private sectors needs to join hands

in this effort. Lack of qualified professionals or treatment facilities and drugs is another hindrance in this process. It should not be forgotten that if a woman is health so is the family healthy and happy. Family being the basic structure of a society is the brick to a healthy foundation and healthy generations. Women should be involved in decision-making, and not be at risk of social exclusion in any facet of life. The key to empowerment is to remove all barriers in choice, influence and control and also empower relations so as to strengthen women's say over their health and their life.

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Aeon –A digital magazine of ideas, philosophy and culture started in London in 2012 by an Australian couplePaul and BrigidHains. https://aeon.co/

Newspaper

Times of India, Bangalore *13.7% Indians are mentally ill, study says*VandanaKamath | Bangalore Mirror Bureau | Updated: Oct 12, 2016, 08.49 AM IST

It is health promotion and disease prevention plan initiated by the US dept. of Health and Human Services, designed to serve as a guide for improving the health of all people in the United States.

This plan is revised every 10 years.

It aims to:

-Promote health

-Prevent illness, disability and early death

CHATPER 11

ROLE OF SOCIAL MEDIA TO MAINTAIN WOMEN HEALTH

Ms. Sangeeta Dhar, Dr. Amit Awasthi

"Social Media"- now has become parallel life for most of the people especially adolescents. They are expressing their happiness-sadness-emotion more via social media rather than they express in real life. The virtual reality now has become more real than reality. Facebook, Watsapp, Tweeter, Instagram, youtube, Tumblr, Pinterest, Flickr, etc have become controlling factors of our life. With the increased popularity of social media, people started comparing their life as portrayal in social media. This increases the expectation from life and as well as build unrealistic expectations. Comparison and parity make people more betterment orient at the same time dissatisfied.

Social media reaches large number of people and good for spreading the massage. In the context of health, it is very effective way to spread awareness. People are becoming more concern about their health and understand when to fetch medical help. Especially for woman, some of them may still not comfortable to discuss the problem and sometime not getting scope of revealing their own problem can get help from social media connectivity. It is also observed that a notable number of populations are ignorant about women health related issues.

A survey conducted on a international dating app related to breast cancer reveal 41% of user is unaware about the importance of regular health examination in women and did not encourage any women (mother, sister, wife, friend or any known woman) to go for a check-up. According to a study conducted by National Cancer Institute's Health Information National Trends Survey (HINTS) in 2005 studied the behaviour of three kind of internet users, 'searching for health information for oneself, participating in a support group for those with similar health or medical conditions, and purchasing medicine or vitamins' reveals that 58% of Internet users usually searching for health information for themselves,

3.8% used online support groups to fetch information, and 12.8% bought medicine or vitamins online in the past years. Multivariate analysis also indicates that women are more frequent user on internet in case of medical help seeking.

The women health-related problem is not only confined to physical problem but it also includes aspect related to mental health. Physical and mental health both is interconnected, able to impact each other. Physical sickness affects mental health. At the same time mental health also hamper physical health. Overall, effect of social media on women health is vast and multi-dimensional aspect where many other factors are also involved.

Factors influencing women towards social media

With the popularity of social media practices, it is increasingly gaining popularity among women. There are various factors which plays direct and indirect role behind this popularity.

Lack of proper association

Now a day most of the people are busy in their own world and becoming more self-centric. A large number of women are facing regular lack of association as they are busy with household chores and their movement outside home is restricted. Social media give a space where they can connect with others from the comfort of their home.

Lack of real-life support

In many cases women are deprived of real-life support system. They may face problem insight as well as outside of known circle and sometimes proper support is not there in known circle when required. This may include physical, mental or other problems which require understanding and troubleshooting with proper concern but it is not readily available. In such cases women reach for support online.

Social media provide ample information and opportunity to get solution in such situation. It helps with relevant connections for help, advice, and support and even with fundraising in many cases. There are various pages on Facebook, Instagram or in other social media dedicated to women related problem and support. It covers health related information, fitness, helpline (abuse, domestic violence etc) and many more. These give platform to express as well as to understand the view of others.

Easy to use

The available social media platforms are user friendly. One can easily operate it without any training. No extra training is required to use it.

Economic and easy accessibility

Social media is a very accessible option. One can be part of this at nominal expenditure. No paperwork, hard labour or money required for usage. It can be used from anywhere anytime. Only internet connection and suitable gadget are essential to use it.

Vast options and dedicated groups for each topic

Social media covers almost every topic and give platform to connect with like-minded people. Hobbies, cooking, housekeeping, reading, fan page, classes, examination preparation or name a thing one wish to be part of, is existing with social media. If not one can easily create one according to his/her own criteria.

Easy option to express and alteration of opinion

It is easier to express in social media. One can express and can react to any post, just by using some signs or emoji. Signs showing grief, happiness, liking, disliking, heart-felt/close to heart, care, etc is available and just few clicks away. The commenting option is also available. The alteration and deletion option of one's expression or comments are also available. If someone has admin authority then one can alter, block, delete or restrict comments as per wish. It is as easy as, if you don't want to hear criticism, block the commenting part in your post and done!

Positive Effects of social media

The use of social media can accelerate the wellbeing activities and bring positive changes. Now a day where people are getting chance of exposure via electronic media and are flooded with opportunities, one can make best out of it according to need and interest.

Physical effects

There are various physical activity pages where one can be part. Many women now a days actively joined and take part of different physical activities which is very important to maintain the health. There are varies women related disease which can be managed with regular physical activities. PCOS (Poly-cystic Ovarian Syndrome), diabetes, obesity is few to name which are prevalent in women and can be managed by regular physical activities.

Nutrition

Healthy diet and regular intake of nutritious food helps to stay healthy. Social media now give an open platform to discuss on this matter. Various pages provide information and recipes depending on one's food preference (vegan, veg, non-veg). Even new and cross- continental recipes along with traditional and tested food are also available to try.

Awareness

The first step of taking preventions or measures against health-related problems is realizing 'this is a problem' and 'knowing the necessity of taking steps at right time.' There are some issues where females are not comfortable while discussing the problem. Breast related problems, pregnancy issues, intimate hygiene problems are few of those. Moreover, in most cases they don't know what symptoms they should discern off. In this type of situation social media acts as awareness creator. Now a day breast cancer can be cured completely if the treatment starts early. For this self and regular checking of breast is required. The information, i.e. how to check breast for abnormal development, is available with social media. Other than this, when to consider any feminine change is abnormal, relevant information are available. It is possible to get access all this information without personal interaction or revealing identity with the help of social media. There are some sensitive issues where social media can act as awareness creator (Table 1).

Table 1: Describes some of these topics where via social media awareness can be created.

Subjects	Topics
Pre pregnancy	Awareness regarding prenatal vitamins, restrictions, tests, food and nutrition
Pregnancy	Myth busting, knowledge against superstition, safety. Food to be avoided, comfort house hold and cloth, food and nutrition Signs of ectopic pregnancy, warning signs Before delivery precautions, child movement watching and emergency signs

Nutrition/balance diet	Data about nutritious food, importance of balance diet
Physical activity	Different physical activities according to health condition
Post pregnancy	Myth busting, general care methods, food and nutrition
Hygiene (urine and mensuration hygiene)	Maintaining process, suggestion, available options and their uses
Feminine problems (UTI (Urinary Tract Infections), Breast related problems, vaginal, genital and menstruation problem)	Home checking methods for abnormalities, signs and symptoms of infection/tumour etc/warning signs
Family planning	Method of contraceptives, safe sex methods, family planning duration
Awareness against mal practices	Forced marriage, Genital mutilation, Child bride
Psychological	Understand the abuse, know the rights
Support forum	In case of any above situation knows where to find help, legal rights

Discussion forum

There are many forums where people can discuss their own experience and can get support from others. Topic oriented discussion forum not only make people realized that she is not the only one but others are also there who is going through same patches of life. At the same time success stories may make other woman inspire, can shows new path and hope. There are many online platforms which provide community based monotypic communications dedicated to woman. Some of the forums available on internet are pregnancy, other health issues, marital discord, new mother's forum, post-menopausal depression, to understand abusive relation etc.

Counselling

The activities of social media not only restricted to awareness generation, discussion but also very helpful for getting proper counselling in different aspects. Woman who unable to go to a counsellor specially for seeking mental health counselling, can easily get access to counsellor via social media services. There are few NGOs and governmental organizations that

provide counselling online, although many new organizations have also come up with same facilities. Counselling regarding physical and mental health maintenance, guidance etc are important and can be provided easily via social media to a large number of users.

Distal suggestion

In many cases the people especially woman cannot able to go to distal part to find proper and best solution. For example, a woman who is living in a village which lack of advance medical facilities can faces the difficulties to get access of treatment and to find a good option. With popularization of social media, it has become easier to fetch the information. Information are also available to consult before considering the option, viewing the reviews of other users are also available. Social media provides an economic and effective solution in distal consultation. After taking decision now the necessitated woman can go for further consultation with concern person with the help of digital media.

Information

Social media serve the purpose of data banking. It is a broad casting media where people can get information or can find relevant information when required. There are many topics where women are depictive of relevant information. Women often don't know what facilities are available.

There are many segments of women wellbeing where not enough information is readily available for females and some of them don't even know the necessity of that information like why and when to use contraceptives, vaccinations details, side effects of different drugs, maintaining hygiene and regarding selection of proper health related products etc. Even after collecting information, they can discuss and post their thought in social media, can get experienced opinion as well as can share their own opinion for optimum benefit.

Socialization and mental health in females

In present lifestyle when most of the people are too busy to spend time with family, there a few house wives or stay at home mothers or agedly ladies are felt aloof and segregated. Limited and controlled use of social media helps them to come out of that situation and also help them to develop communication skill, thus enhance their mental health.

Creating consciousness and positivity

The positive image portrayal in social media made people conscious and concerned. Now this actually motivate some women to achieve better in terms of physical change, motivate for exercise, develop new hobbies, do better in in life or regaining mental strength. All these positively boost physical and mental health.

Realizing the truth and get rid of unrealistic expectation

There are few truth revealing trends are there in social media which actually shows the reality of life. These stories or trends helps woman to stay encourage and positive which creates a positive impact on body and mind. 'Accepting body as it is', 'post pregnancy depressions are normal', 'accepting flaws', 'reality of influencers and celebrities' are few of them which motivate and create positively by easing unrealistic expectations.

Negative effects of social media

Along with the positive sides there are some negative aspects of social media on women's health. Social media can hamper mental peace and can affect the physical health. These side effects are mainly due to misuse of media and because of the availability of wrong information or false portrayal.

Physical effects

There are few companies who are showcasing these concepts of unrealistic beauty standards. There are some who are advertising and selling products without proper testing. As because these products are not tested and well formulated, use of these products can lead to severe side effects. Even there are fake reviews are also available to increase trustworthiness. Due to popularization of social media now it is even easier to sell and advertise these kinds of products. The excessive use of social media increases screen time which causes many other unwanted physical effects. This reduces physical activities, disrupt even cause sleep deviation. Increased screen time causes eye strain.

Mental effects

Human's expectation from life and self is now manipulated by the media. The constant portrayal of perfect female figure, fairness, desired physical traits etc, put pressure on mind as it creates inferiority complex. For example, the passion of looking slim and prefect body shape cause self-dissatisfaction and sometimes trigger tendency of unnatural trying to

achieve desire body type which may harms health. This unnatural concept of beauty should not be encouraged. It is very important to reveal the actual truth behind the photos. The huge use of photo editing tools, software and multiple clicks are the reality behind the unreal photogenic pictures. The comparison with other's life and bait towards more luxurious lifestyle cause dissatisfaction. These altered outlook causes depression.

FOMO (Fear of Missing Out)

Absorption in virtual world may create an artificial crowded place which give all time involvement and social interaction. This excessive interaction gives a feel-good factor as well as develops FOMO (Fear Of Missing Out) along with unrealistic expectations. The life of user may concentrate to likes, numbers of followers, comments or to any related attributes. Any drop or reduction in such social media interaction may induce insecurities and FOMO.

Anxiety

Social media shows and exhibit too many factors of life. One who is involved in social media use may get affected by it. Let's say, a perfect physics, luxurious lifestyle, perfect family, high success in career, spending at holiday destination along with this a popular and influencer in social media is too much for one to be. These all factors, expectations and thinking ultimately create anxiety which is not good for physical and mental health.

Medical errors

Social media and forums provide ample information at finger tips. General people especially woman who feels uncomfortable to discuss their medical problems with outsider and sometime due to lack of proper scope try to find medical solution over internet and take help of social media face some problems. Self treatment can go wrong and even can worsen the situation. Without verification the distal medical suggestion also can go wrong.

Loneliness

Skill of communicating in digital media and communication in real life is a bit different. Females who have become more addictive towards social media often suffer from loneliness. For example, a beautiful lady who is having lots of followers in facebook and instragram and spent lots of time on internet can be alone in real life because she will be having less time for interacting with real human.

Detachment from practical life

The unreal expectation can detach a female from real life. Let us consider a fact, a woman who is always appreciated for her look and even get hundreds of likes for posting a picture of breakfast definitely will start looking at the life from different prospect and can become overwhelm due to attentions. Do we expect the same situation in real life that one is sitting with breakfast and surrounded by 100s of people who appreciate her while eating! The false image of celebrity feeling create a virtual world and in long term definitely it overpowers the practical aspects of life.

Effect on young and teen

The teens and young populations are major user of social media. They spend more time on social media than others and this is even increases after Covid time. During this long home isolation period, they become more addicted to it. It has become trend. Some of them using it judiciously and getting benefit out of it whereas some of them due get so absorbed that they are facing all negative effect quickly (Table 2).

Table 2: Alarming points for young female social media user

Some alarming points for young female social media users
• Felling prays to online trap of money laundering, relationship, career trap etc. • Relying and using untested and untrusted products health and beauty. Such as skin tone enhancer, mensuration flow controlling supplements, menstrual cramp reliever etc. • Putting self to dangerous situation to meet social media friend, going for a job interview to an unknown place etc. • Ignoring reality to follow and become popular in social media. • Compromising health, daily work, education for social media. • Ignoring or getting detached with real-life friends, families for social media association. • Changing self to be as par with social media standard.

Improving impact of social media

The use of social media comes with good and bad aspects. To screen the good from the bad there are few steps which can be applied which can also be applied to women's health related issues.

Rectification of social media use pattern and time limitations

Balancing time between virtual connectivity and real life is important. It is necessary to segregate the virtual world from reality. The limited and relevant use of social media is one of the most important ways for combating the mal effects. These may reduce the relative negative effects.

Reliable source finding

It is very important to validate the activity and reliability of any social community before joining. Even after participation all posts and activities should not be believed. Previous to take any consultation the authenticity of source should be verified. Obviously social media is a platform of public interaction but at the same time it does not provide the authenticity of users. So, it is important to check the authenticity of information available on media, at the same time the access by others should be restricted and meaningful. Any information available on social media should not be believed blindly, should be reconsidered and cross checked before trusting it.

Legal steps

The legal control is important to check the unauthorized use social media. Every post should be screened and there should be internationally accepted laws and punishment system for unscrupulous activities. The liability of maintaining the veracity also lies upon the users.

The users should be aware and cautious about any malefic activity, should report any suspicious activity to the admin. Still in developing countries woman are not aware about positive aspects of social media uses. It should be felicitated with proper use protocol. The precautions should also be taken to prevent mal effects of social media use.

Still there are not enough studies solely dedicated to women and social media use. It is a comparatively untouched area of study where more and more studies are needed. The health-related issues of women and effects due to use social media now has become an important aspect of research and still neglected. While using social media, there are many do’s and don’ts that should be taken care of women and highlighted in table 3.

Table 3: Do's and Don'ts while using social media at a glimpse

Do's	Don'ts
• Connect with family and friends. Follow real heros and valued influencers. Stay connected with like-minded people. • Do share and exchange thoughts with like-minded people. • Involve yourself in real life activity. • Post, share good content and learn from informative and motivating posts. • Do practice few self-imposed boundaries to maintain usage extend of social media. • Be yourself. • Find relevant details or product, but consult honest reviews before including it. • Make a separate real practical life and live it. • Reach for help if you got troubled while using social media. Complaint to respective authority, reach for cyber crime help if required.	• Don't just add everyone to your private account. • Don't share sensitive information like home address, present location, daily schedule with location, bank details etc. • Do not ignore your health, education, work, real life. • Don't post any content which may hurt others. • Don't become absorb in trends. • Don't change or pretend to behave like someone else. • Don't use untested or untrusted product. • Don't let social media become your whole world. • Do not ignore if you fell prey to any trap.

Conclusion

Social media can act as potion or poison depending on its usage. It is important to understand the necessity and mode of use. Especially for women, it is a helpful medium because without facing any unwanted question, or travel to distal place, one can gather required information.

At the same time female should aware about data redundancy. The judicious use can lead to positive changes. This is the era of social media and in upcoming time the social media use will increase many folds. It is not possible to converting everything to positive but the restriction of malicious effects is possible. This is mainly depending on personal preference of usage which actually matters. Women health is a sensitive issue which covers and related with vast areas with multiple factors. The proper use of social media in these aspects may act as boon for many.

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CHATPER 12

MANAGEMENT OF STRESS IN DIABETES: AN ACHIEVEMENT TO HEALTH AND HAPPINESS

Dr. Punita Sharma, Dr. Amit Awasthi, Dr. Mamta Bansal

The management of stress among women suffering with diabetes has majorly focused on four essential aspects viz. diet, exercise, meditation and checking blood glucose level regularly. 'Controlling Stress' is essential for treating diabetes, is largely overlooked. For the successful control on diabetes, stress needs to be controlled, avoid and kept to a minimum.

What is stress?

Stress is defined as a mental state of emotions, rigidity or tension that occurs when a person is not able to manage pressure. The prolonged stress can cause serious and extensive harm to heart, kidney, eyes, nerves and can further cause depression and mental health issues.

Causes of stress

Stress is caused by even minute things of daily life like

- Pressure related to work, job and extended office hours
- Marriages, friendships and peer pressure
- Financial insecurity
- Health related problems, ailments and diseases

Symptoms of stress

1. Sleeping too much or too less
2. eating too much or too less (Change in appetite)
3. Significant weight loss or weight gain
4. Frequent crying for no reasons
5. Trouble with recalling things, lack of concentration and/or anxious thoughts
6. Feeling low, depressed, uneasy, irritable and /or faint

7. Stomach problems and headache
8. Profuse sweating and teeth grinding
9. Loss of interest in sex

Types of stress affecting diabetes

Stress can be both physical and mental (Fig. 1). If a person is told about any chronic diseases like diabetes, thyroid or blood pressure causes lots of mental stress. This stress make more harder to control over blood sugar level, which in most cases become a reason weak immunity, frustration and many other diseases (Fig. 2).

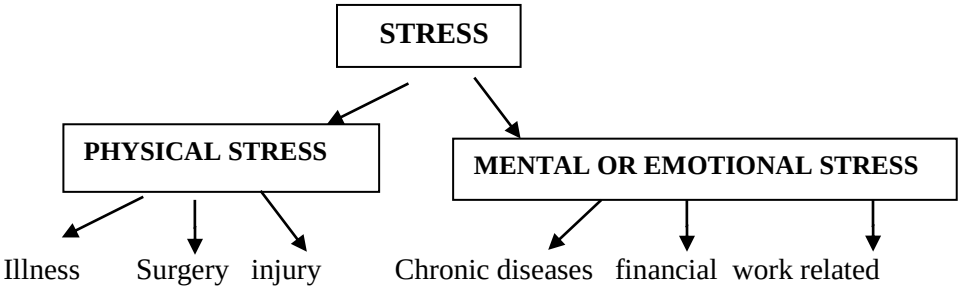

Fig. 1: Type of stresses

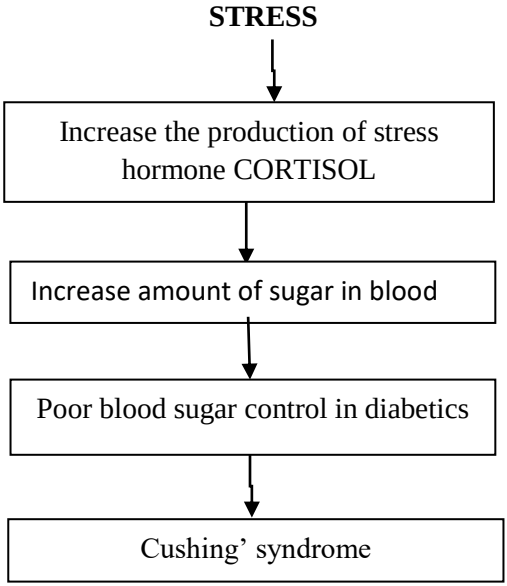

Fig 2: Stress affecting diabetes

Diabetes and stress

There are two types of diabetes that can alter blood sugar level.
In type 1 diabetes: stress increase or decrease blood sugar level.
In type 2 diabetes: stress generally increase blood sugar level.

Diabetes burnout

People with mental stress start ignoring their blood sugar level by

1) Forget to regular monitor
2) Adopt poor lifestyle habits, for example
   a) no or limited exercise
   b) Eating junk or processed food
   c) Drinking more alcohol
   d) Start Smoking

These situations are commonly known as diabetes burnout.

Fight-freeze or flight response in diabetes among women

At the time of stress, human body quickly responds by liberating important hormones which facilitates the frame to escape from danger. This instinctive response to recognize threats is known as the "Fight-freeze or flight reaction/response".

This is an evolutionary mechanism which automatic or reflex reaction where person act quickly to deal with stress, threat or fear. At this time when a person is threatened, hormones are released, that helps a person to get ready for

- Either fight the threat
  Or
- Quick flight/escape

In diabetic women, this Fight-freeze or flight response does not really work well, because insulin hormone is required to store glucose into the body's cells. But the process is hampered because insulin is not effectively used by the body or in certain cases it is not even produced. This results in the accumulation of excess glucose in the blood. There are lots of physical changes occurs in women in Fight-freeze or flight response as shown in Fig. 3. As at this moment women need more energy to combat with stress, glucose level in the blood increases to provide more energy to the body. Blood pressure also increases so that body can intake more fresh oxygen to work muscles well.

Few hormones level like adrenaline increases in blood to gain alertness.

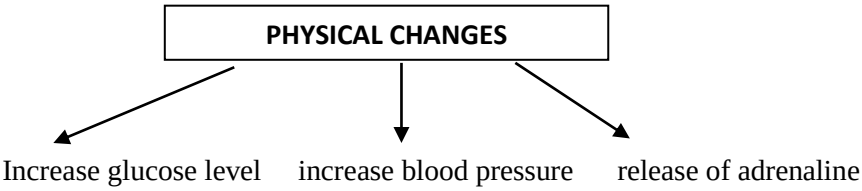

Fig. 3 Physical changes occur in women during Fight-freeze or flight response

"Diabetes" is a Greek name. But Romans noticed that the bees like a urine of diabetics, so they added the word "mellitus" which means sweet as honey.

Well, it is now known, the sugar is found in the urine of the diabetics.

Family and friend support in Diabetes to manage stress

A diagnosis of diabetes can be a stressful event for families as a whole. According to the opinion of a family, diabetes is a woman's' individual disease. But it is really a family matter because everyone in the family involved when a loved one has any physical or mental health issue. Family members can easily understand patient's feelings as woman with diabetes sometimes, go through stages of mental stress like grief, anger, sorrow, denial, depression. They also feel stressed, concerns and worries about the disease. Family members should talk to woman about her expectations from the members so that misconceptions corrected, real prospects sets and her feelings understood. Studies have proven that one of the excellent predicators of the way properly a person takes care in their diabetes is the amount of assist they get from their own circle of relatives and friends. A study by Miller and Matteo in 2013 has shown significant and positive relationships between support from the family and friends and

treatment observance among patients with diabetes. Result indicated the diabetes related stress is buffered when patient get social support.
Even, sometime family members want to support the diabetic person, but they do not know 'HOW to do this'. For this first of all, they need to

- Complete understand the concept of diabetes
- Meet doctor frequently with patient
- listen to them completely
- support them in making healthy changes in life style

The most important suggestions for the friends and family members are a) to have realistic information about diabetes and blood glucose levels and b) not to blame them unnecessarily because it is very important to understand that a woman spend more time in kitchen, and cannot always control blood sugar levels even if she follows all diabetic diet and sugar care plan. So, blaming the women with diabetes for blood sugar level (high or low) will never help, rather hurt emotions and feelings, leads to arguments and serious conflict. The genuine support is real problem solving decisions and stopping doing blames. Sometime, few words of sympathy, a hug or a rose for a diabetic woman is helpful. This kind of support at the time of frustration can be very helpful. So, it has become very important to understand the kind of support a diabetic woman need. To make support a real workable, diabetic woman also needs to set her mind to decide

- What, when and how support is required and from whom.
- Then her need to be told clearly to the people who are involved.

For this to happen, a diabetic women need to

- Aware the family members about the diabetes.
- Should clearly indicate and define how she wants family members, colleague and friend to help.

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Tips to coping with stress in diabetes

In diabetes, stress can significantly affect the ability to control blood sugar level by skipping the meals or forget to take medicines on time. So, learning to deal with the stress is very important in diabetes. Although it is difficult to remove stress completely from the life, but there are several ways to reduce it, so that diabetes can be better controlled. There are many methodologies, strategies and skills for keeping stress at a lowest or minimum. Here are some tips.

1. **Positive Attitude**
   If any situation in a family, a diabetic woman feels, going in a wrong direction, try to find the solution with calm mind and extract something good from that situation. Talk to trustworthy person in a family or friends. Think positive for work, talents, abilities, awards and health. By thinking about the positive things and good ideas would help to manage stress in diabetes.
2. **Make a habit to accept changes**
   For the stressful situations, issues and problems which are out of control and that cannot be changed, try to develop a simple plan of action. Sit at calm place and ask the following questions from the mind:
   - Will this problem be same and important one or two years from now?
   - Can control over this situation?
   - Am I in the position to change this situation?
     If the answer is 'NO', then make some plan to go for a holiday or simply listen to the favourite music.
3. **Talk to Someone About the stress**
   Talk about the stress with the family and friends and seek help from them. But, if it is difficult for you to talk with a friend or family member, there are counselors that provide support and insight. And if you want to talk with a counselor, ask the doctor for the help and recommendations.
4. **Practicing yoga, meditation and exercise**
   To reduce the stress, the effect of yoga, meditation and exercise are well known, particularly if a woman is suffering with chronic diseases like blood pressure, diabetes or thyroid. These activities can regulate blood sugar level by regulating insulin production and consumption that helps to maintain accurate blood pressure and burn calories. These activities also relax mind and body from the pressures of everyday life.
5. **Mindful living: An Antidote to Stress**
   Proper knowledge and awareness of what, why and how you are doing a particular work and being aware of what is good and what is not good for you is always a mindful approach to cope with diabetes and other diseases. That will also prevent unnecessary stress and inevitable stress at the minimum. Woman with housewife or in job, should always aware of what is happening around them and what is happening within self.

6. Stop stress at the source

Every woman should practice to prevent or stop the situation or things that stress either by leaving that place or by ignoring it. There are certain changes in your mind or in the lifestyle that can make to minimise the stress:

- **Traffic jam** – you can avoid heavy traffic by commuting on a different route or by leaving home/work at a bit early.
- **Work load** – a female can talk to the employer or colleague about making amendments to the work load, that will help to cope the work related stress. If woman still struggling to cope, she should apply for a transfer.
- **Relationships conflicts** - try taking the first steps to resolving any long lasting conflicts and mending your relationships.
- **Try something new** - it is very helpful technique for diabetic female to cope with diabetes to take up a new hobby or learning a new craft that introduce a positive factor into the life, which could be just what you need.
- **Managing diabetes** - Try joining a support group where and meet with other people in the same situation. Pick up tips on how to reduce the burden of diabetes-related stress.

Stress is word which is not fully understood but used widely. We all experience stress but we react in different way. A situation which is causing stress to one person but not for other. It is a link between the MIND & BODY. So, stress itself is not harmful, but one should know the way of dealing with it.

Conclusion

Diabetes care is 95% self care, even though a woman has the support of family, friends and the guidance of health care providers. Stress can actually be unhealthy and dangerous, particularly if a woman suffering with diabetes. So, don't give power to stress over your health. Reduce stress and take charge of your life. By learning the use of simple technique of stress busters, helps to better manage the diabetes. It solves

- Many problems
- Increase concentration
- Make emotionally smart
- Recognize and reframe unhealthy beliefs
- Harness the power of your imagination
- Use humor to stay positive and healthy
- Help to live grounded and mindful

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CHAPTER 13

WOMEN'S HEALTH-CAUSE OF CONCERN: LEGISLATIVE ASPECTS

Dr. Promil Diwan, Dr. Richa Lasta

Introduction

Right to life is considered one of the fundamental rights, and health is one of the vital indicators reflecting quality of human life. In this context, it becomes one of the primary responsibilities of the state to provide health care services to all its citizens. India, despite being a signatory to the Alma Ata Declaration of 1978, which promised 'Health for All' by 2000, is far from realizing this objective. On paper, India has an excellent health care structure that has the potential to reach a large section of the population. Yet, despite this elaborate structure and the rapid advancement of medical sciences, the reality is deplorable. The percentage of population covered by the public health care services is reportedly a mere 30 per cent. Although programmes are being constantly reviewed and revised, the problems persist and continue to worsen.[1]

Women are the precious creation of God because they have ability of procreation. But it is irony that they not given right over their own body. Decision related to their life and body is still taken by the men folk and it is assumed that they will follow them.[2]In India women have subordinate position and it has direct impact on the health of the women. Women hardly have access to health facilities. There is lack of awareness among women requiring quality of health and facilities available for good health. Women bear an inconsistent burden because of health issues and fail to

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[1] Women's right to health, Available at: https://nhrc.nic.in › sites › default › files › Women's, visited on28 Aug 2022

[2] Sayeda Urooj, "Reproductive right of women: Issues and Challenges", Women's Link, 2(Vol.18, No. 4 Oct.- Dec. 2012)

participate effectively in the development endeavour. The social, cultural and political structure of the society prevent women from enjoying basic right i.e. right to health.[3]

The health of Indian women is intrinsically linked to their status in society. Research on women's status has found that the contributions Indian women make to families often are overlooked, and instead they are viewed as economic burdens. There is a strong son preference in India, as sons are expected to care for parents as they age. This son preference, along with high dowry costs for daughters, sometimes results in the mistreatment of daughters. Further, Indian women have low levels of both education and formal labor force participation. They typically have little autonomy, living under the control of first their fathers, then their husbands, and finally their sons.

All these factors exert a negative impact on the health status of Indian women. Poor health has repercussions not only for women but also their families. Women in poor health are more likely to give birth to low weight infants. They also are less likely to be able to provide food and adequate care for their children. Finally, a woman's health affects the household economic well-being, as a woman in poor health will be less productive in the labor force. While women in India face many serious health concerns, this profile focuses on only five key issues: reproductive health, violence against women, nutritional status, unequal treatment of girls and boys.[4]
Health care services for women imply more than just access to the medical care and medicine. It requires to provide all facilities to the women, which are very essential for the protection and to maintain the health of women. Health denotes the conditions of a person's body or mind t include physical and mental health and healthy environment for protecting the around health of a human being.[5]

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[3] Sushma Sharma, "Reproductive rights and women's health : A new perspective", International Journal of Humanities and Social Sciences Invention, 51(Vol.4,Issue4, April-2015)
[4]Women's health in India, Available at:
https://www.census.gov/content/dam/Census/library/publications/1998/demo/wid98-3, visited on 29 Aug 2022.
[5] P S Jyoti, Mohan S Singhe "Understanding the concerns and promoting reproductive health right among women*", International Journal of Current Advanced Research*, 1012(Vol.5,Issue6, June 2016)

Right to health is the path through which we can achieve the other basic rights, which are related to the economic, social, cultural etc. Health is basic human rights and fundamental freedom. It is the result of living in accordance with the law of nature relating to the body, mind, and environment. It is not static phenomenon. It fluctuates within a range varying from optimum function to various levels of dysfunction. It is multidimensional physical, mental, and non medical.[6]

According to the WHO" Health is a state of complete physical mental and social well being and not merely the absence of disease.[7]

International provisions related to Healthcare for women United Nation Charter, 1945

The United Nation charter declares that the promotion of human rights and fundamental freedom for all without discrimination based on race, sex, language, or religion in one of its fundamental purposes.[8] This charter also provides provisions for higher standard of living, full employment, conditions of economics, social progress, and development and solutions of international economics, social, health and related problems.[9]

Universal Declaration of Human Rights, 1948

In 1948 the General Assembly of the United Nations adopted and proclaimed the *Universal Declaration of Human Rights*. It was meant to be a common standard of achievement for all peoples and all nations and their universal and effective recognition and observance is meant to be secured – both among the peoples of Member States themselves, which India is since 1945, and among the peoples of territories under their jurisdiction. According to the Right to Health of Women and Children, Article 25 affirms:

(1) Everyone has the right to a standard of living adequate for the health and well being of himself and of his family, including food, clothing, housing and medical care and necessary social services

(2) Motherhood and childhood are entitled to special care and assistance. All children, whether born in or out of wedlock, shall enjoy the same social protection (Universal Declaration of Human Right).

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[6] Jai S Singh, "The Human Right to health and social Justice: A survey", *Nyaya Deep*, 21(Vol. IX, Issue 4 Oct. 2008).

[7] Preamble to the Constitution of World Health Organization as adopted by the International Health Conference, New York.19-22 June 1946.

[8] Article 1(3) of the United Nations Charter, 1945.

[9] *Ibid.*, Article 55.

International Instruments

Between 1969 and 1997, the Government of India has ratified principal international human rights treaties dealing with Economic, Social and Cultural Rights; Civil and Political Rights; Racial Discrimination; Torture; Discrimination against Women and the Rights of the Child. (Office of the UNHCHR) Regarding to the 'Right to Health' of women and children, the *International Covenant on Economic, Social and Cultural Rights* (CESCR); the *International Convention of the Elimination of All Forms of Racial Discrimination* (CERD); the *Convention on the Elimination of All Forms of Discrimination against Women* (CEDAW) and the *Convention on the Rights of the Child* (CRC) are the most convenient.

International Covenant on Economic, Social, and Cultural Rights, 1966

The human rights are for all equality for men and women to the enjoyment of all economics, social and cultural rights set forth in the covenant.[10] Article 12 provides that the health right of all human kind including women; it provides that: the state parties to the present covenant recognize the right of everyone to the enjoyment of the highest attainable standard of physical and mental health.[11] Special provisions should be accorded to mother during a reasonable period before or after child birth.[12] The provision for the highest attainable standard of physical and mental health also mentioned in this covenant.[13]

The International Convention on The Elimination of All Form of Discrimination against Women. 1979

This convention provides international legal provisions for the protection and promotion of the women's rights. This convention has 30 articles pertaining to different aspects of the political, social, economic, health, employment, legal and right to marriage of the women and mechanism for the implementation of these rights. This convention also describes the reproductive rights of the women that in order to prevent discrimination

[10] Article 3 of the International Covenant on Economics, Social and Cultural Rights, 1966.
[11] *Ibid* , Article 12(1)
[12] *Ibid*, Article 10(z)
[13] *Ibid*, Article 12

against women on the ground of marriage and maternity and to ensure their effective right to work, state parties shall take appropriate measures.[14]

National Provisions related to Health Care for Women

Constitutional provisions

Part IV of the Indian Constitution which deals with the Directive principles of state policy imposes duty on state regarding the healthcare services for women in India. There are many provisions in the Directive Principles of State Policy which directly or indirectly deals with the health care services. Sate has the power to give orders for the promotion of welfare of the people but without public health we can't achieve it.[15] It is a obligation of the state to ensure good health under directive principle of state policy contained in articles 39(e) and (f), 42 and 47 of the Indian constitution. It is a duty of the state to provide public assistance basically for sick and disabled persons.[16]

Constitution makes provisions to protect the health of infant and mother by maternity benefits.[17] Article 47 deals with the primary duty of the state to improve public health, securing of justice, human, securing of justice, human condition of works, intention of sickness, old age, disablement and maternity benefits.[18] Article 48 A ensure that the state shall endeavor to protect the environment from pollution for good health.[19] Not only state, Panchayat and Municipalities are also liable to improve and protect public Health.[20]

Legislative Protections related to women healthcare

The Indian Penal Code 1860,

The Indian Penal Code 1860, is concerned with different offences that may be committed against the body, marriage, honor, modesty and property of women along with other offences. Sections 312 to 318 of IPC deal with

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[14]Article 11(2) of The International Convention on The Elimination of All Form of Discrimination against Women. 1979.

[15] The Constitution of India, Art. 38.
[16] *Ibid*, Article 41
[17] *Ibid*, Article 42
[18] Ratlam Municipal Corporation v Vardicahnd, AIR 1980 SC 1622.
[19] The Constitution of India, Art. 48A
[20] *Ibid*, Article 243G

unlawful miscarriages and also the infanticide as well as feticide. For example: causing of miscarriage[21], injuries to unborn children[22], of the exposure of infants[23] and of the concealments of births[24].

The National Commission for Women Act, 1990

An age long agitation against the discrimination of women inside and outside the parliament has been continued by the parliamentarians by common men and women, by organizations and societies. Several Commissions have been set up by the govt. to look into the matter of status of women in the Indian Society. All the Commissions reported about unequal status of women in every sphere of life. Realizing the need of setting up an agency to fulfill the surveillance functions as well as to facilitate redressal of the grievances of women, the government decided to set up a commission for women and enacted the National Commission for Women Act, 1990.[25]

The Maternity Benefit Act, 1961

The Maternity Benefit Act, 1961 was enacted to regulate the employment of women in certain establishments for certain periods before and after child birth and to provide for maternity benefit and certain other benefits. The Act was intended to achieve the object of doing social justice to women workers. Therefore, the Court held in *B. Shah* v *Labour Court, Combatore*[26] that in interpretation the provisions of this act the court has to adopt the beneficent rule of construction which enable the woman not only to subsist but also to make up her dissipated energy, nurse her child, preserve her efficiency as a worker and maintain the level of her previous efficiency and output.

The Medical Termination of Pregnancy Act 1971

In recent years health services are available everywhere and the service hospitals with qualified doctors are available to all cases of people. The government proposed to liberalize certain existing provisions relating to

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[21] The Indian Penal Code, 1860, sections 312 and 314
[22] *Ibid*., Section 315 and 316
[23] *Ibid*., Section 317
[24] *Ibid*., Section 318
[25] The National Commission for Women Act, 1990
[26] AIR 1978 SC 12

termination of pregnancy to avoid loss of the mother's health, strength, and life. The Government enacted The Medical Termination of Pregnancy Act 1971, to provide for the termination of pregnancy by registered medical practitioners where its continuity would involve a risk to life or grave injury to her physical or mental health.[27]

The Pre Natal Diagnostic Techniques (Regulation and Prevention of Misuse) Act, 1994

In olden days it was impossible to determine the sex of the baby in the womb of mother until it was delivered. As medicine advanced, new techniques were devised for preventing the genetic, chromosomal disorders of the child in the womb but this advancement turned into a curse towards female child and her health. Instead of using these techniques for the medical purposes, the medical practitioners started using them only for the purposes determine the sex of the child in the womb. This Act has to provide for the prohibition of sex selection, before or after conception and for regulation of pre natal diagnostic techniques or chromosomal abnormalities or certain congenital malformations or sex linked disorders and for the prevention of their misuse for sex determination leading to female feticide and for matters connected therewith or incidental thereto.[28]

The Surrogacy (Regulation) Act 2021

The Act aims to regulate surrogacy in India by establishing a National Surrogacy Board at the central level and state surrogacy boards and appropriate authorities in states and union territories. Under the Surrogacy (Regulation) Act, 2021, a woman who is a widow or a divorcee between the age of 35 to 45 years or a couple, defined as a legally married woman and man, can avail of surrogacy if they have a medical condition necessitating this option. It also bans commercial surrogacy, which is punishable with a jail term of 10 years and a fine of up to Rs 10 lakhs. The law allows only altruistic surrogacy where no money exchanges hands and where a surrogate mother is genetically related to those seeking a child.[29]

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[27] The Medical Termination of Pregnancy Act 1971

[28] The Pre Natal Diagnostic Techniques (Regulation and Prevention of Misuse) Act, 1994

[29] The Surrogacy (Regulation) Act 2021

Besides, Indian Parliament has enacted several laws that are indirectly related to the healthcare for women in India. These include the The Drugs and Cosmetic Act, 1940, the Equal Remuneration Act, 1976, Working Hours and Conditions for Women Workers, the Fatal Accident act 1855, The Transplantation of Human Organ Act 1994, the Prevention of Food Adulteration Act, 1954, the Immoral Traffic (Prevention) Act 1956, the Dowry Prohibition Act 1961,etc.

Judicial approach towards the health care facilities provided for women in India

The courts during judicial explanation have expanded the scope of article 21 to bring within its fold the right to healthcare for women. In ***State of Punjab* v *Ram Lubhaya Bagga***[30]observed that right of one-person correlates to a duty upon another individual, employer, government and authority. Hence the right of a citizen to live under art. 21 cast an obligation on state. This obligation is further reinforced under art.47; it is for the state to secure health to its citizens as its primary duty. The court held that the state can neither urge nor say that it has no obligation to provide medical facilities. If that were so, it would be ex facie in violation of art. 21.

The Supreme court in ***Paschim Banga Khet Mazdeer Samity and others* v *State of West Bengal and anothers***[31]held that it is the primary duty of the government to secure the welfare of the people. The government discharges this obligation by providing healthcare services to the persons seeking to avail of those facilities.

In the case ***Pt. Parmanand Katara* v *UOI and others***[32]SC held that every doctor whether at a government hospital or otherwise has the professional obligation to extend his healthcare services with due expertise for protecting life. No one can intervene to avoid/delay the discharge of the paramount obligation cast upon the medical professionals.

In ***Badhua Mukti Morcha* v *UOI***[33]it was held that it is the fundamental right of everyone in this country, assured under the interpretation given to

[30] (1998)4 SCC 117
[31] (1996)4 SCC 37
[32] AIR 1989 SC 2039
[33] AIR 1984 SC 802

art. 21by the court in ***Francis Mullins case***[34]to live with human dignity, free from exploitation. ***In CBSE Ltd* v *Subhash Chandra Bose***[35]court held that term health, implies more than an absence of sickness.

The court, while reiterating its stand for providing health facilities in ***Vincent* v *UOI***[36]held that a healthy body is a very foundation for all human activities. Therefore, it is obligation of the state to ensure the creation and the sustaining of condition congenial to good health.

The ***Justice K S Puttaswamy* v *UOI*** [37] judgment specifically recognized the constitutional right of women to make reproductive choices, as a part of personal liberty under Article 21 of the Indian Constitution. The bench also reiterated the position adopted by a three-judge bench in ***Suchita Srivastava v Chandigarh Administration***[38] which held that reproductive rights include a woman's entitlement to carry a pregnancy to its full term, to give birth, and to subsequently raise children; and that these rights form part of a woman's right to privacy, dignity, and bodily integrity.

Conclusion

A healthcare service means the privileges of women to attain the highest standard of healthcare services and at the same time attaining full participation in the social and economic life. The healthcare services for women is very indispensable part of the society but due to lack of concern and awareness among the government, executive and public it turns into a big issue of the today's scenario. Therefore, there is a requirement to have access to suitable, reasonable and excellence health care facilities and related services for women. There is a necessity for appropriate legal outline to address and identify the healthcare services for women in India. Heath care services should emphasis more on women's health including reproductive health. The need of the hour is to have proper implementation of laws and programs that has made for the health.

[34] AIR 1980SC 849
[35] (1992) 1 SCC 461
[36] AIR 1987 SC 994
[37] (2017) 10 SCC 1
[38] (2009) 14 SCR 989

ABBREVIATION

AIDS- Acquired Immuno Deficiency Syndrome

AMFAR- American Foundation for AIDS Research

ANC - Ante-natal Care

ART - Anti Retroviral Therapy

ASHA - Accredited Social Health Activists

BAI - Body adiposity index,

BDI - Body density index

BRCA - Breast Cancer gene

CBT - Cognitive Behaviour Therapy

CD4+ - Cluster of Differentiation 4

CDC - Centers for Disease Control and Prevention

CHCS - Community Health Centers .

CVD: CardioVascular disease

DALYs - Disability-Adjusted Life Years

DRC - Depression Related Suicide

FDA - Food and Drug Administration

FOMO - Fear of Missing Out

FNAC - Fine Needle Aspiration Cytology

HDL - High-density Lipoprotein

HINTS - Health Information National Trends Survey

HIV - Human Immuno Deficiency Virus

HPV - Human Papilloma Viruses

HRT- Hormone Replacement Therapy

HSV - Herpes Simplex Virus

GAMA- Global Action for Measurement of Adolescent health

IBD - Inflammatory Bowel disease

IBS - Irritable Bowel Syndrome

ICDS - Integrated Child development Services

IVF - In Vitro fertilization

IUD - Intrauterine device

JSY- Janani Suraksha Yojana

KS - Kaposi Sarcoma

LDL - Low-density Lipoprotein

MTCT - Mother to Child Transmission

MMWR - Morbidity and Mortality Report

NACO - National AIDS Control Organization

NACP - National AIDS Control Programme

NCDs - Non-Communicable Diseases

NFHS- National Family Health Survey

NICU - Neonatal Intensive Care Unit

NIH -National Institutes of Health

NIMH - National Institute of Mental Health

NIMHANS - National Institute of Mental Health and Neurosciences

NRHM - National Rural Health Mission

NSSO- National Sample Survey Organization

OCD- Obsessive-Compulsive Disorder

OIs - Opportunistic Infections

PCOD- Polycystic Ovary/Ovarian Disease

PCOS - Polycystic Ovary/Ovarian Syndrome

PEPFAR - President's Emergency Plan for AIDS Relief

PHC - Primary Health Care

PLHIV - People Living with HIV/AIDS

PMD - Potentially Malignant Disease

PMDD - Premenstrual Dysphoric Disorder

PrEP - Pre-Exposure Prophylaxis

REBT - Rational Emotive Behaviour Therapy

RRE - Red Ribbon Express

SACS - State level AIDS Control Societies

SAD - Seasonal Affective Disorder

SIMS - Strategic Information Management Services

STD - Sexually Transmitted Diseases

STIs. - Sexual Transmitted Infections

TB – Tuberculosis

UNAIDS- United Nations Programme on HIV/AIDS

UNESCO-United Nations Educational, Scientific and Cultural Organization

UNFAP- United Nations Fund for Population Activities

UNICEF- United Nations International Children's Emergency Fund

UN Women- United Nations Women

UTI - Urinary Tract Infections

VIA - Visual Inspection with Acetic Acid

VTCT - Voluntary Counselling and Testing

WHO – World Health Organization

WFP- World Food Programme

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About Book

The health of a woman is of prime importance to create a healthy society and health world. This book on 'Women Health-Vol I' is prepared in accordance with the current scenario of women which is broader in scope and extensive in contents. The vision and mission of this book is to provide commitment to improve the health and well- being of women and girls globally. This book will help students, researchers, health care workers and other readers to understand the current woman health understandings. It will also be helpful to meet the millennium development goals in the Women health perspectives. Starting with the introduction and definitions, we have thoroughly discussed all components of Women health extensively as individual chapters in the book. The wider approach to Women health and related chapters assess more about a complete woman-centered view rather than only a disease-centered opinion. The book highlights the importance of quality of life rather than survival, disease or mortality due health issues. Emphasis is given on the issues and challenges and on how to tackle and overcome the challenges which sometime is not easily accessible to the readers. The book does not only evaluate treatments and interventions of diseases but also greatly focused on the way how a woman can spend their life in a better way. The significant knowledge, information and communication regarding health of a woman ensure their safety, improve health of a family and population, mental health, facilitate the delivery of government healthcare services.

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